

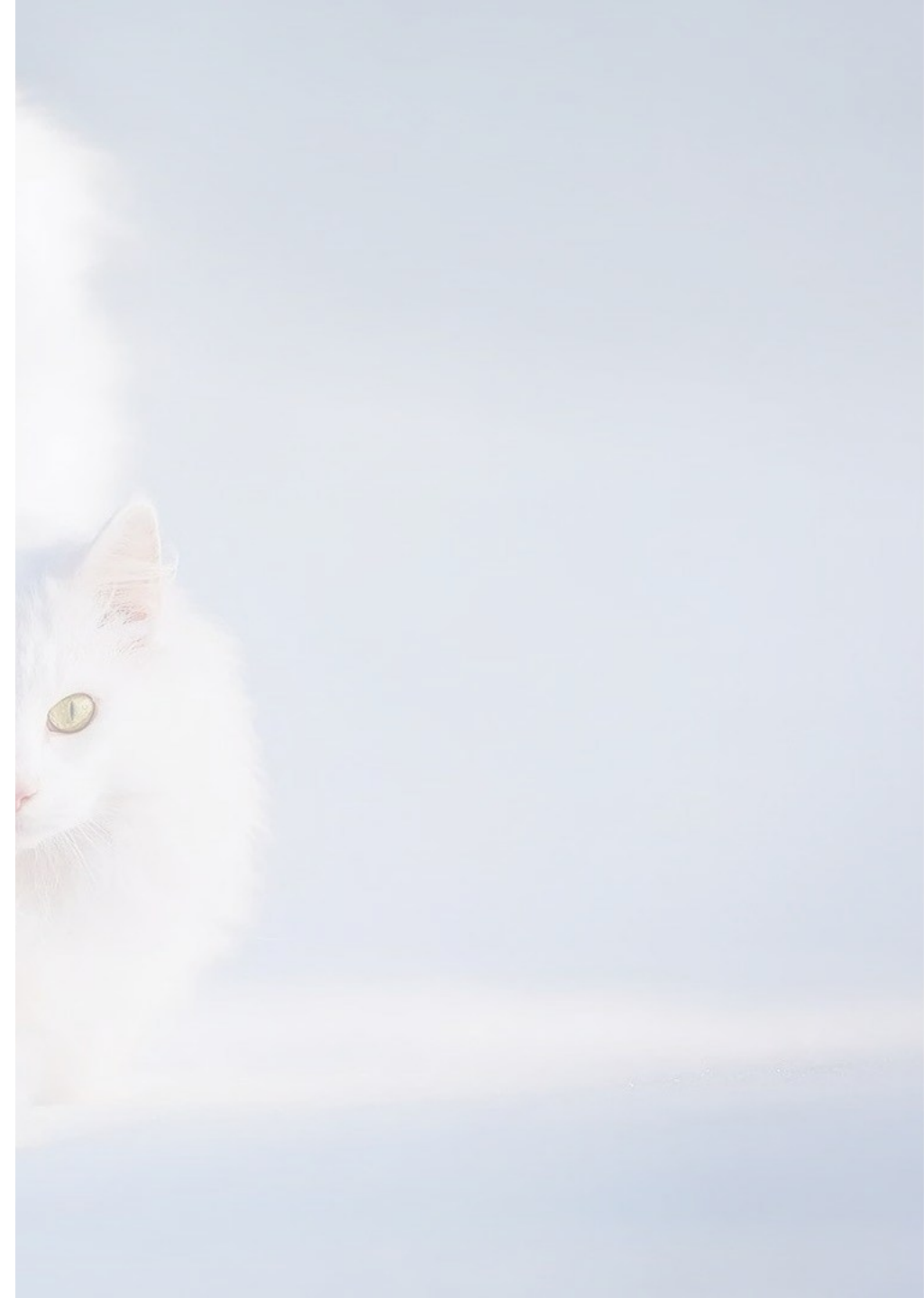




# Módulo 2 – Machine Learning

## Bloque Temático 3 y 4

- Repaso -> Feedforward.
- Gradiente Descendiente.
- Backpropagation.

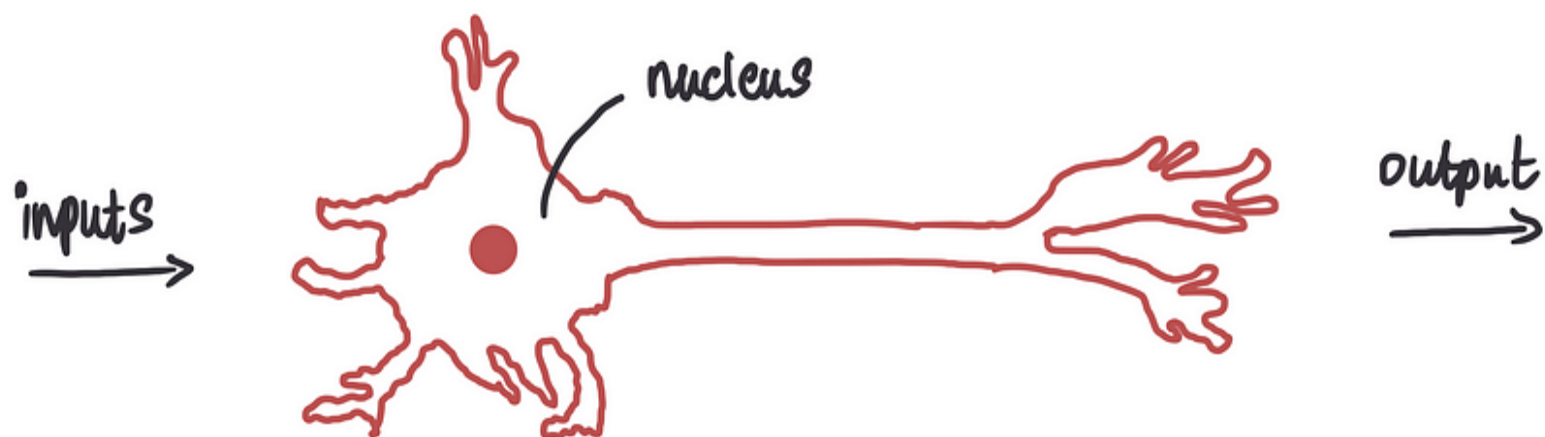


# **Introducción a Redes Neuronales**

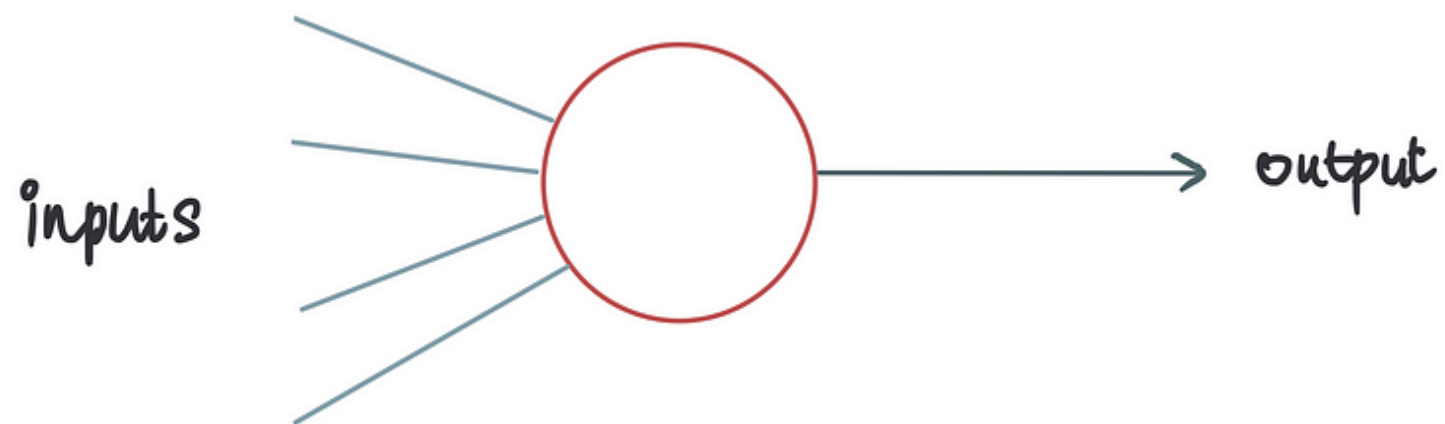
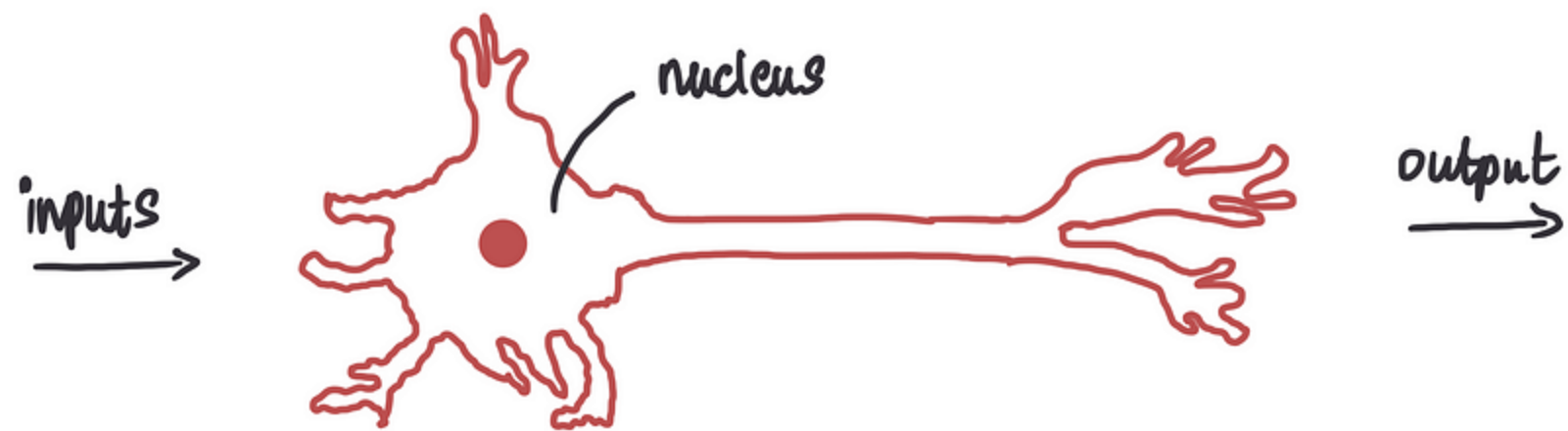
## **FeedFoward**





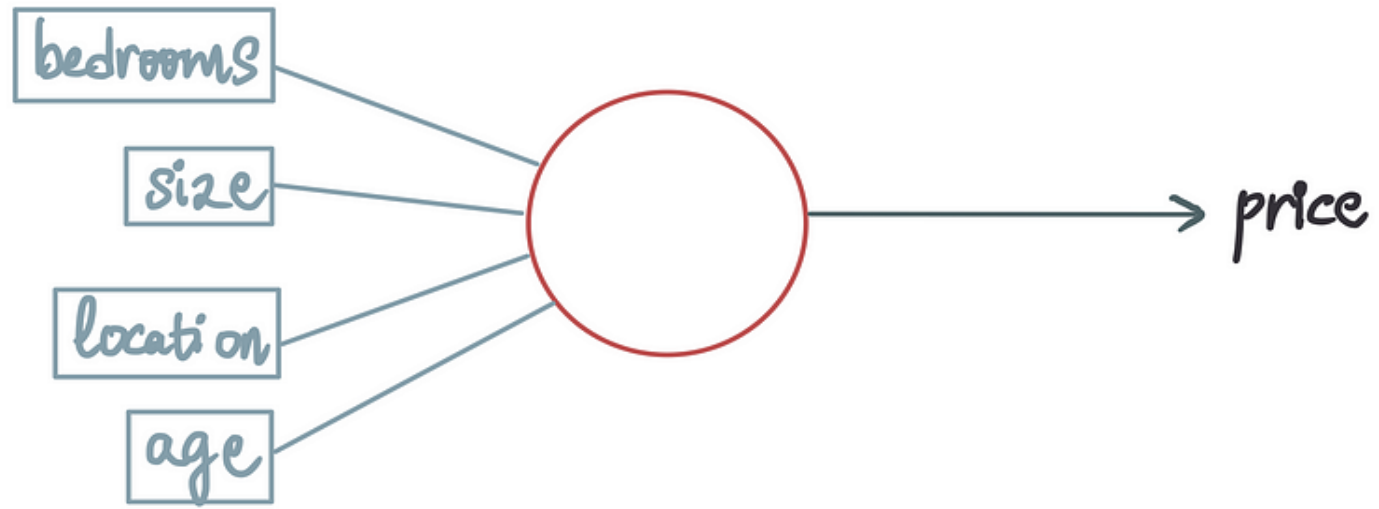




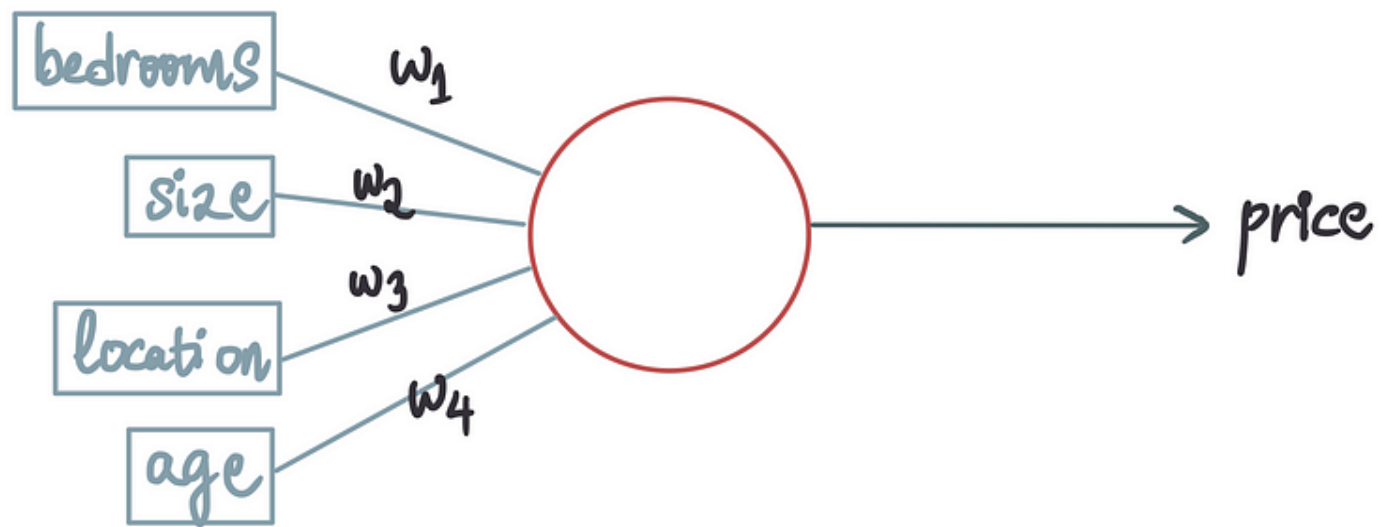


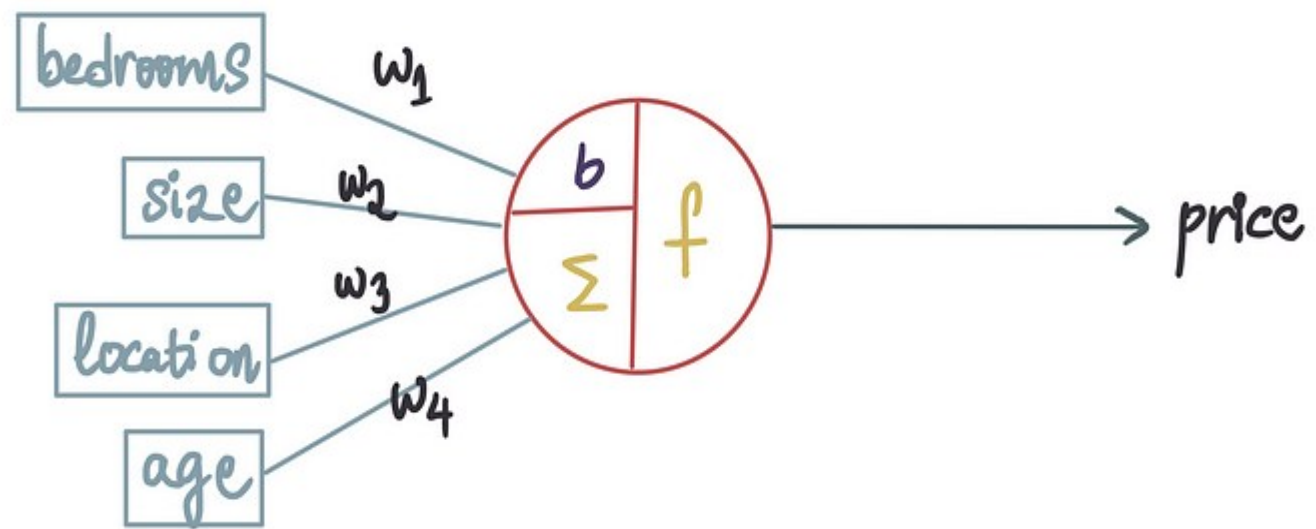


| bedrooms | size | location | age | price |
|----------|------|----------|-----|-------|
| 4        | 2680 | 2        | 8   | 1.75M |
| 5        | 700  | 1        | 27  | 750K  |
| 3        | 2360 | 1        | 28  | 1.2M  |
| 5        | 1280 | 2        | 98  | 1.05M |
| 5        | 3480 | 3        | 21  | 2.3M  |

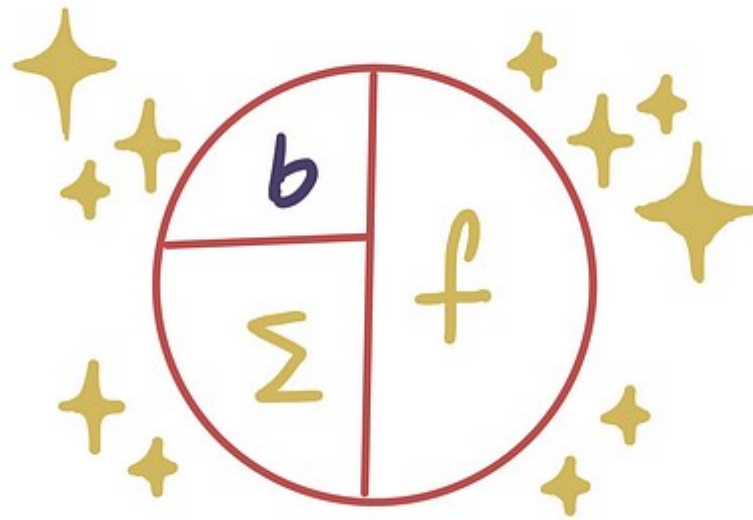


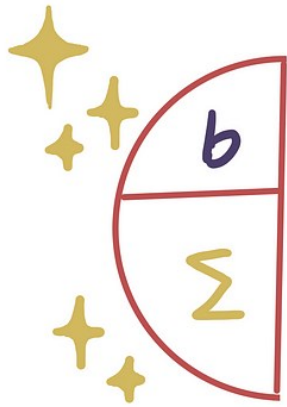


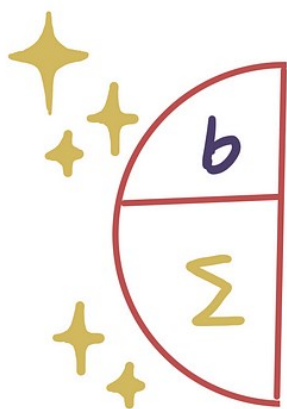






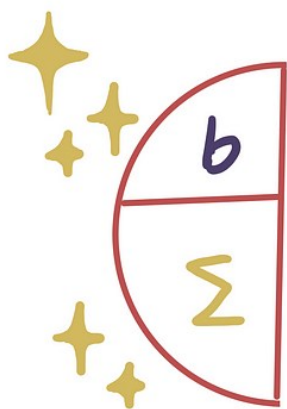






$$w_1 \cdot \text{bedrooms} + w_2 \cdot \text{size} + w_3 \cdot \text{location} + w_4 \cdot \text{age}$$





$$w_1 \cdot \text{bedrooms} + w_2 \cdot \text{size} + w_3 \cdot \text{location} + w_4 \cdot \text{age}$$

$$w_1 \cdot \text{bedrooms} + w_2 \cdot \text{size} + w_3 \cdot \text{location} + w_4 \cdot \text{age} + \underset{\text{bias}}{\overset{c}{b}}$$

magic, part 1

$$\begin{aligned} &w_1 \cdot \text{bedrooms} + w_2 \cdot \text{size} + w_3 \cdot \text{location} + w_4 \cdot \text{age} + b \\ &= \sum_{i=1}^n w_i x_i + b \end{aligned}$$

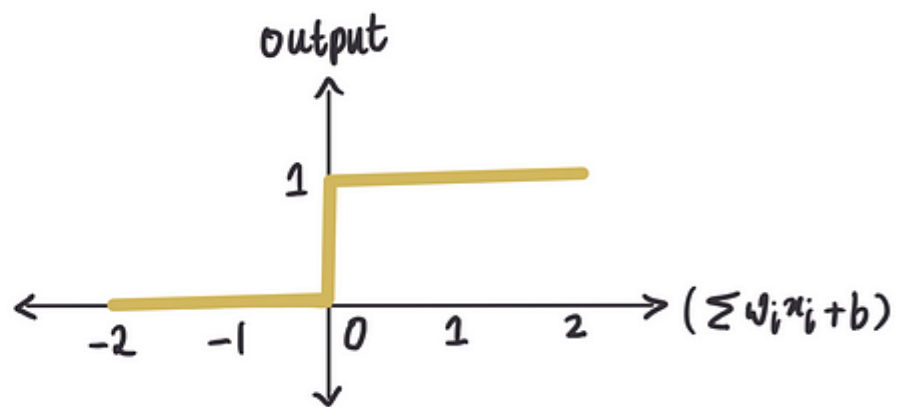
magic, part 2

$$f\left(\underbrace{\sum_{i=0}^n w_i x_i + b}_{\text{input}}\right) = \text{output}$$

binary step

$$f(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

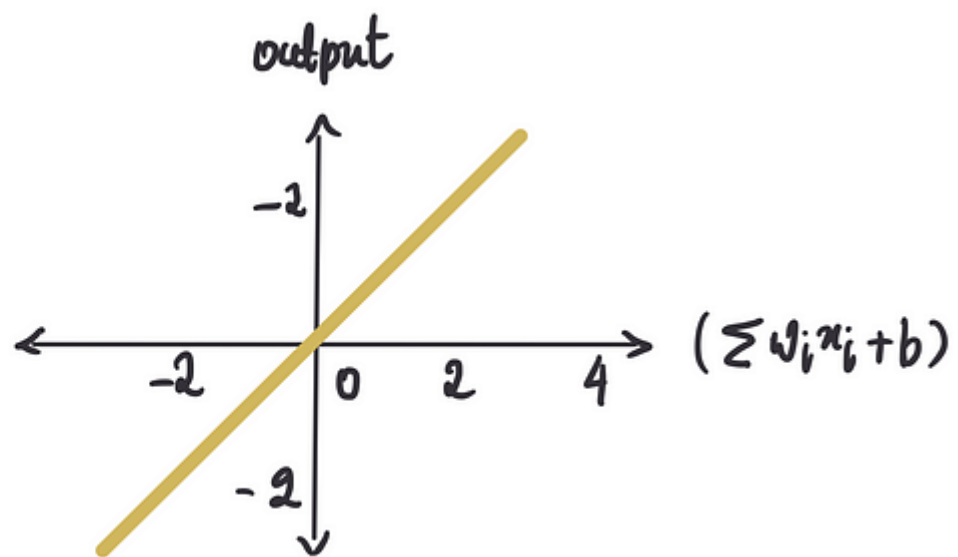
Some  
input





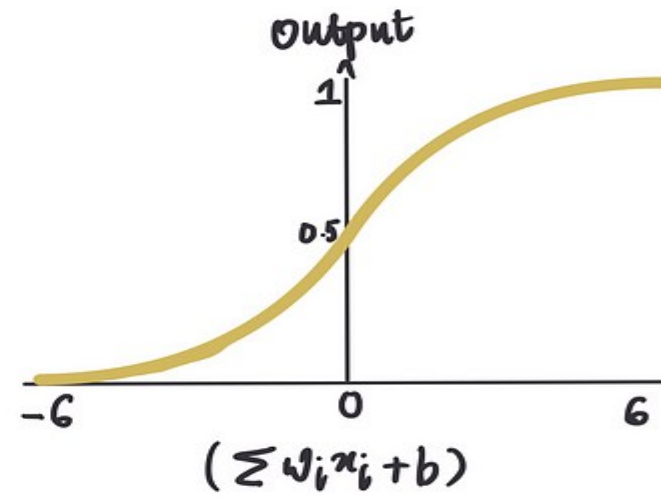
linear activation

$$f(x) = x$$



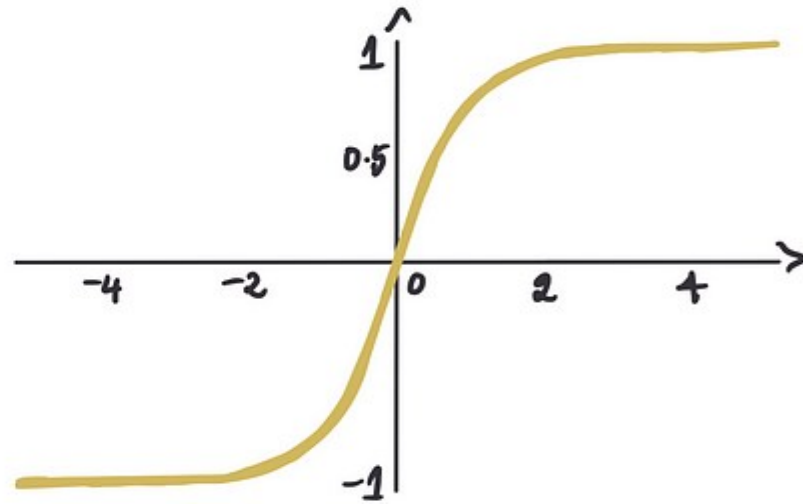
Sigmoid

$$f(x) = \frac{1}{1 + e^{-x}}$$



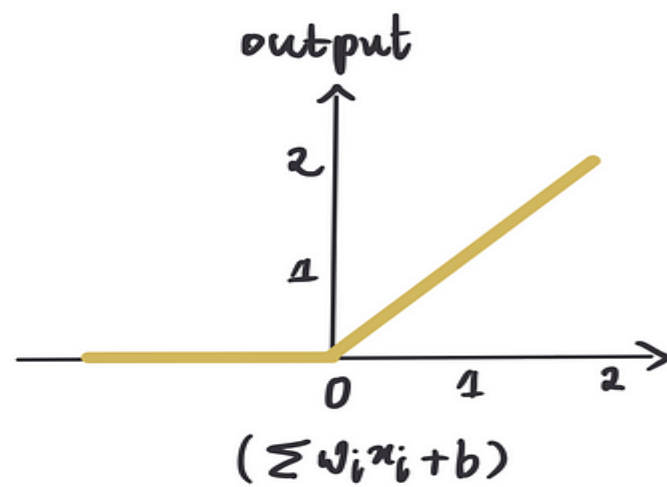
hyperbolic tangent

$$f(x) = \frac{1 - e^{-2x}}{1 + e^{-2x}}$$



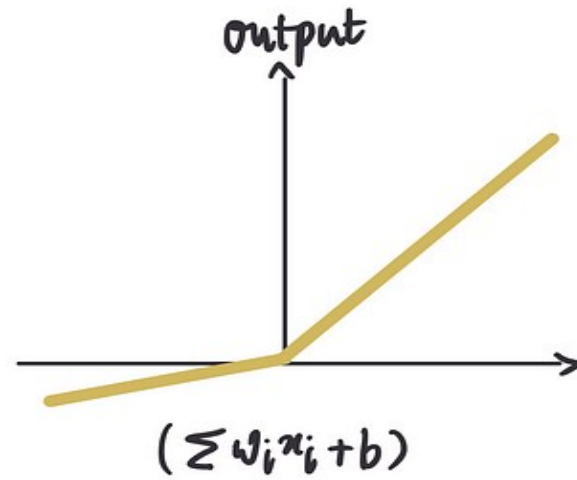
rectified linear units

$$f(x) = \max(x, 0)$$



leaky ReLU

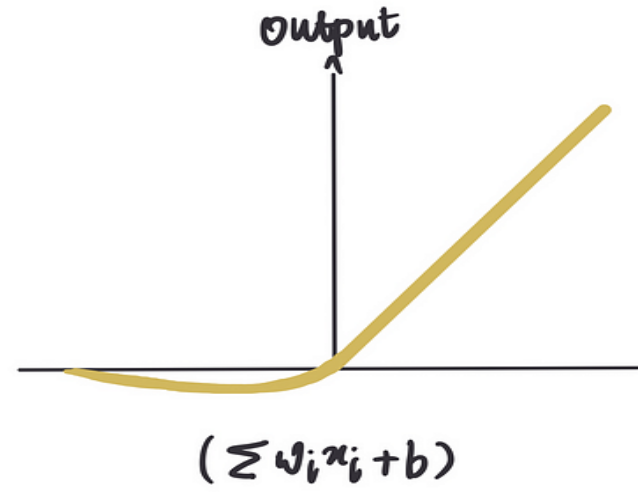
$$f(x) = \begin{cases} 0.01x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$

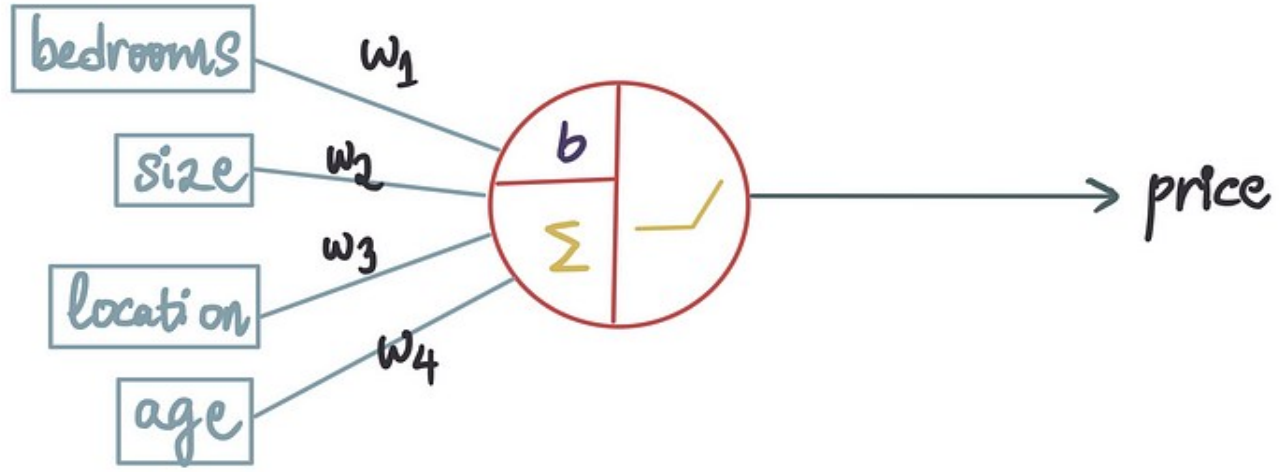


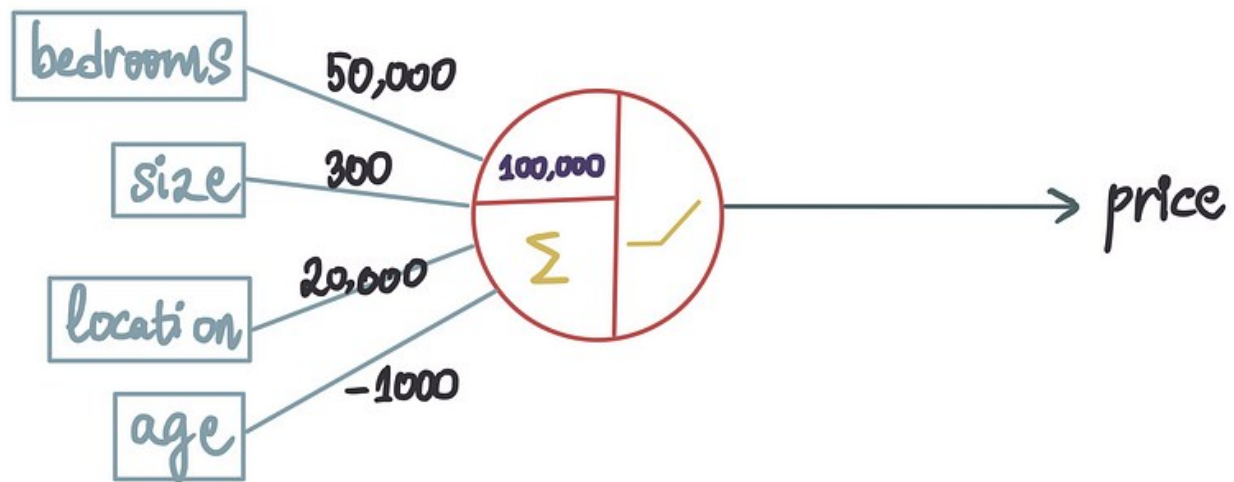


swish

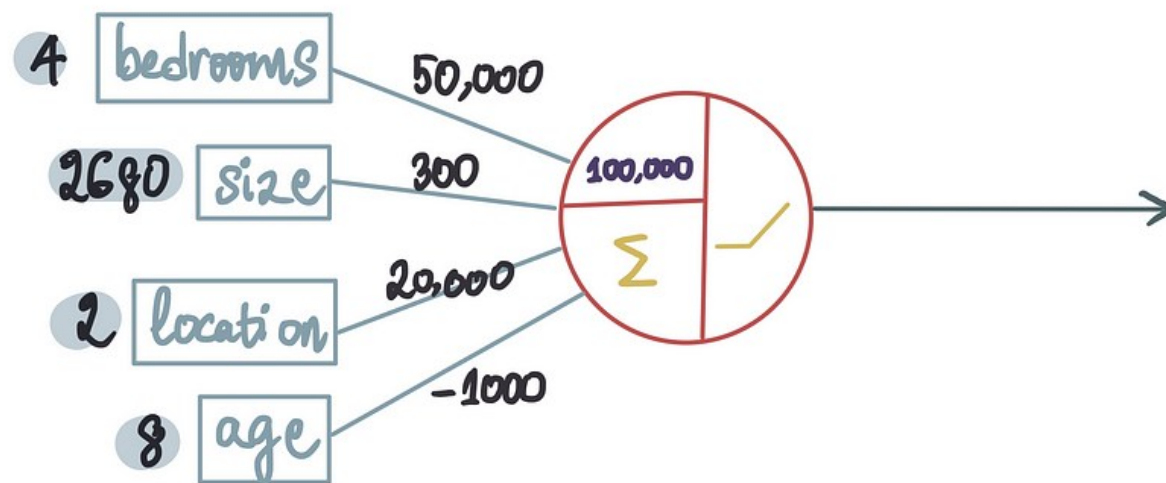
$$f(x) = \frac{x}{1 + e^{-x}}$$







| bedrooms | size | location | age | price |
|----------|------|----------|-----|-------|
| 4        | 2680 | 2        | 8   | 1.75M |



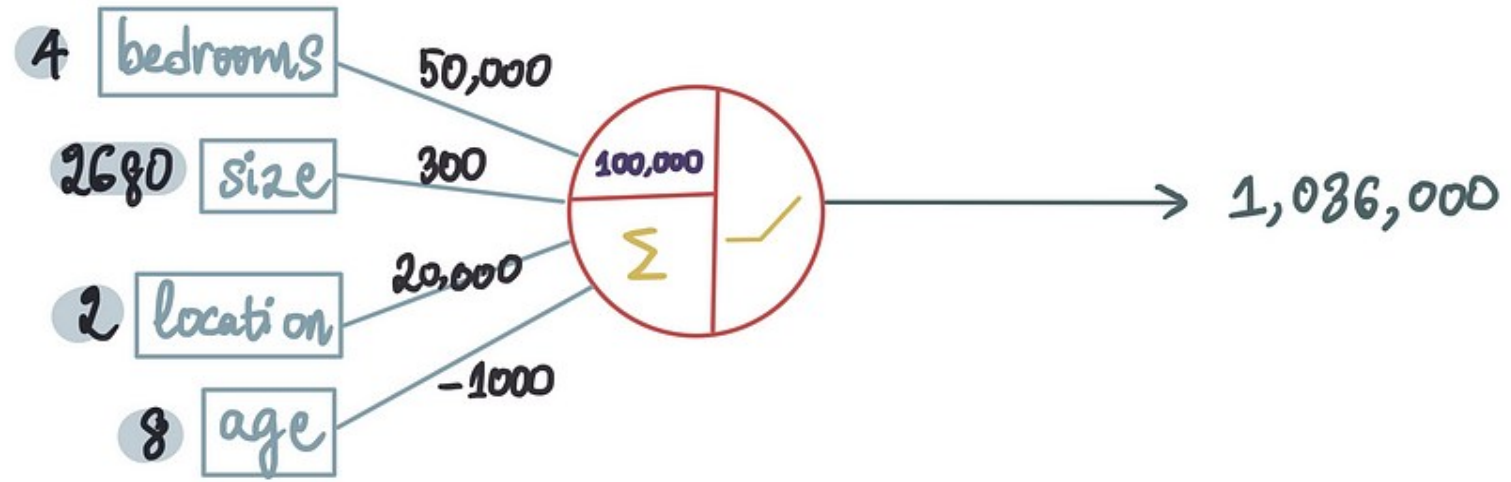
$$\sum_{i=0} w_i x_i + b$$

$$= 50,000 \cdot 4 + 300 \cdot 2680 + 2 \cdot 20,000 + 8 \cdot -1000 + 100,000$$

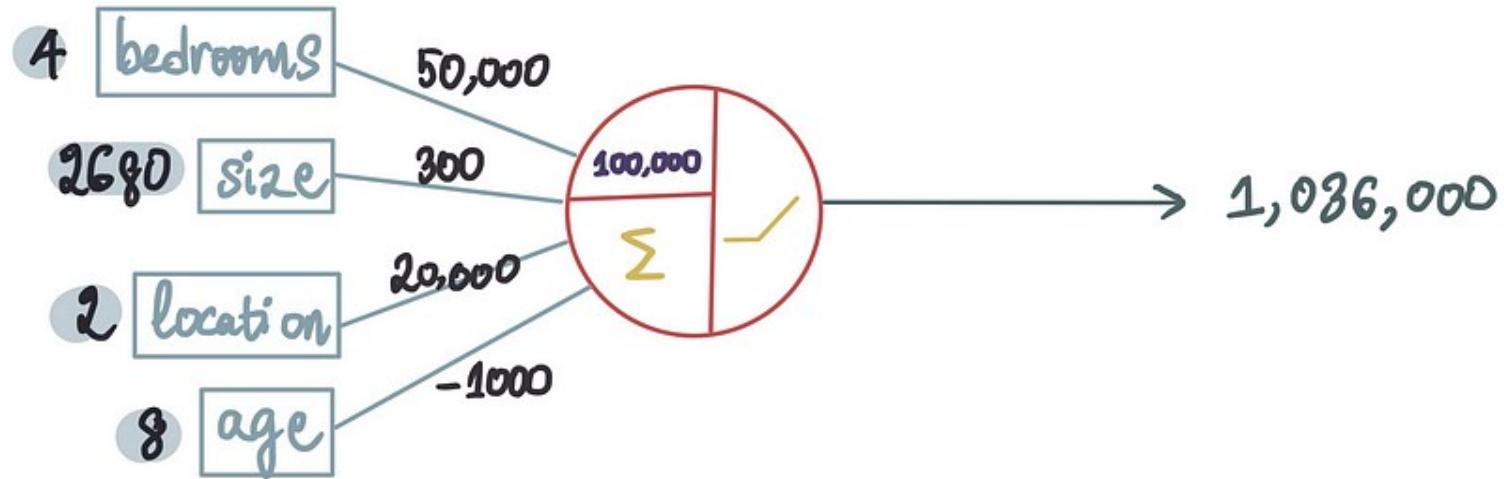
$$= 1,036,000$$

$$f(x) = \max(x, 0)$$

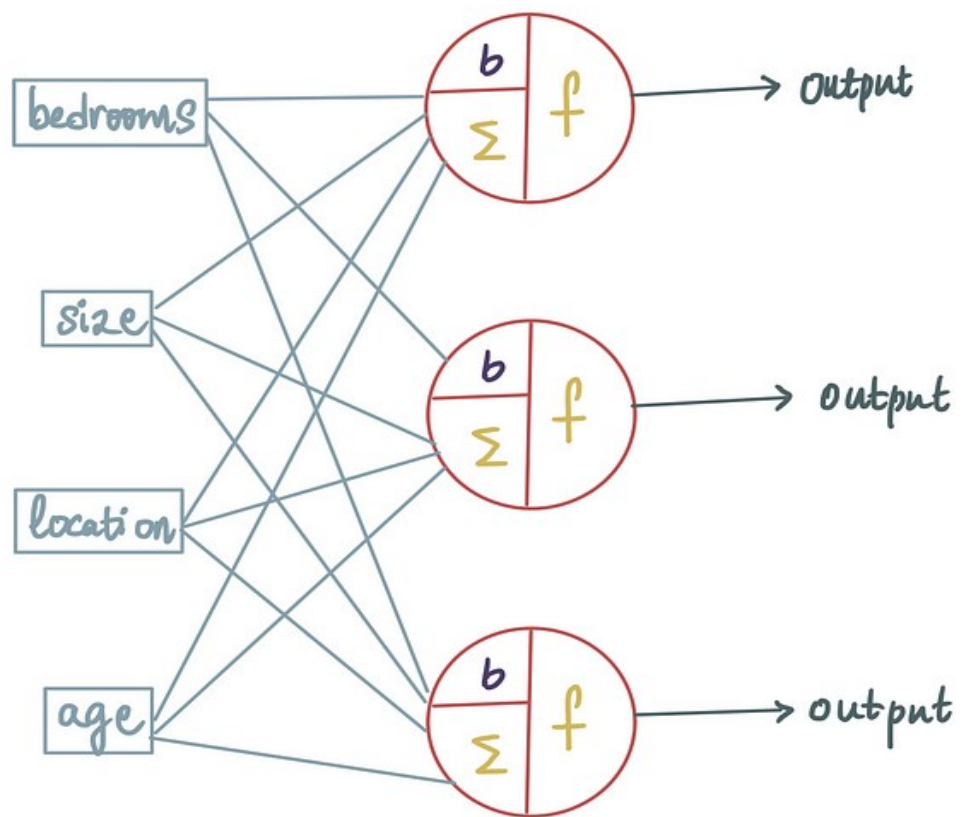
$$\Rightarrow f(1,036,000) = 1,036,000$$

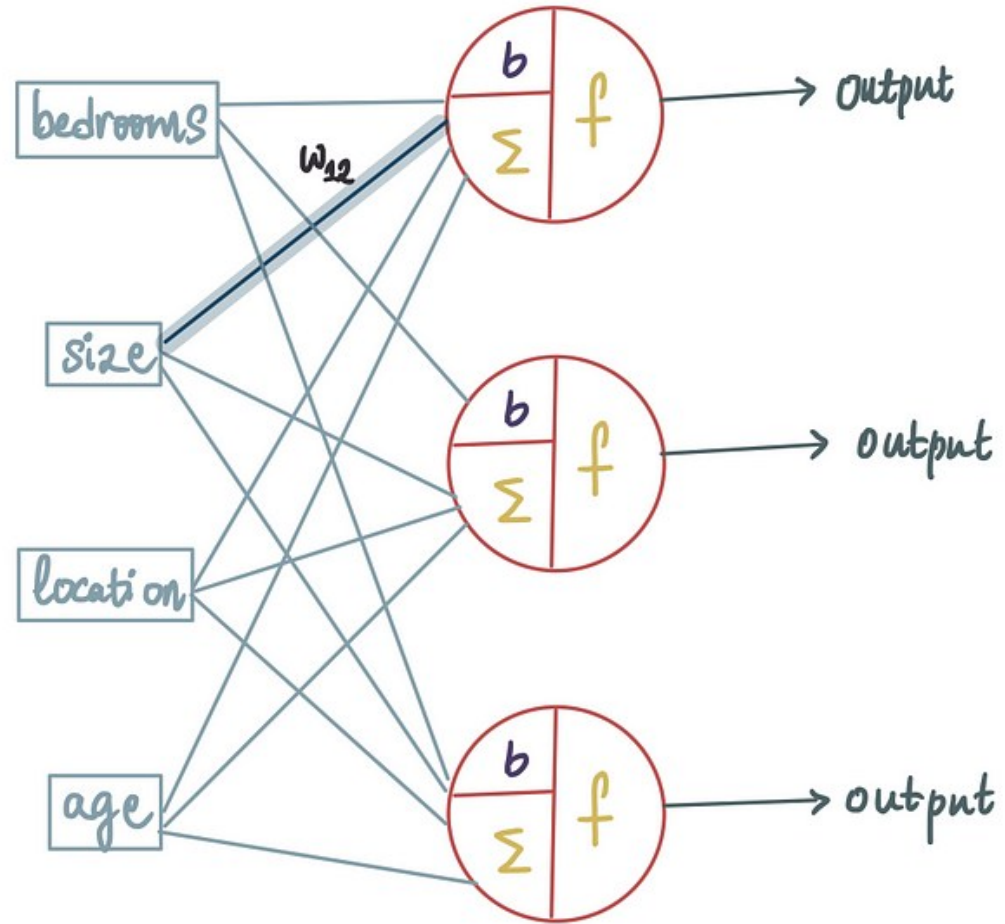


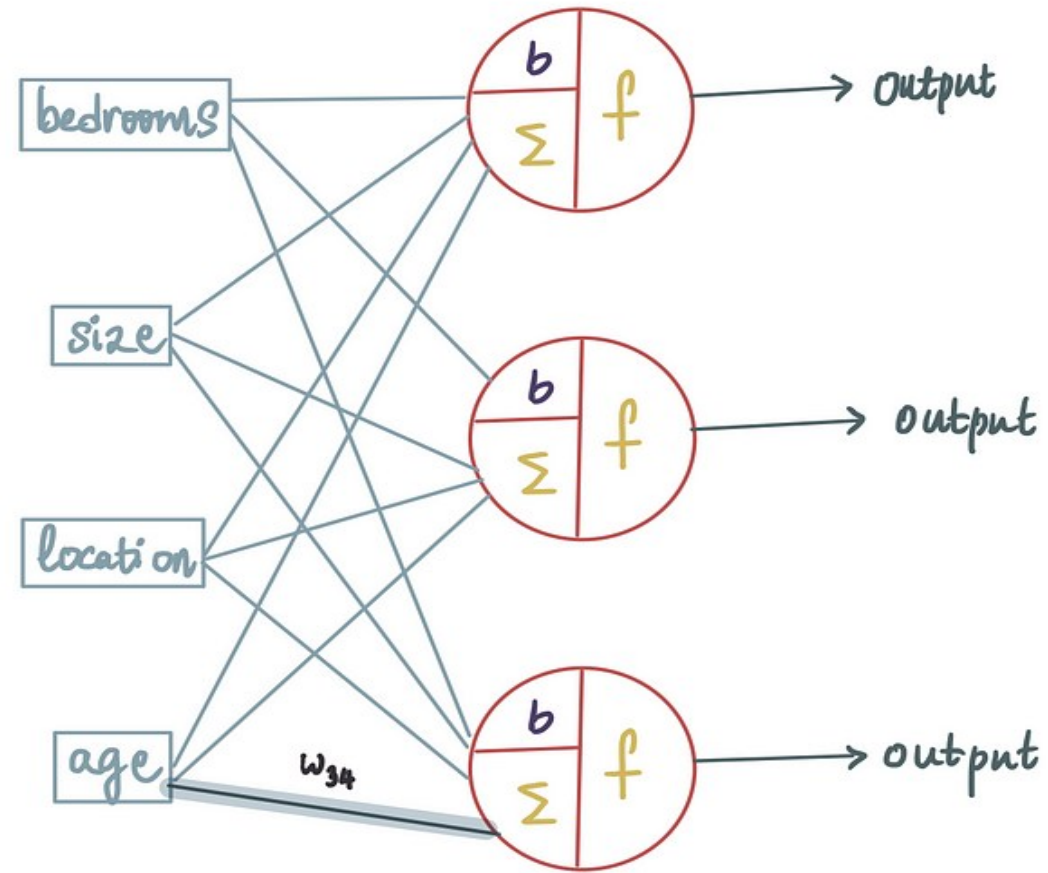


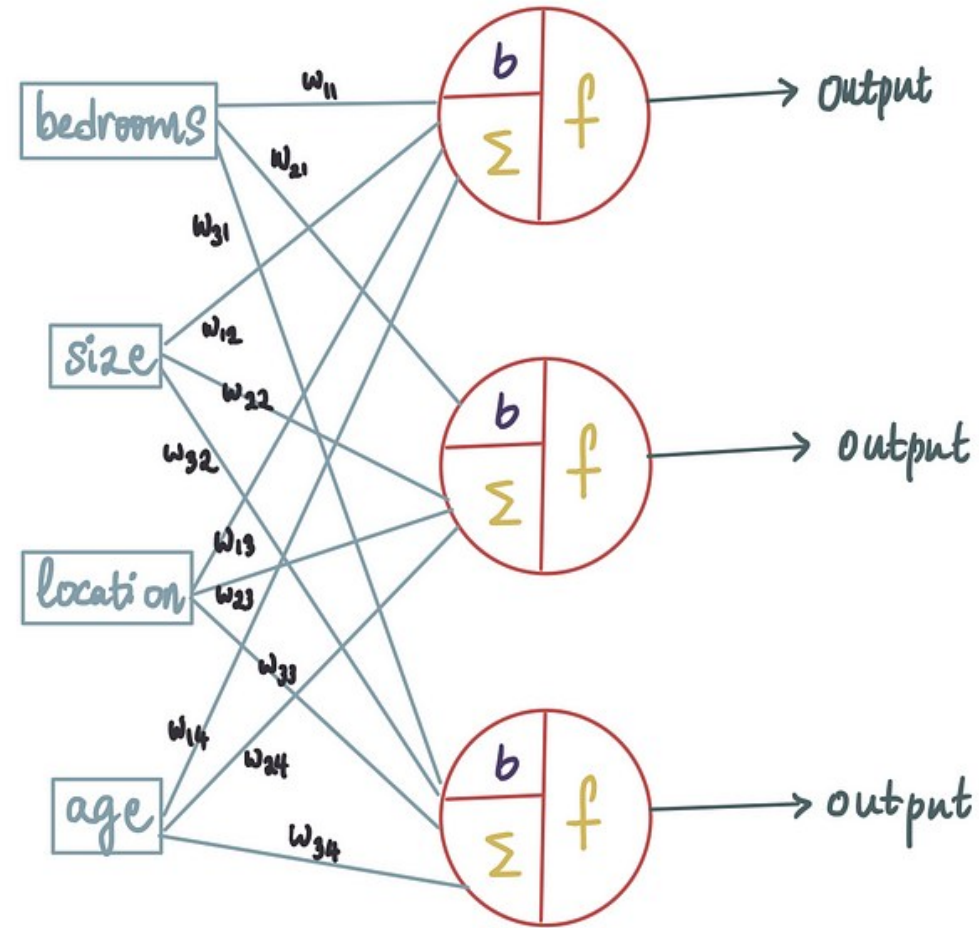


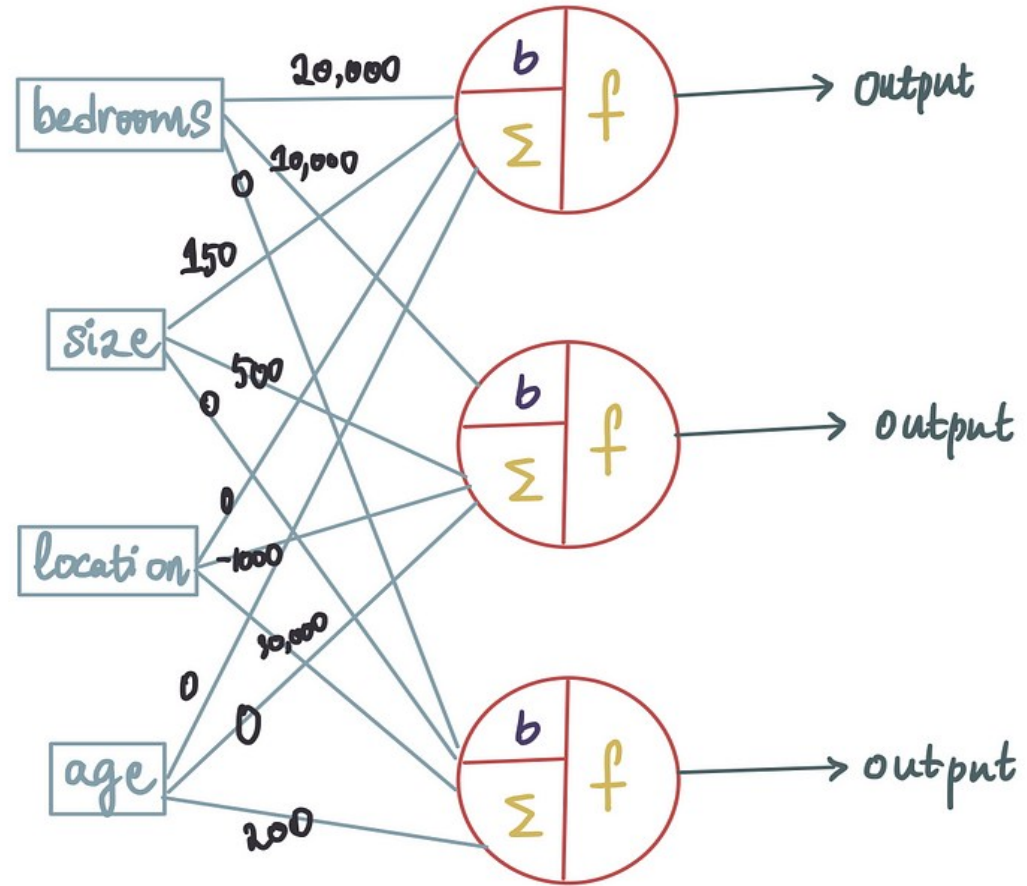
| bedrooms | size | location | age | price | predicted price |
|----------|------|----------|-----|-------|-----------------|
| 4        | 2680 | 2        | 8   | 1.75M | 1,036,000       |
| 5        | 700  | 1        | 27  | 750K  | 453,000         |
| 3        | 2360 | 1        | 28  | 1.2M  | 870,000         |
| 5        | 1280 | 2        | 98  | 1.05M | 676,000         |
| 5        | 3480 | 3        | 21  | 2.3M  | 1,333,000       |

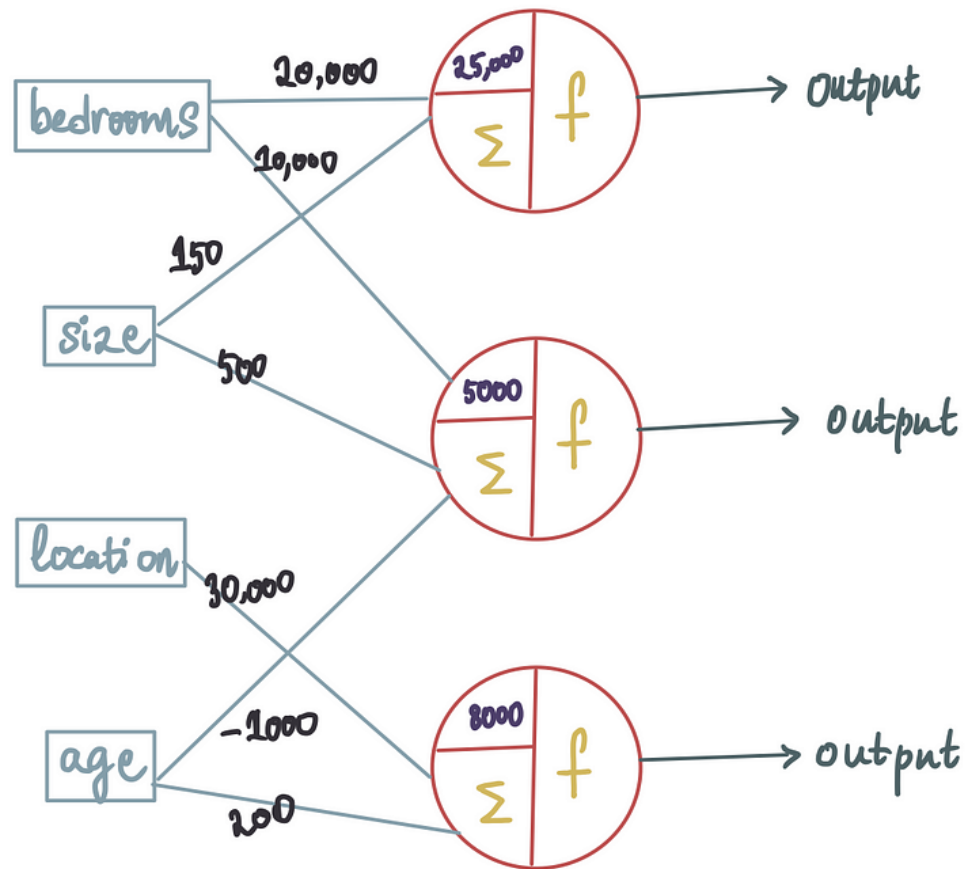




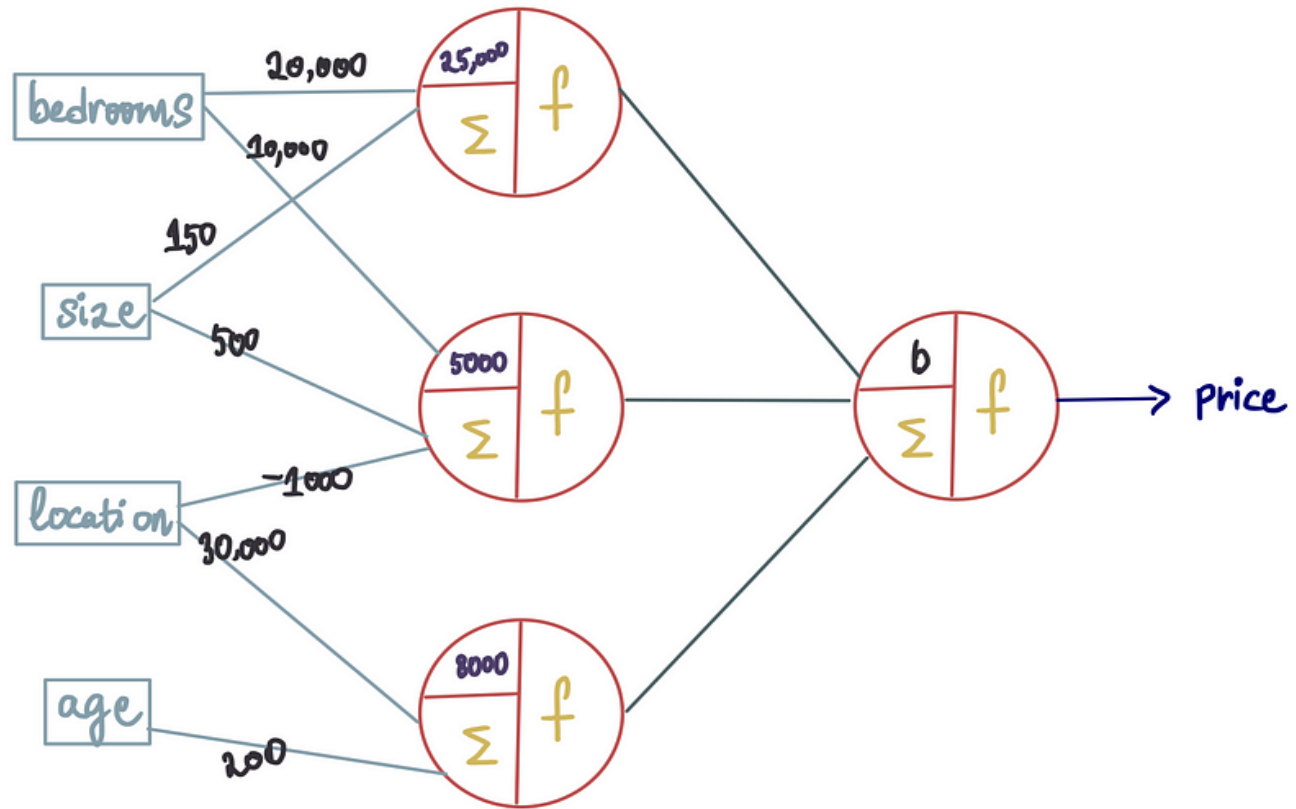


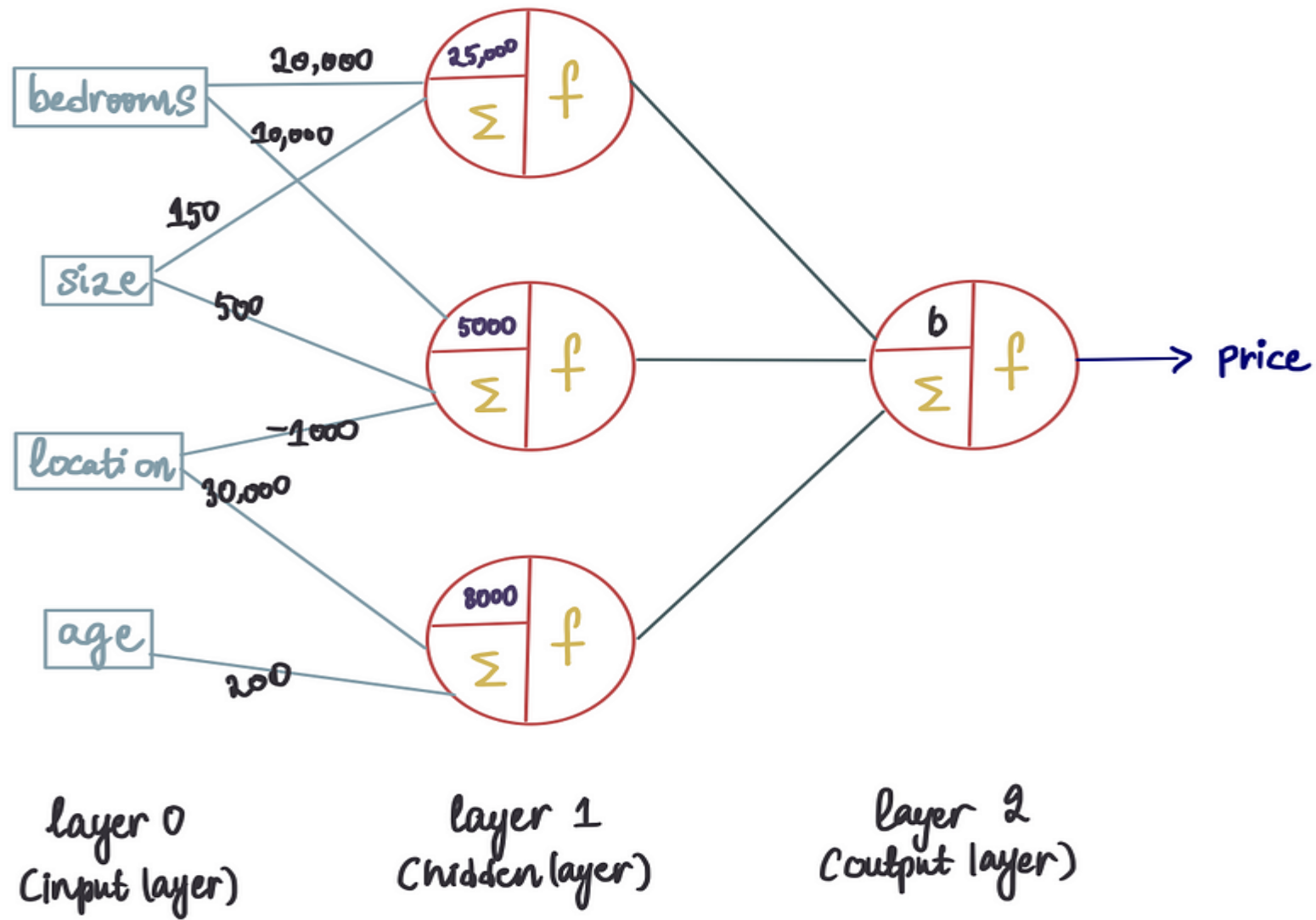


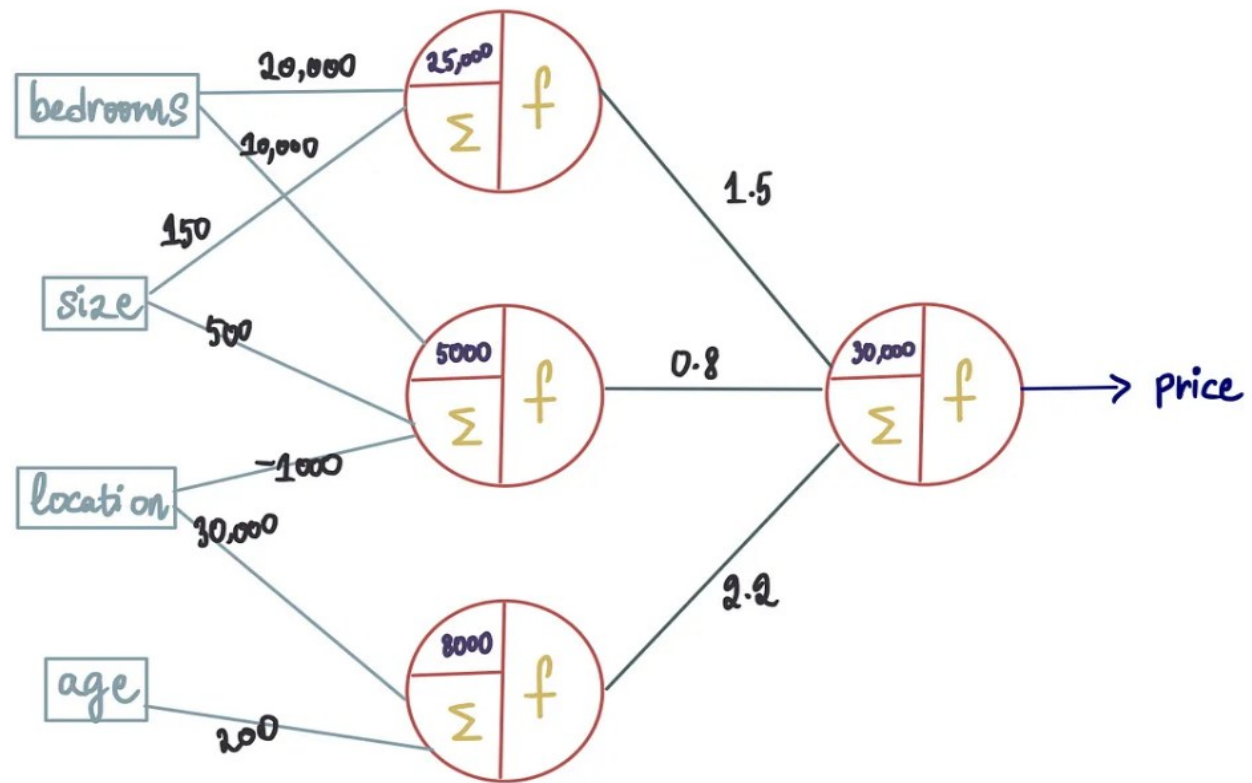


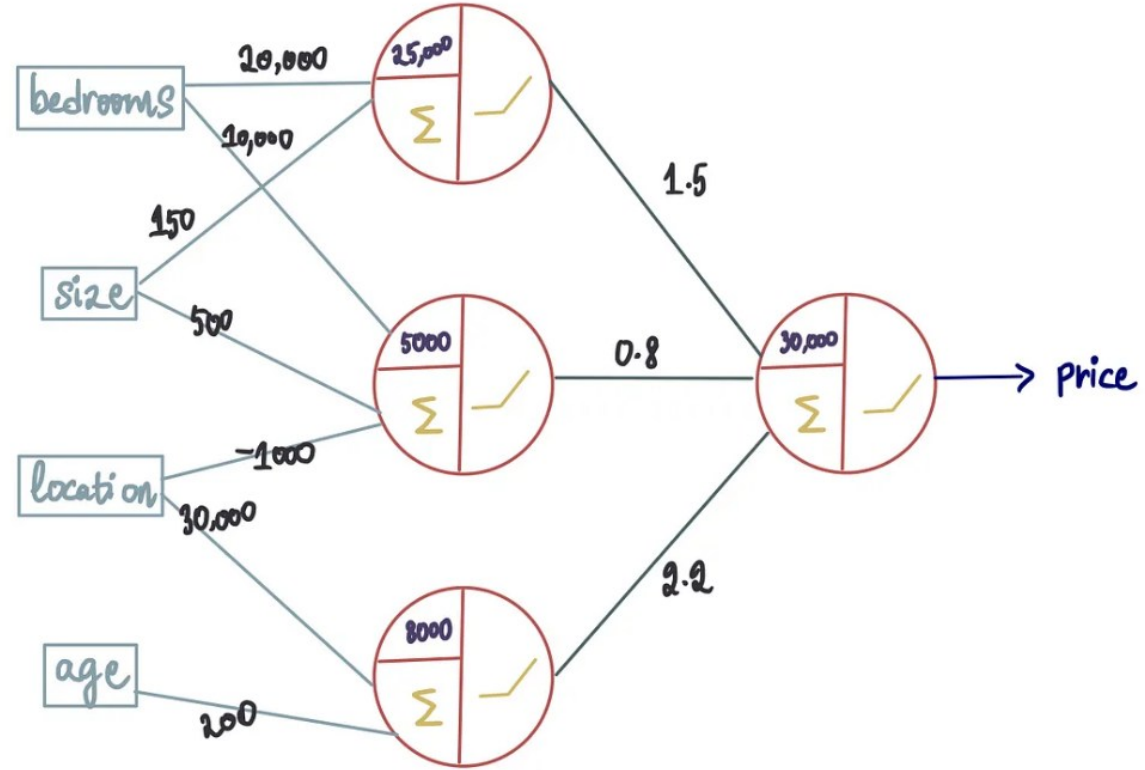


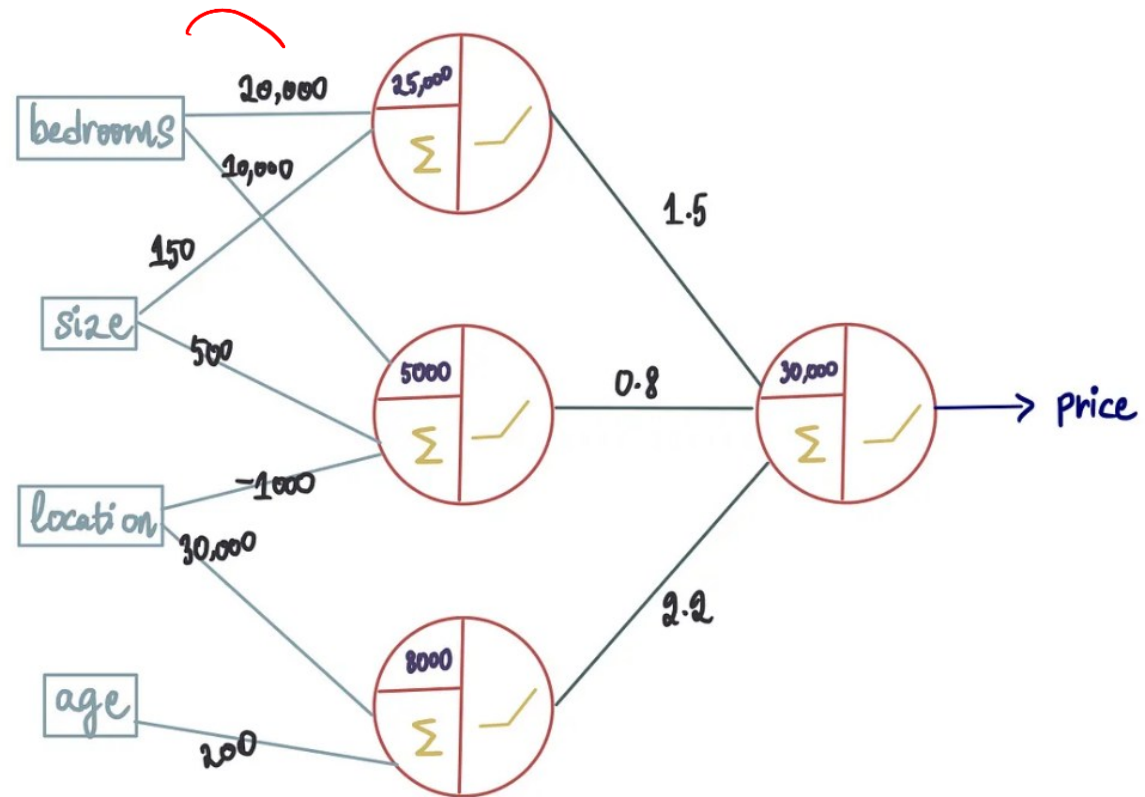




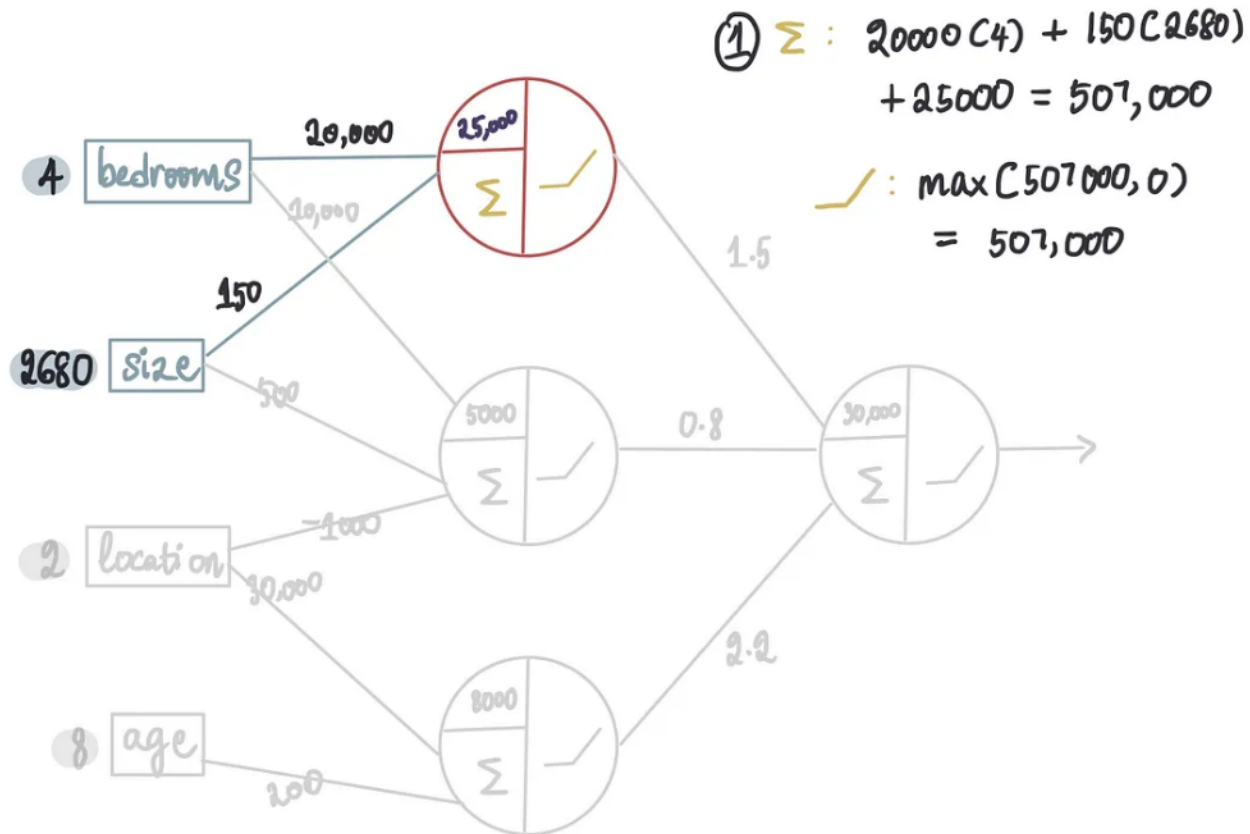




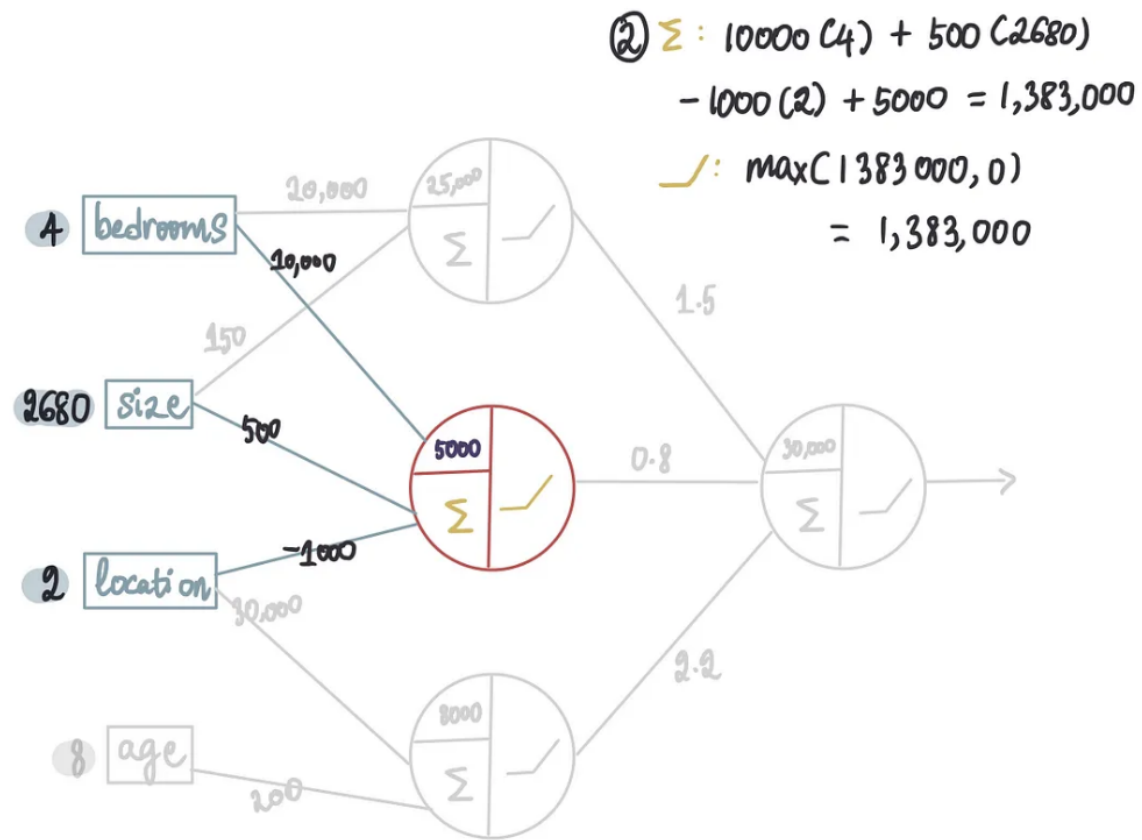




| bedrooms | size | location | age | price |
|----------|------|----------|-----|-------|
| 4        | 2680 | 2        | 8   | 1.75M |



step 1 — first neuron

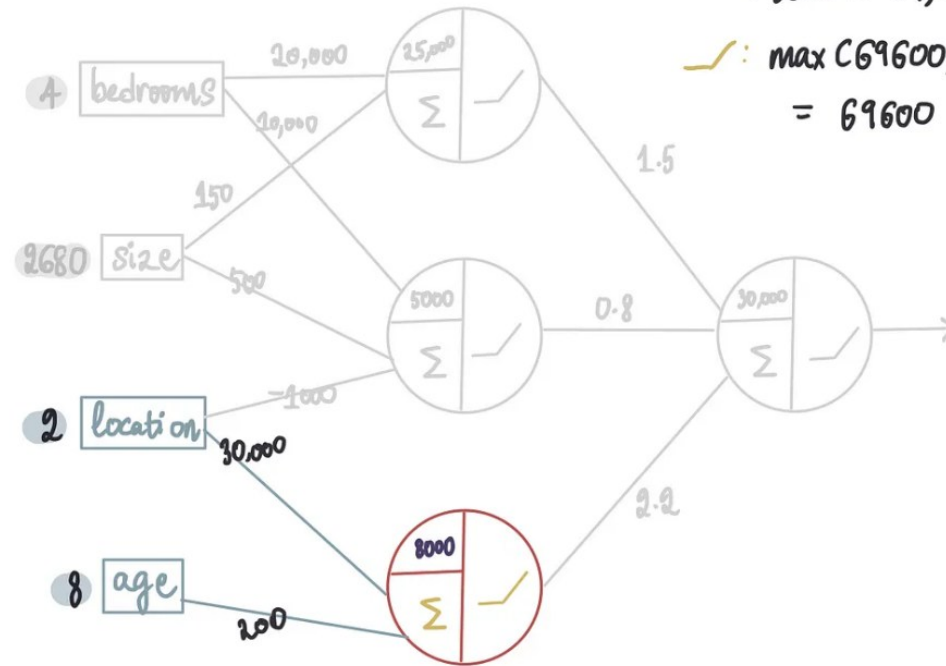


step 2 — second neuron

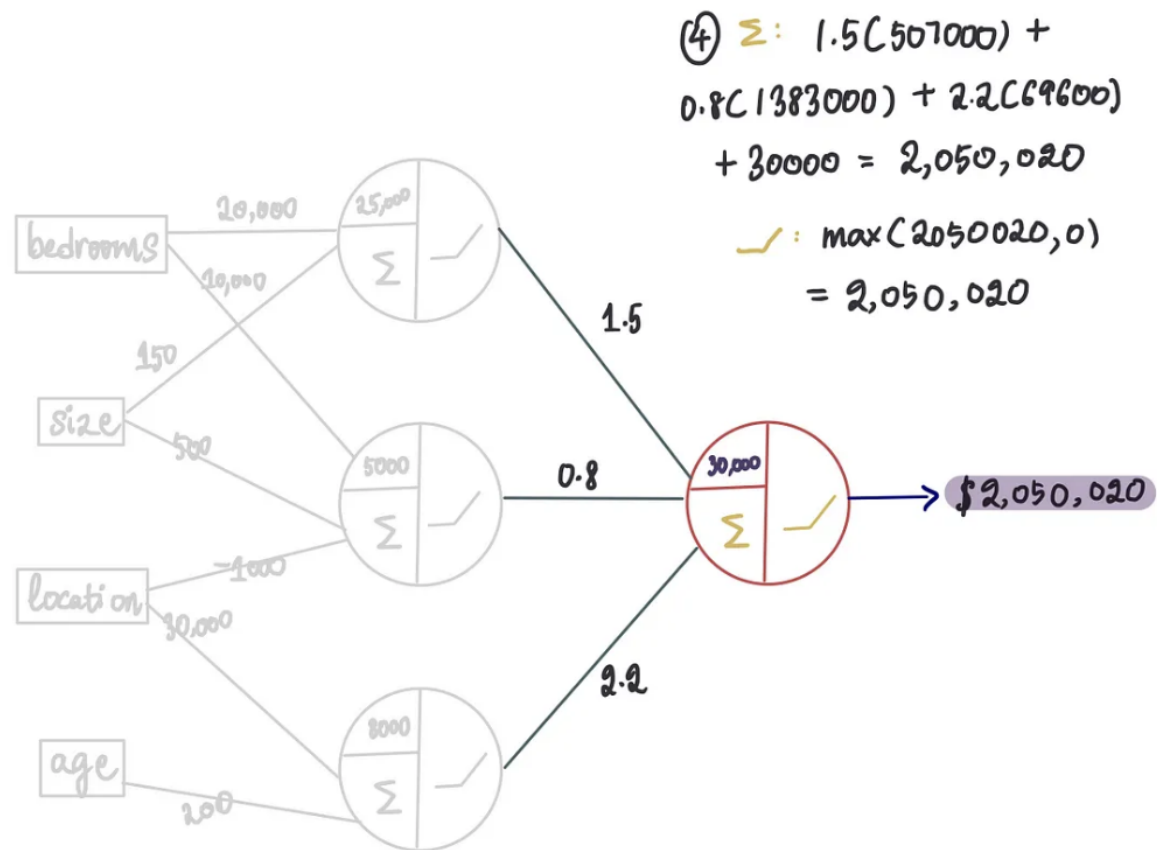


$$\textcircled{3} \Sigma: 30000(2) + 200(8) + 8000 = 69,600$$

$$\text{✓}: \max(69600, 0) = 69600$$



step 3 — third neuron



step 4 — final neuron

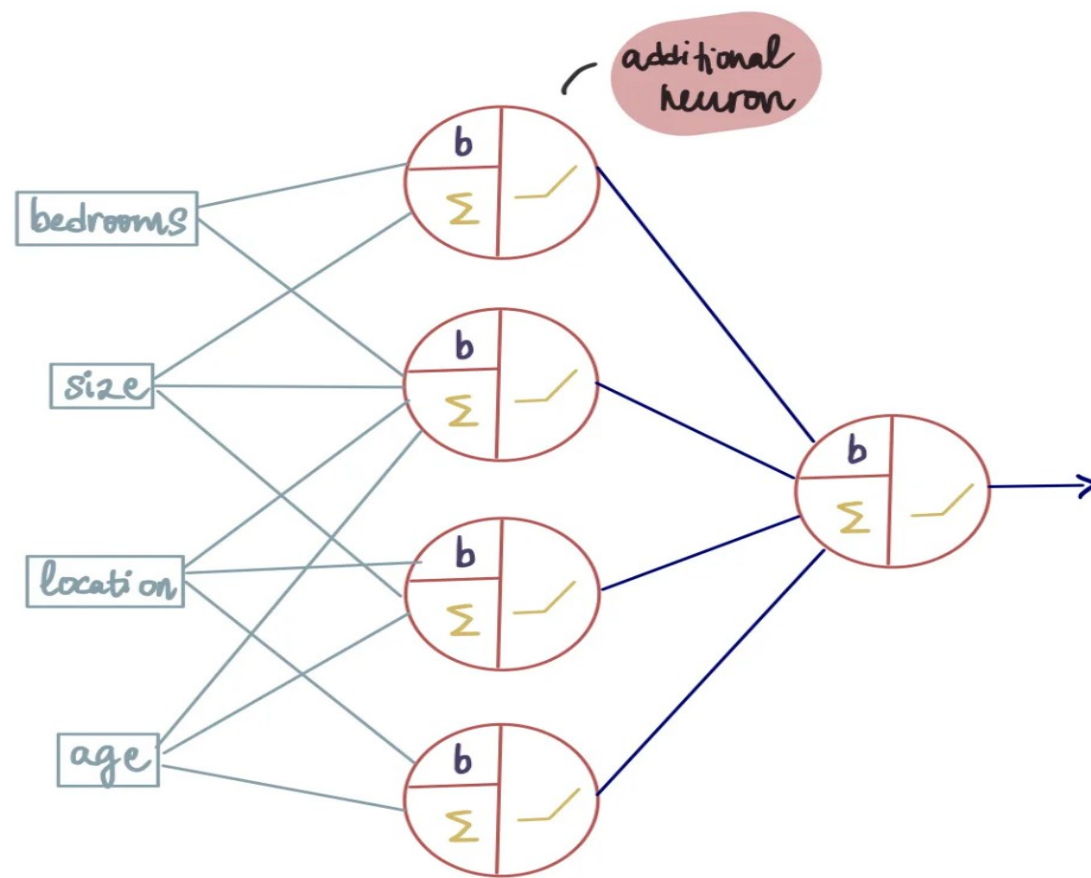
| bedrooms | size | location | age | price | predicted price |
|----------|------|----------|-----|-------|-----------------|
| 4        | 2680 | 2        | 8   | 1.75M | 2,050,020       |
| 5        | 700  | 1        | 27  | 750K  | 793,680         |
| 3        | 2360 | 1        | 28  | 1.2M  | 1,820,820       |
| 5        | 1280 | 2        | 98  | 1.05M | 1,252,620       |
| 5        | 3480 | 3        | 21  | 2.3M  | 2,658,940       |

| actual<br>price | old<br>predicted price | new<br>predicted price |
|-----------------|------------------------|------------------------|
| 1.75M           | 1,036,000              | 2,050,020              |
| 750K            | 453,000                | 793,680                |
| 1.2M            | 870,000                | 1,820,820              |
| 1.05M           | 576,000                | 1,252,620              |
| 2.3M            | 1,333,000              | 2,658,940              |

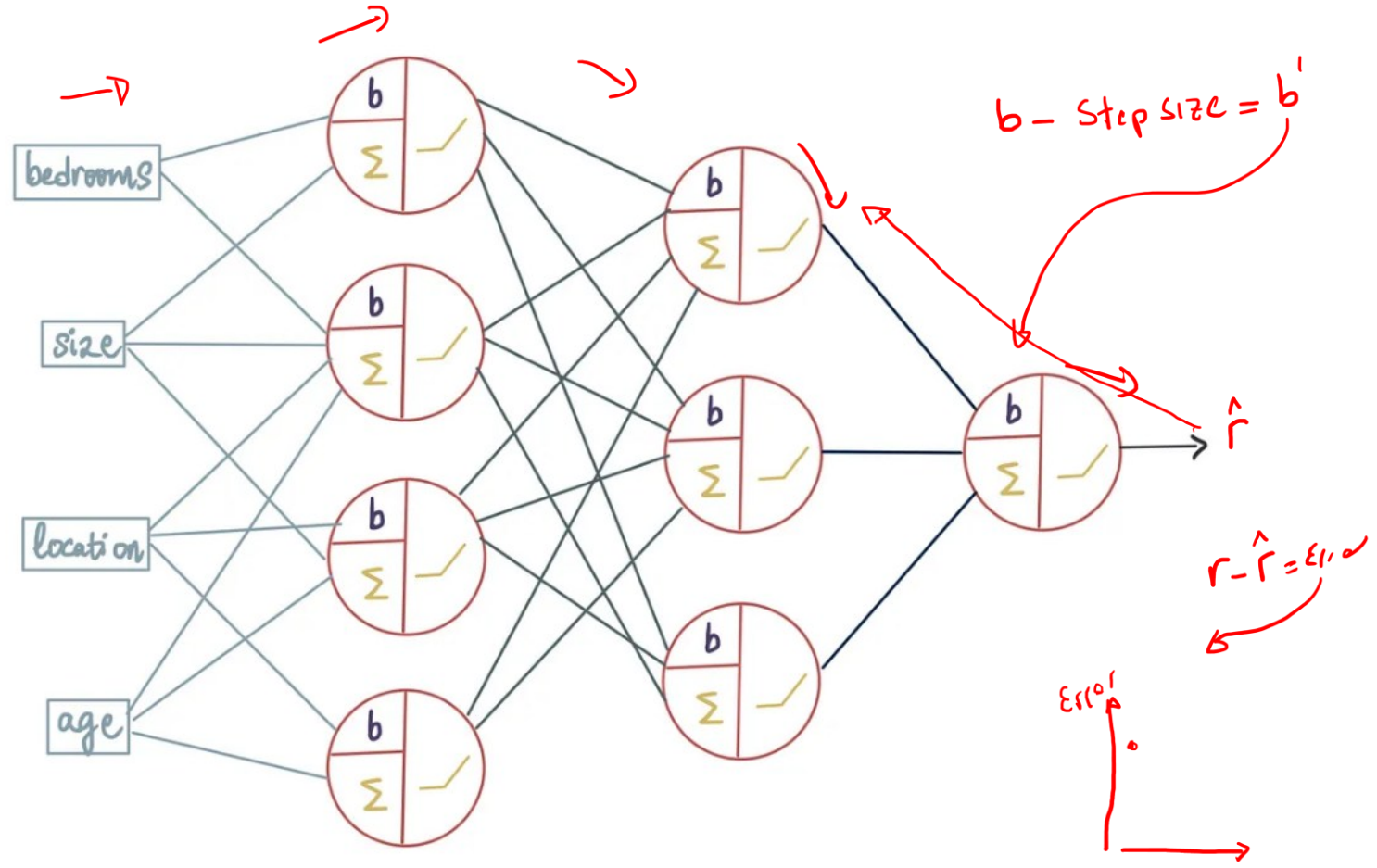
$$\text{MSE} = \frac{1}{n} \sum (\overset{\text{actual value}}{y} - \overset{\text{predicted value}}{\hat{y}})^2$$

$$\begin{aligned} \text{MSE}_{\text{old model}} &= \frac{1}{5} \left[ (1.75\text{M} - 1,036,000)^2 + (750\text{K} - 453,000)^2 \right. \\ &\quad \left. + \dots + (2.3\text{M} - 1,333,000)^2 \right] \\ &= 1.867 \times 10^{12} \end{aligned}$$

$$\text{MSE}_{\text{new model}} = 6.47 \times 10^{11}$$

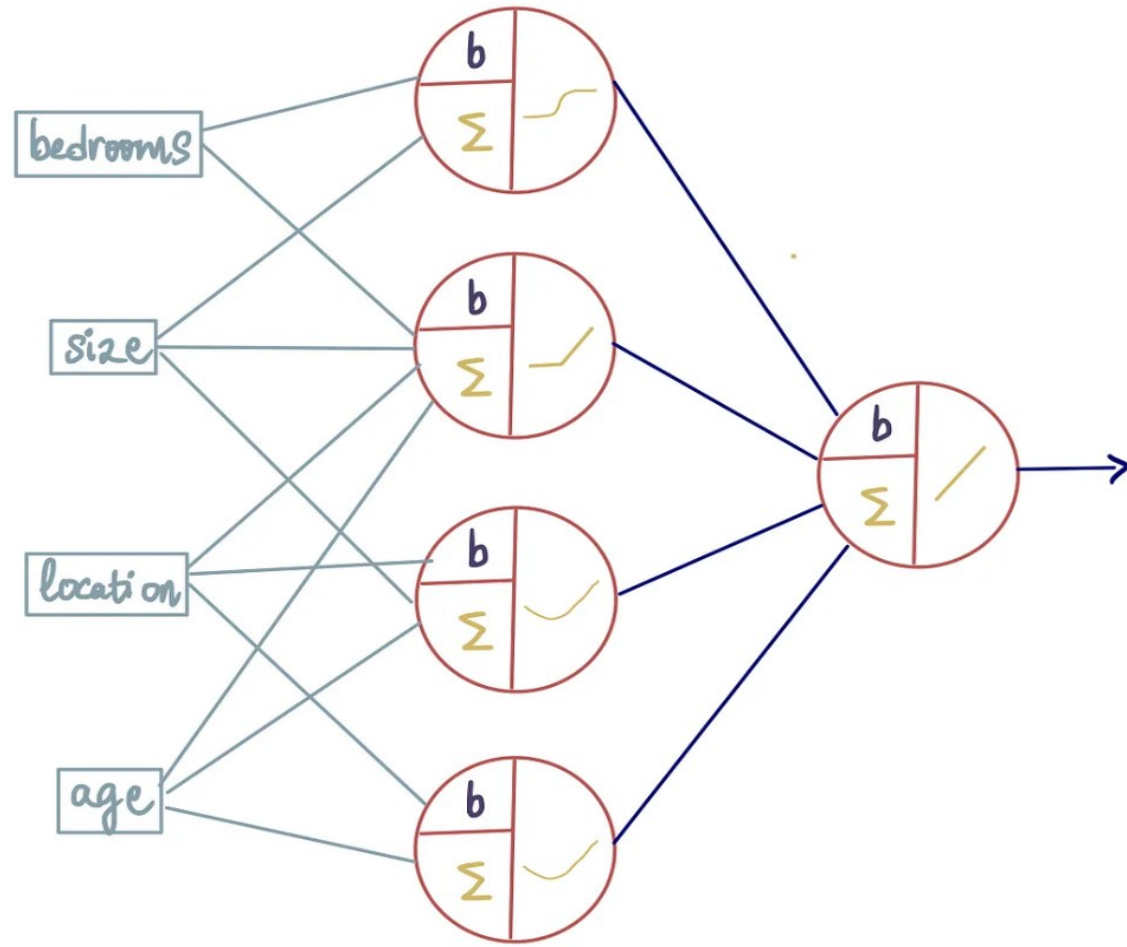


added a fourth neuron to the hidden layer



added a second hidden layer with 3 neurons





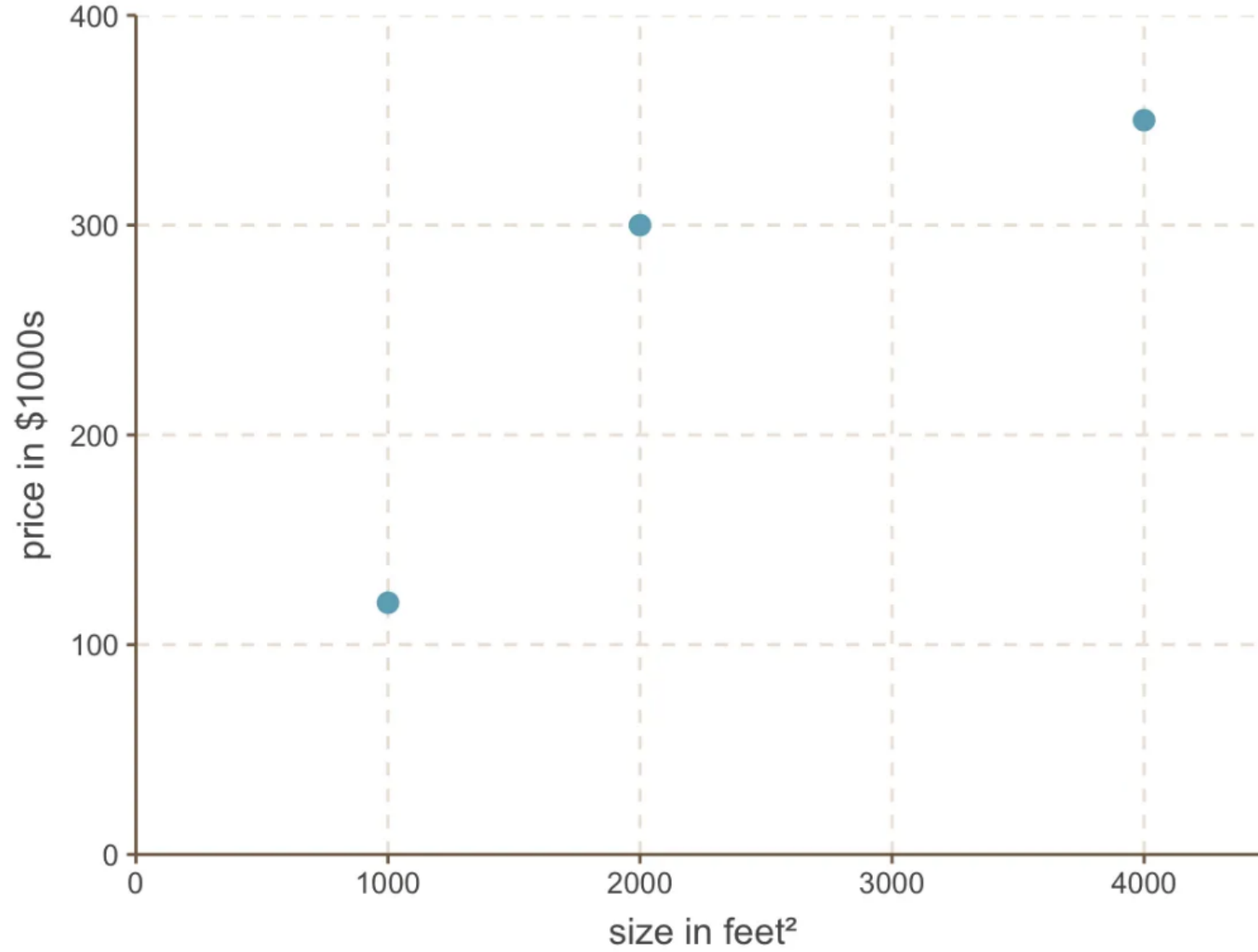
A Visual And Interactive Look at Basic Neural Network Math  
– Jay Alammam – Visualizing machine learning one concept  
at a time.

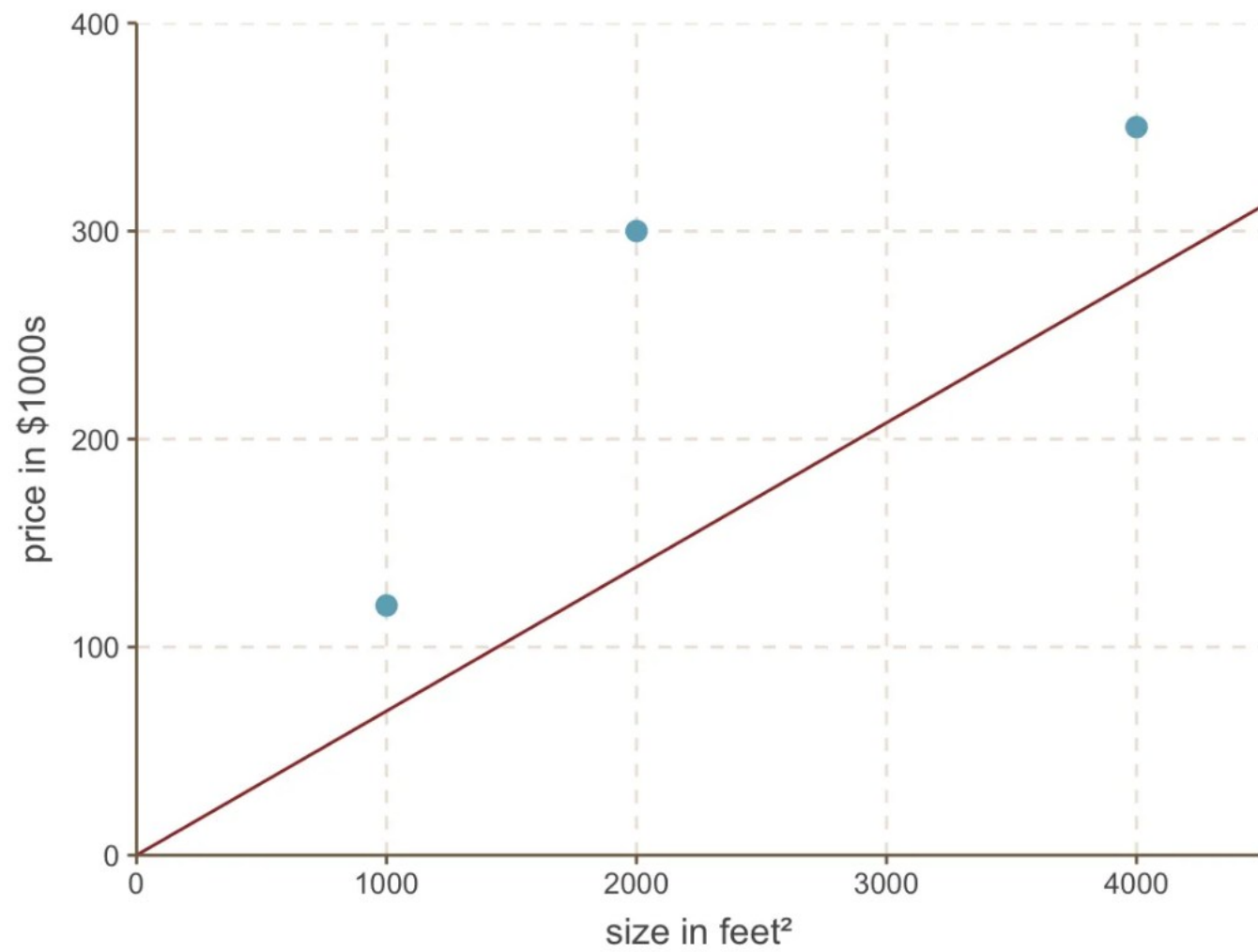
# **Introducción a Redes Neuronales**

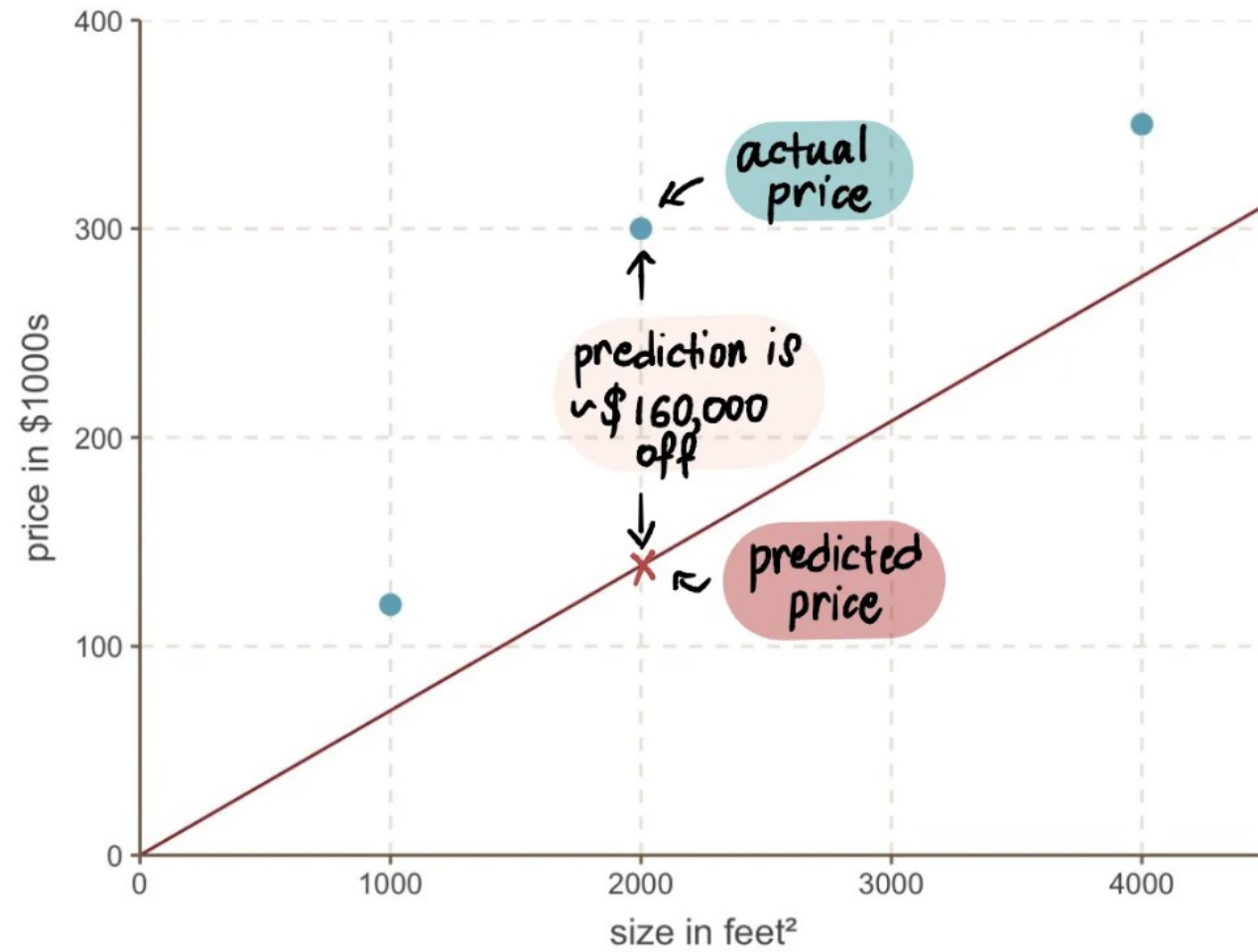
## **Gradiente Descendiente**

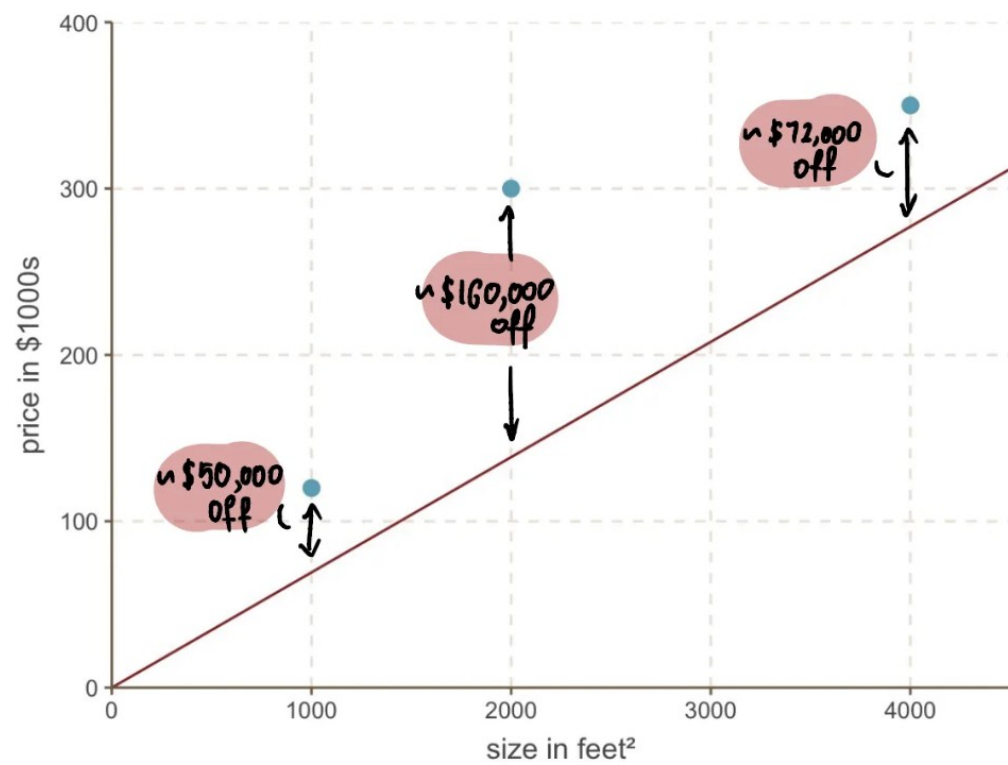


| <u>house size</u>      | <u>price in \$1000s</u> |
|------------------------|-------------------------|
| 1000 feet <sup>2</sup> | 120                     |
| 2000 feet <sup>2</sup> | 300                     |
| 3000 feet <sup>2</sup> | 350                     |

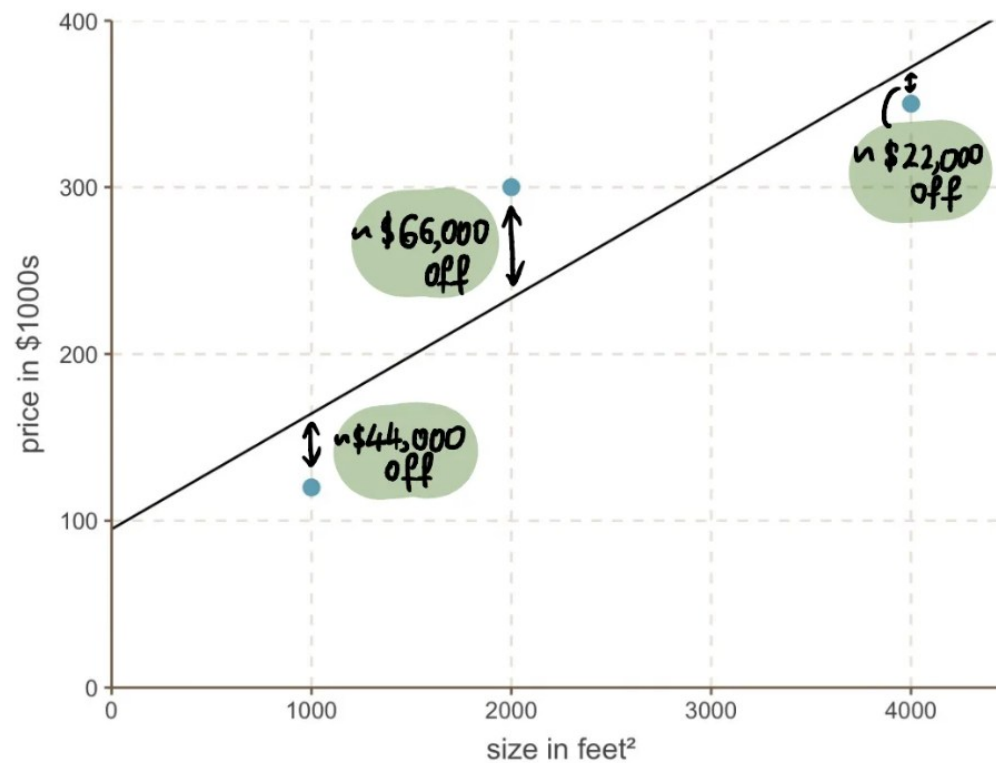
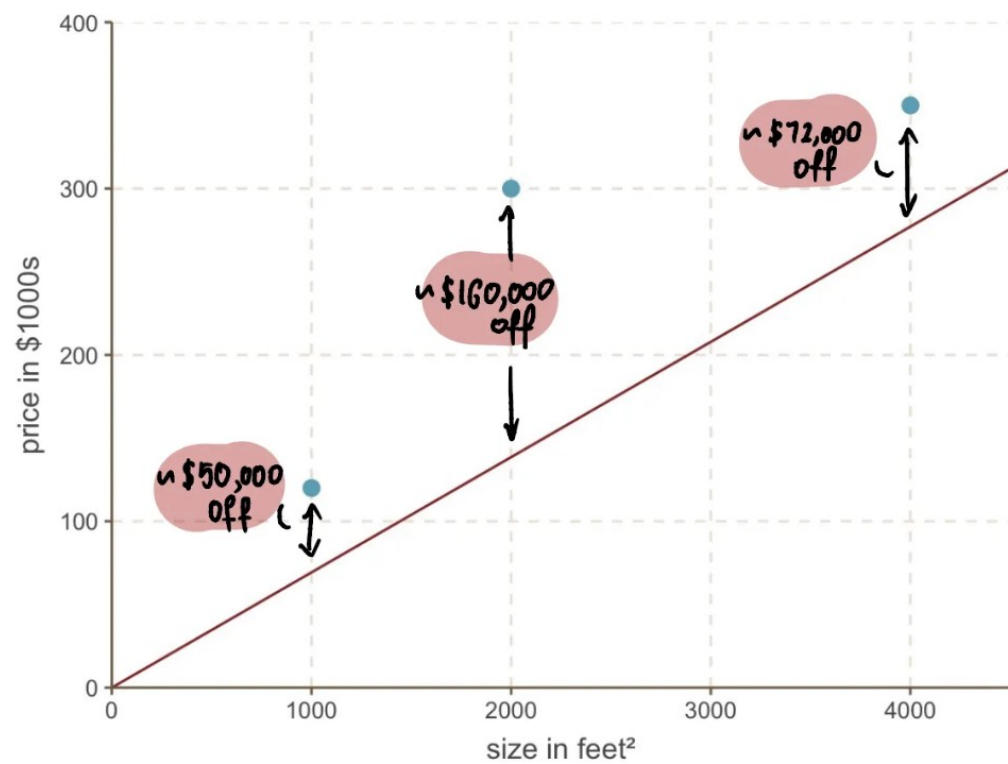












② the absolute difference between the actual and predicted house price

① predicted house price

$$MAE = \frac{1}{n} \sum_{i=1}^n |y - \hat{y}_i|$$

④ calculate the average

③ sum of all the absolute differences

② the absolute difference between the actual and predicted house price

① predicted house price

$$MAE = \frac{1}{n} \sum_{i=1}^n |y - \hat{y}|$$

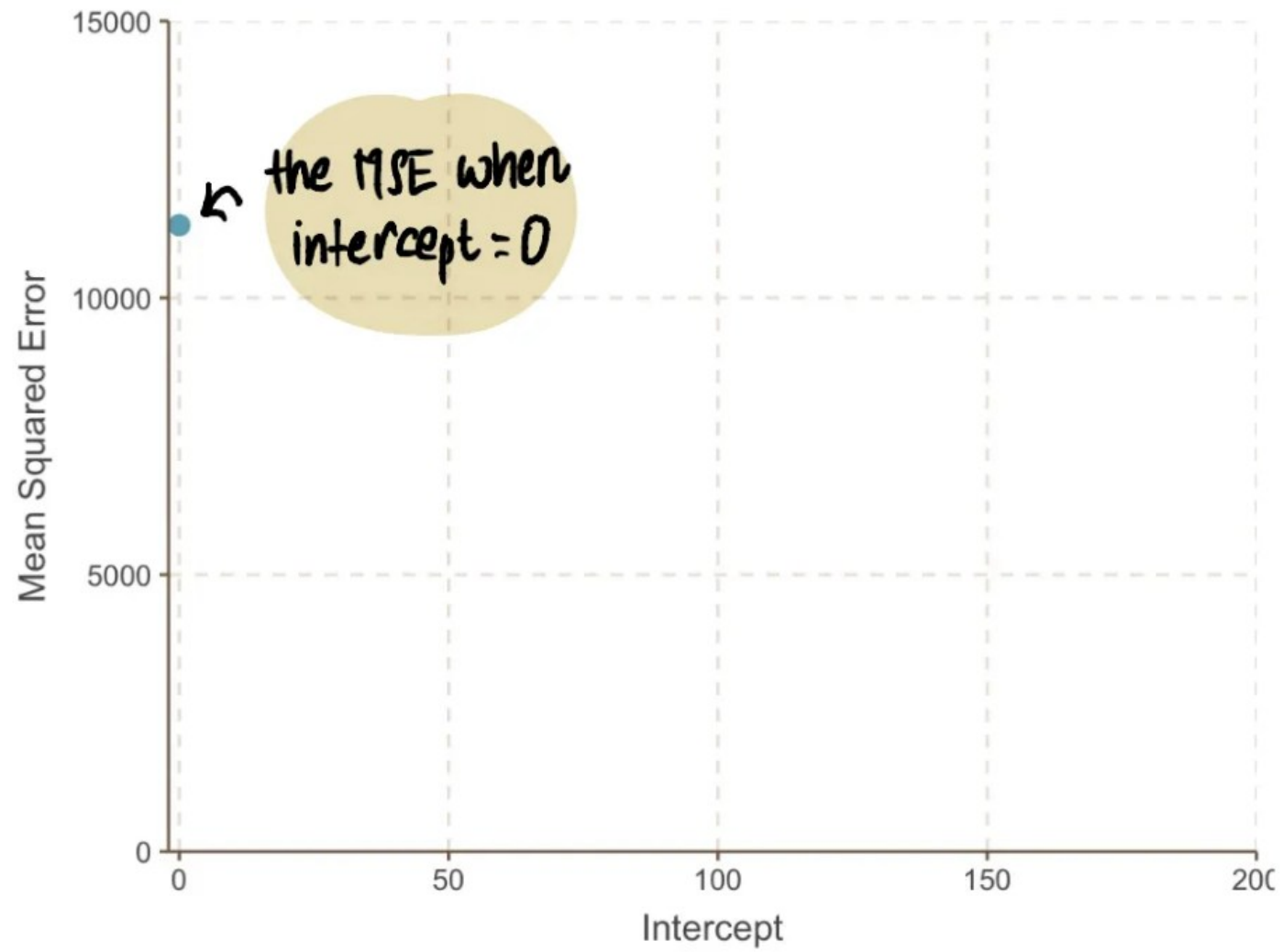
④ calculate the average

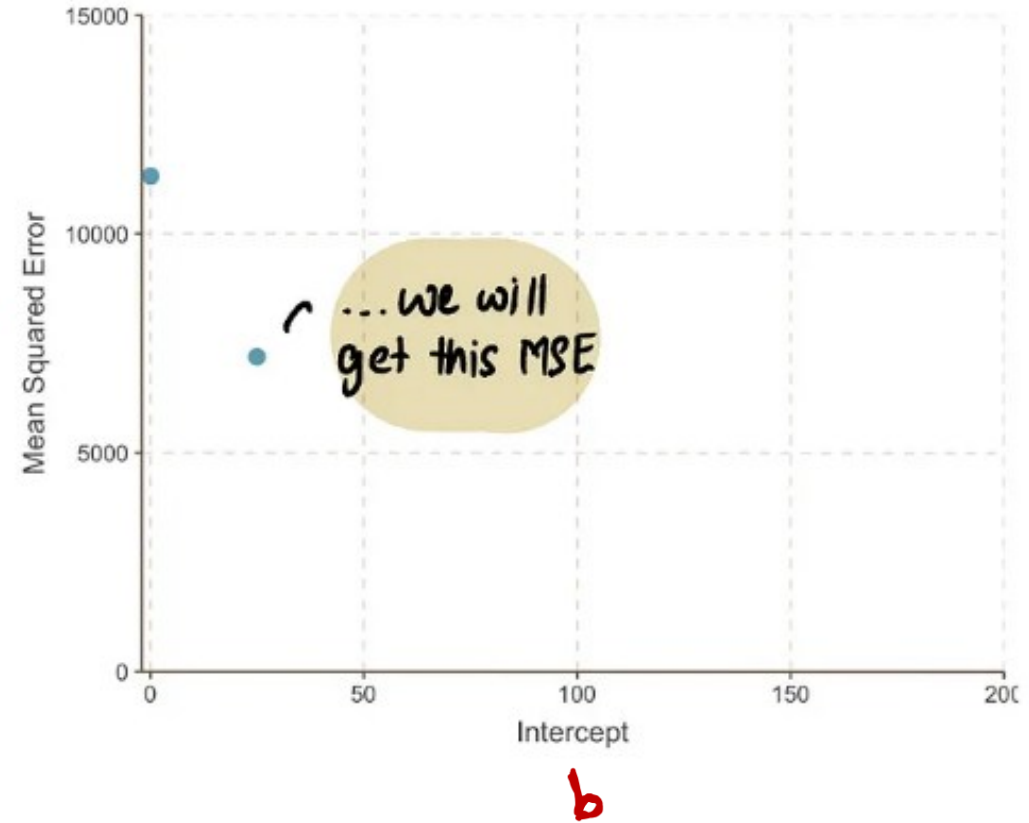
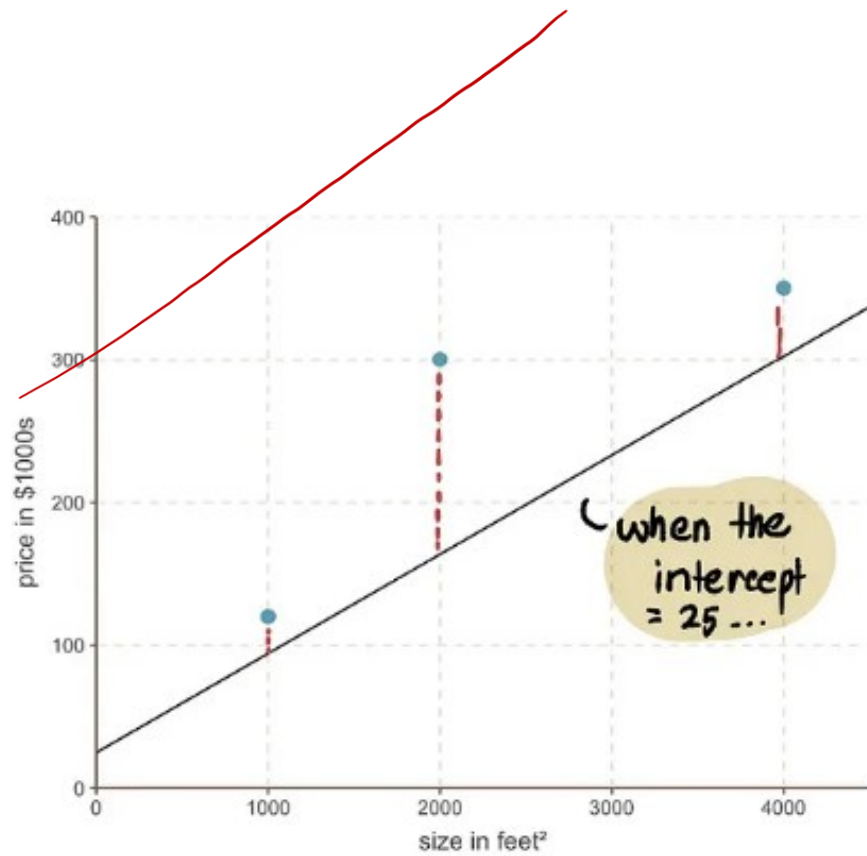
③ sum of all the absolute differences

actual house price

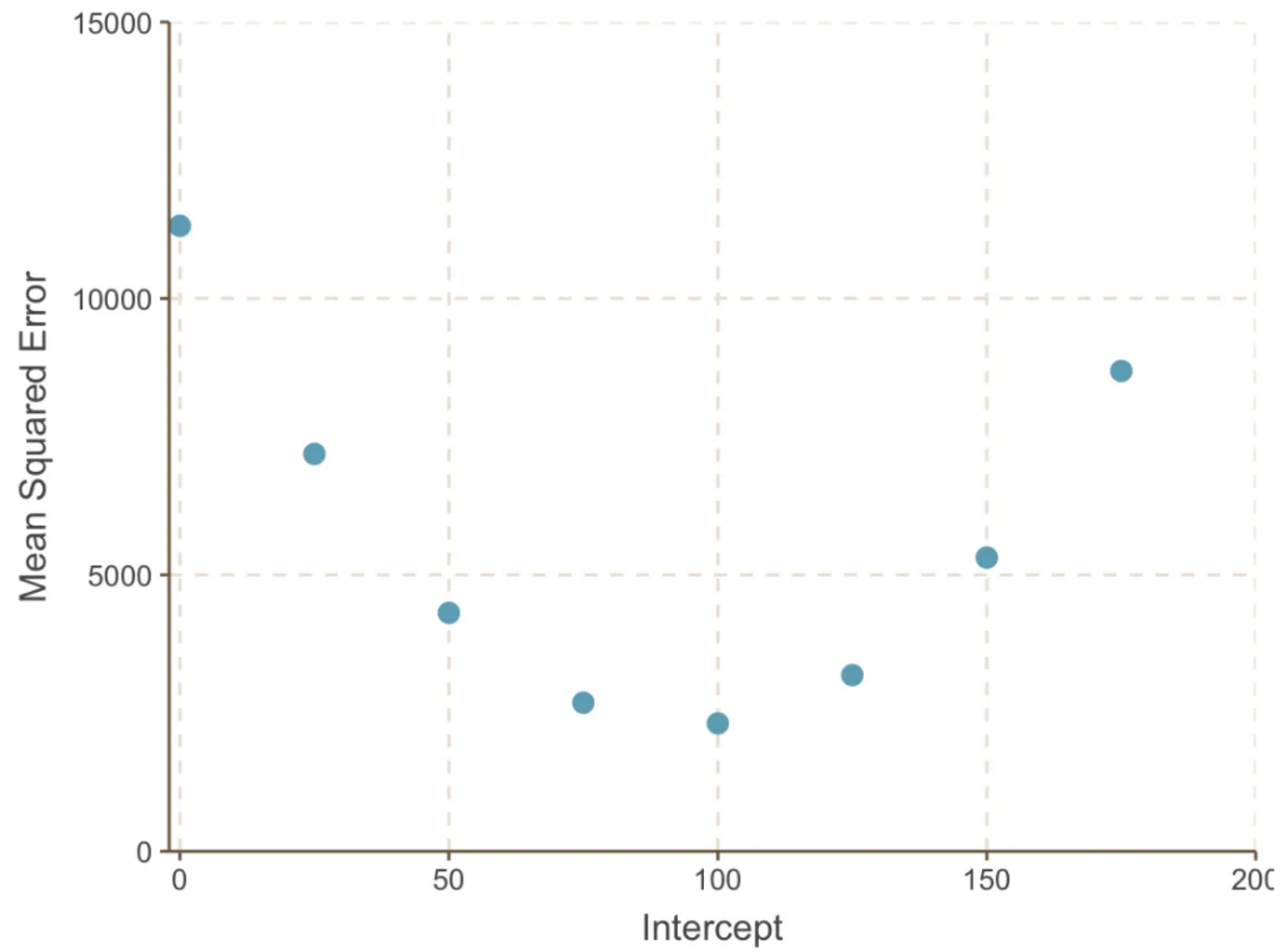
predicted house price when Intercept = 0

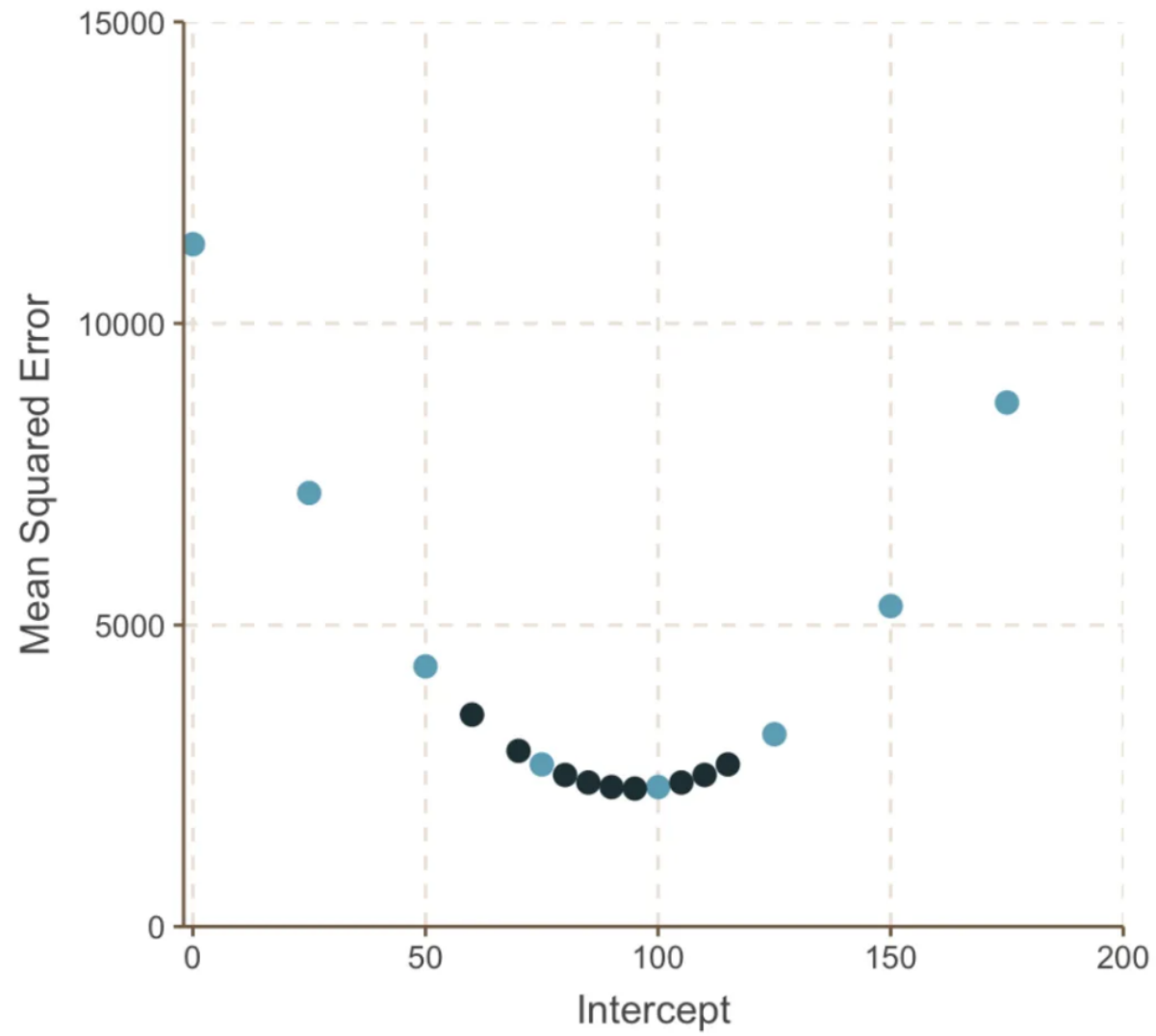
$$\begin{aligned} MSE &= \frac{1}{3} [(120 - (0 + 0.069 \times 1000))^2 + (300 - (0 + 0.069 \times 2000))^2 \\ &\quad + (350 - (0 + 0.069 \times 4000))^2] \\ &= 11313 \end{aligned}$$



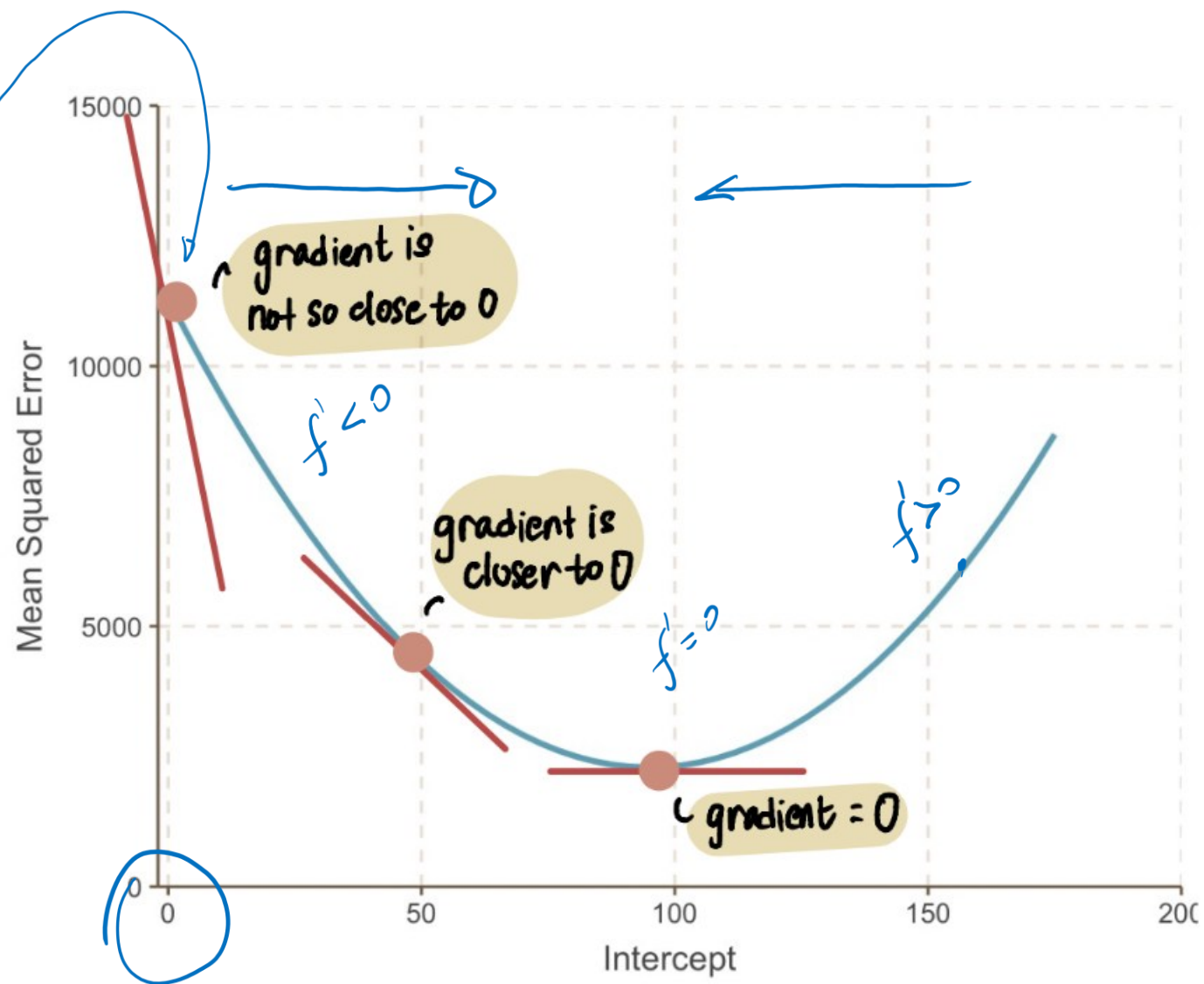
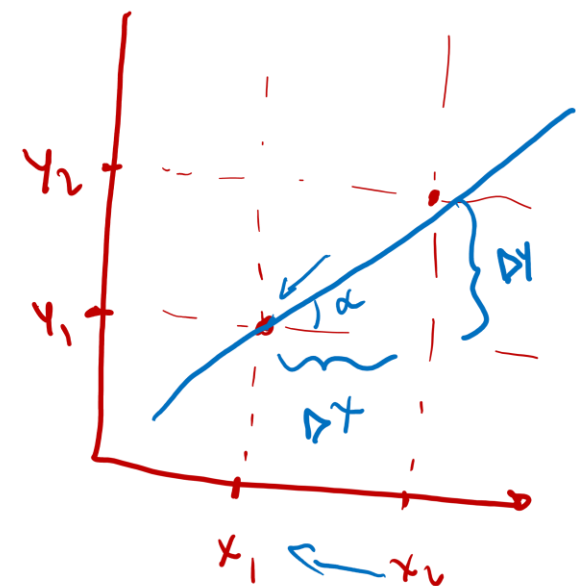


$$\begin{aligned} \text{MSE} &= \frac{1}{3} [(120 - (25 + 0.069 \times 1000))^2 + \\ &\quad (300 - (25 + 0.069 \times 2000))^2 + \\ &\quad (350 - (25 + 0.069 \times 4000))^2] \\ &= 7188 \end{aligned}$$

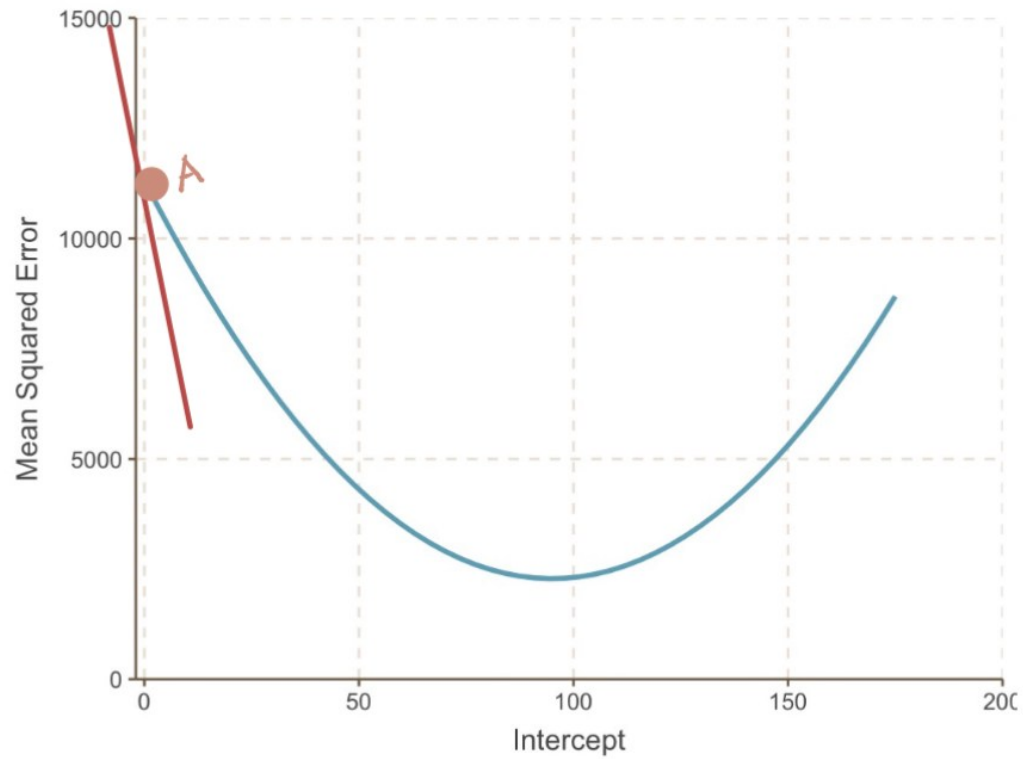




$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$







① derivative is denoted by 'd'

② derivative with respect to the intercept

③ derivative at a specific value of the intercept

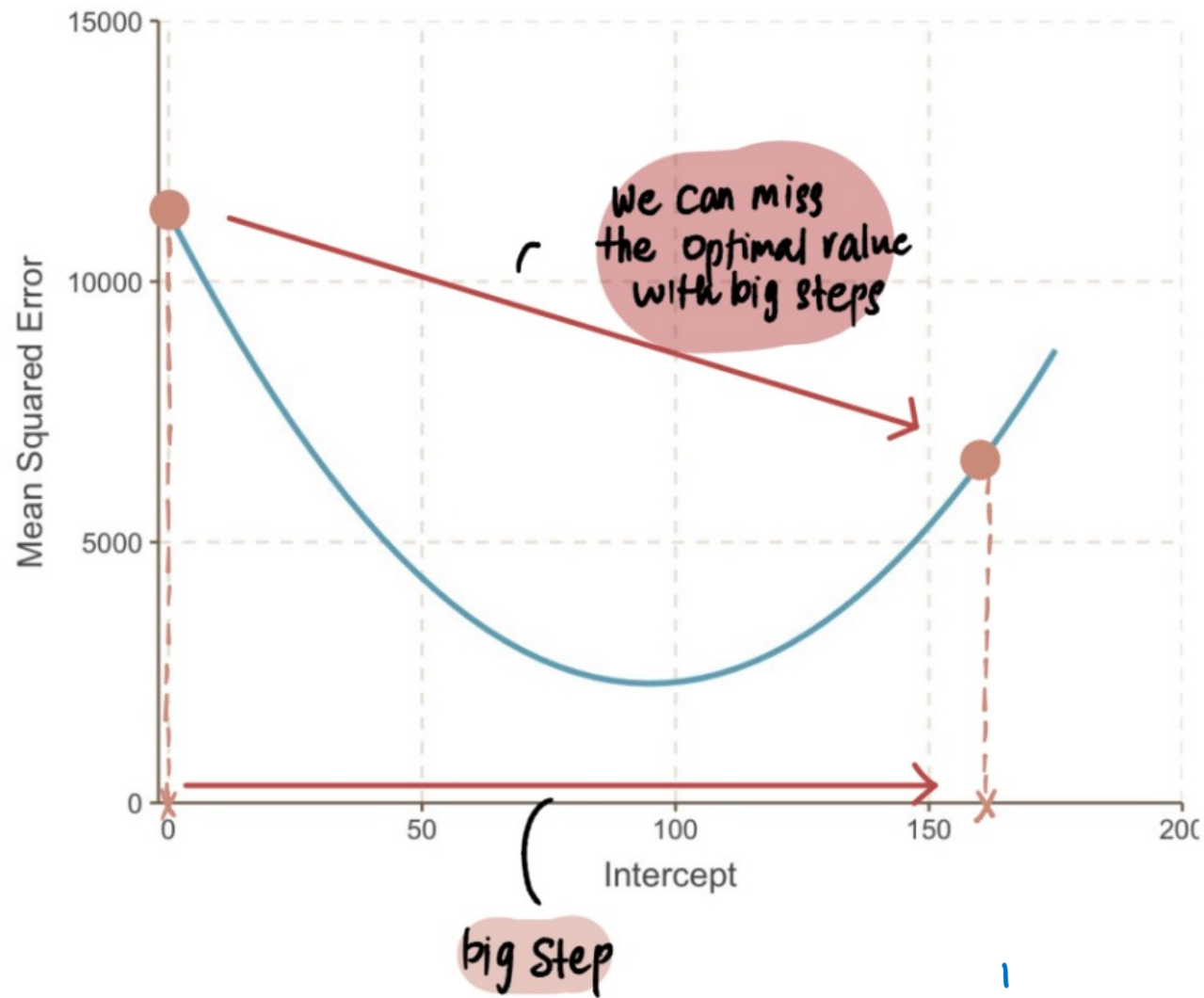
$\frac{d \text{ MSE}}{d \text{ intercept}} \bigg|_{\text{intercept}}$

$$\frac{dMSE}{d \text{ intercept}} = \frac{d}{d \text{ intercept}} \left[ \frac{1}{3} \left[ (120 - C \text{ intercept} + 0.069 \times 1000)^2 + (300 - C \text{ intercept} + 0.069 \times 2000)^2 + (350 - C \text{ intercept} + 0.069 \times 4000)^2 \right] \right]$$

$$= \frac{1}{3} \left[ -2(120 - (\text{intercept} + 0.069 \times 1000)) - 2(300 - (\text{intercept} + 0.069 \times 2000)) - 2(350 - (\text{intercept} + 0.069 \times 4000)) \right]$$

$$\frac{dMSE}{d \text{ intercept}} \Big|_0 = \frac{1}{3} \left[ -2(120 - (0 + 0.069 \times 1000)) - 2(300 - (0 + 0.069 \times 2000)) - 2(350 - (0 + 0.069 \times 4000)) \right]$$

$$= -190$$



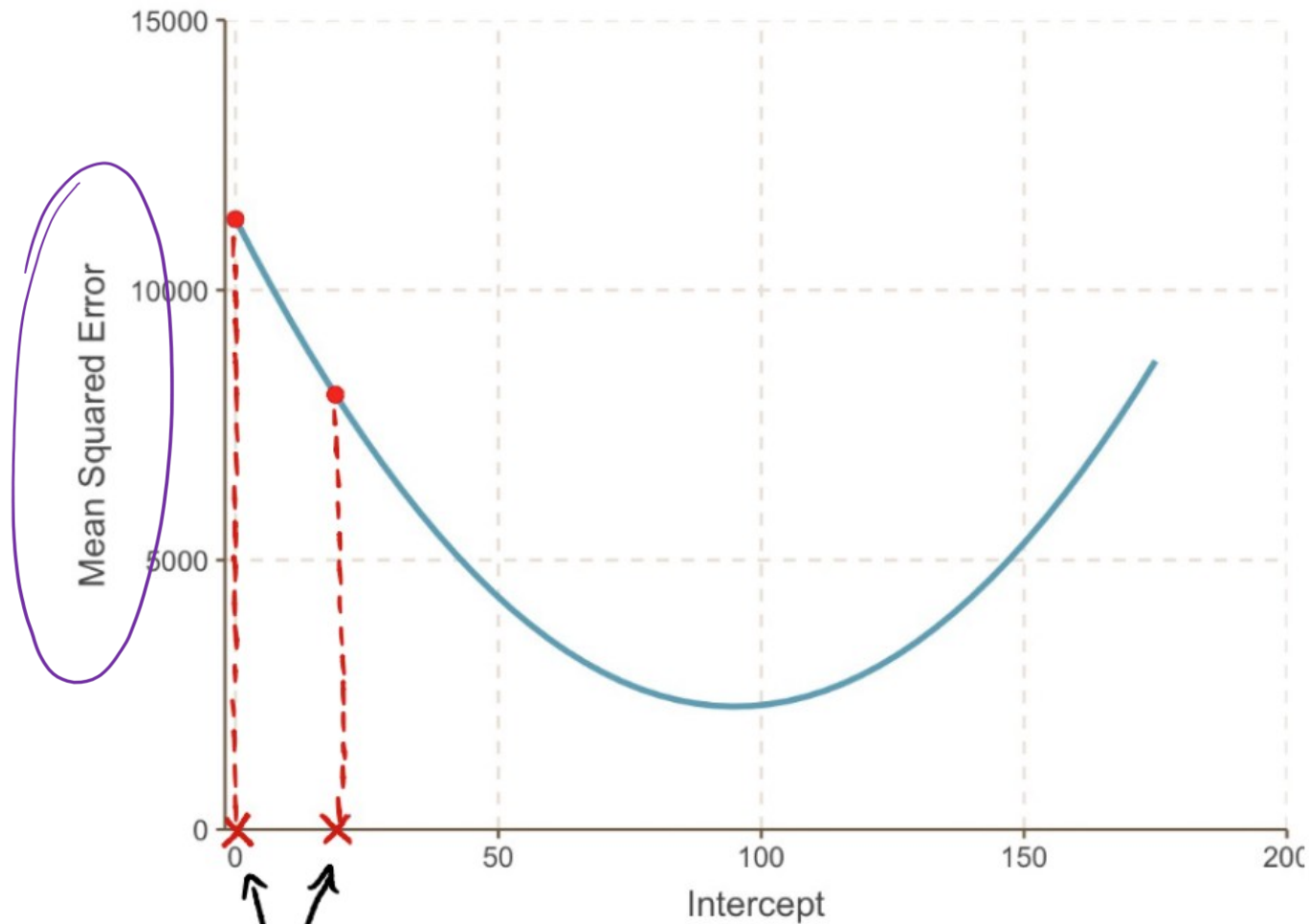
$$\text{Step Size} = \text{Learning Rate} \times \text{gradient}$$

$$\text{Step Size} = 0.1 \times -190 = -19$$

$$\text{New Intercept} = \text{Old Intercept} - \text{Step Size}$$

0

$$\text{New Intercept} = 0 - (-19) = 19$$

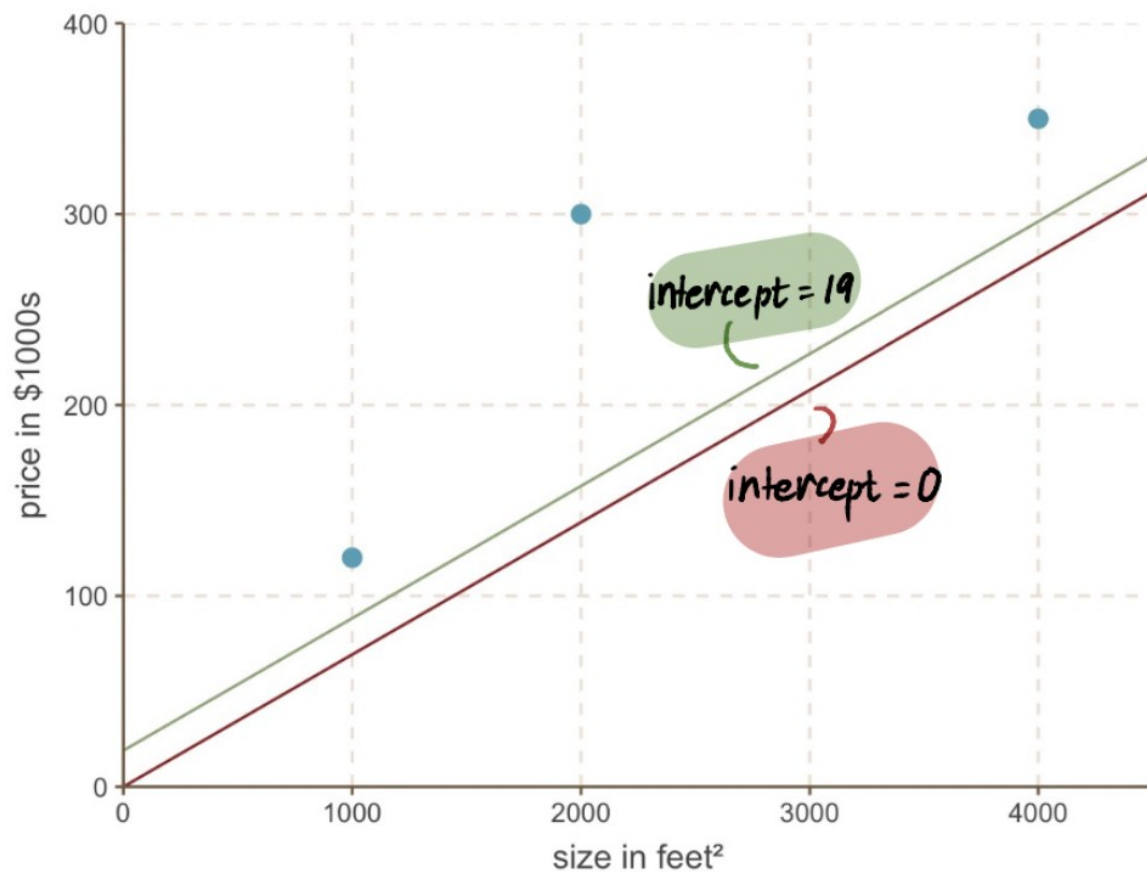


in one big Step,  
we moved much  
closer to our  
optimal value

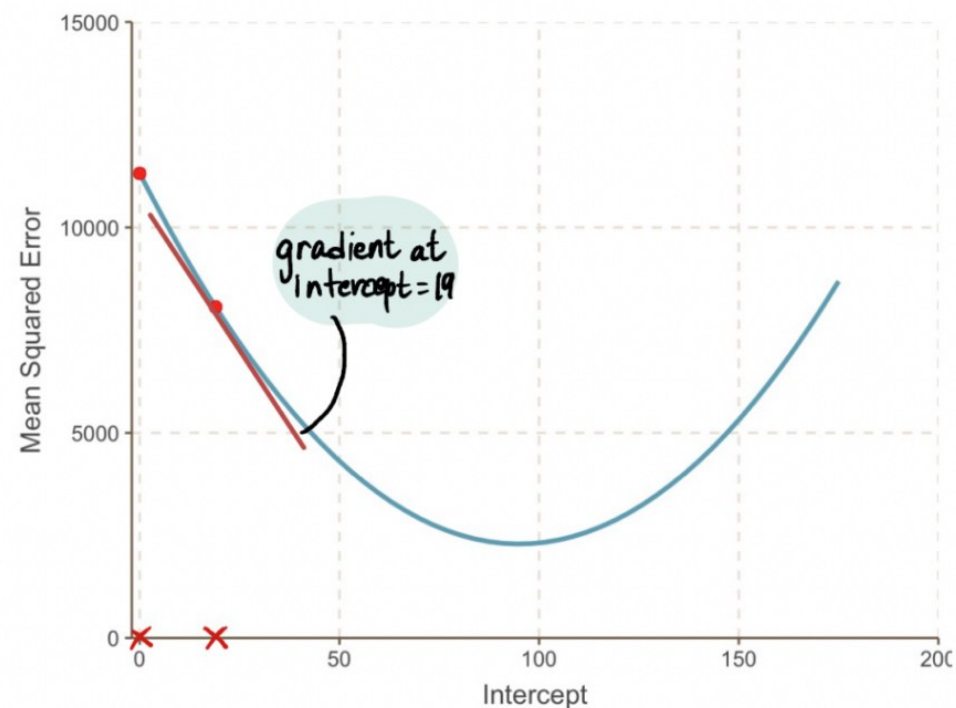
b

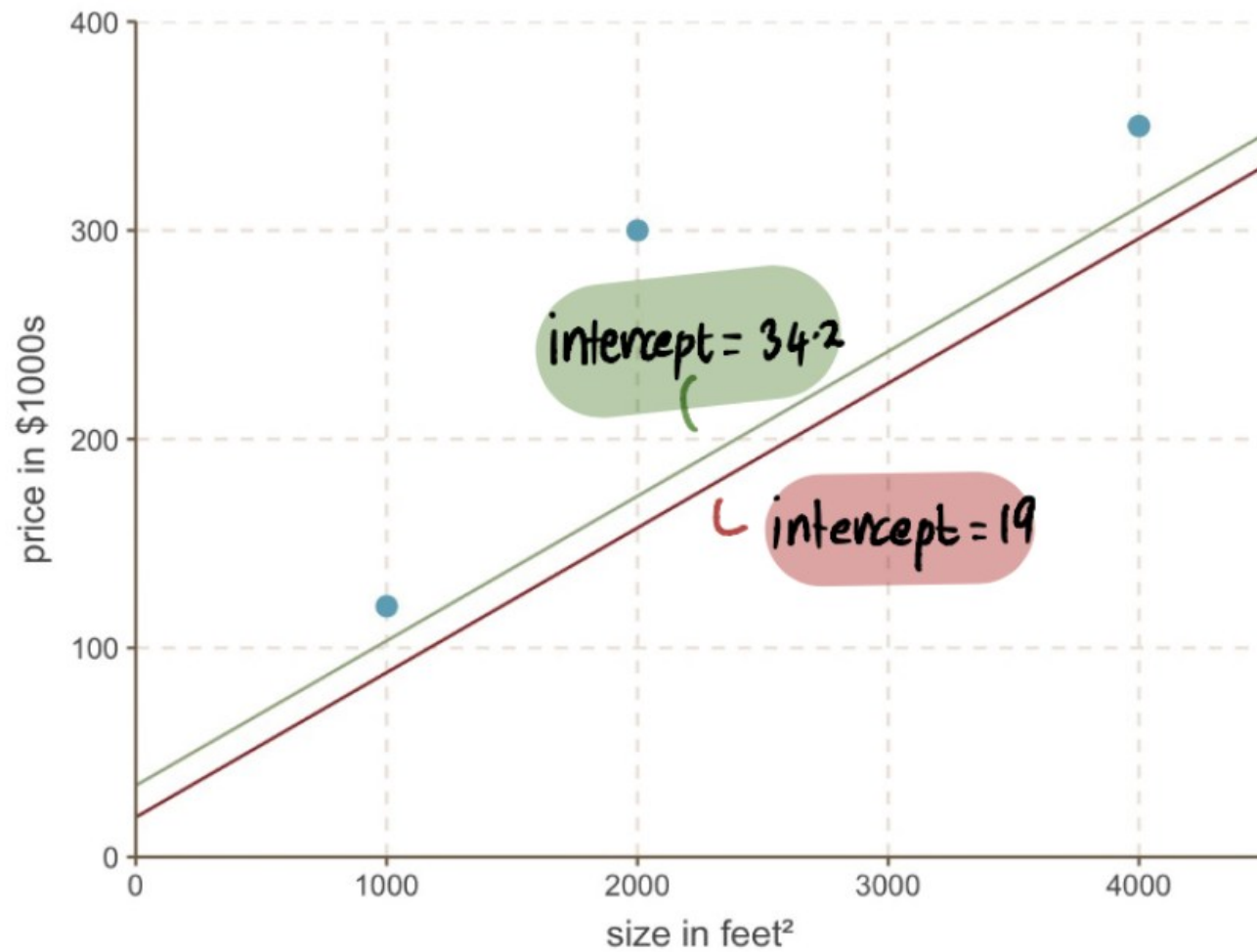
$$\begin{aligned} \text{New Intercept} &= 0 - C - 19 \\ &= 19 \end{aligned}$$





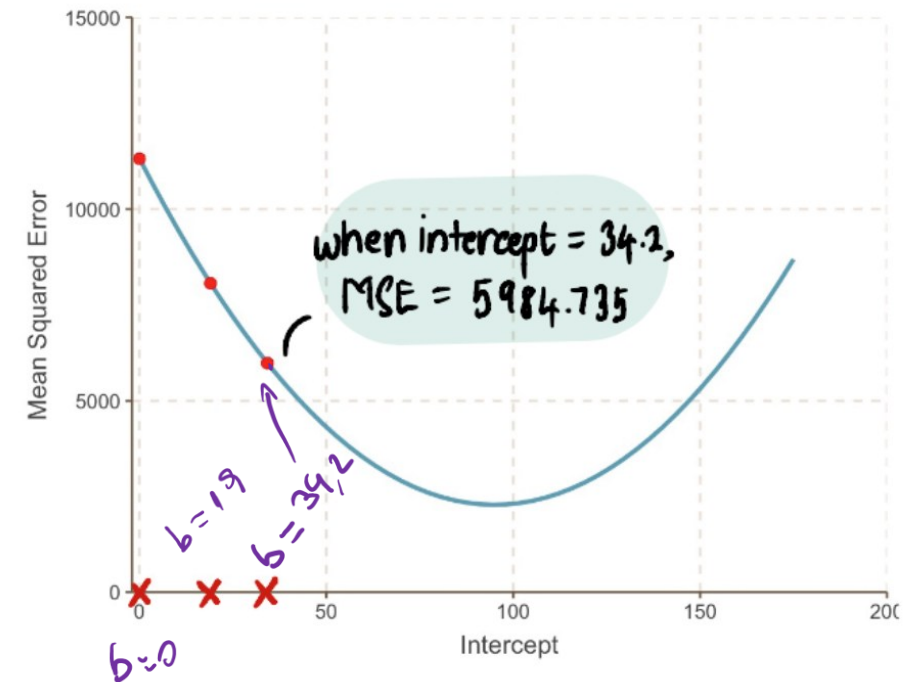
$$\frac{dMSE}{d \text{ intercept}} \bigg|_{19} = \frac{1}{3} [-2(120 - (19 + 0.069 \times 1000)) - 2(300 - (19 + 0.069 \times 2000)) - 2(350 - (19 + 0.069 \times 4000))] = -152$$

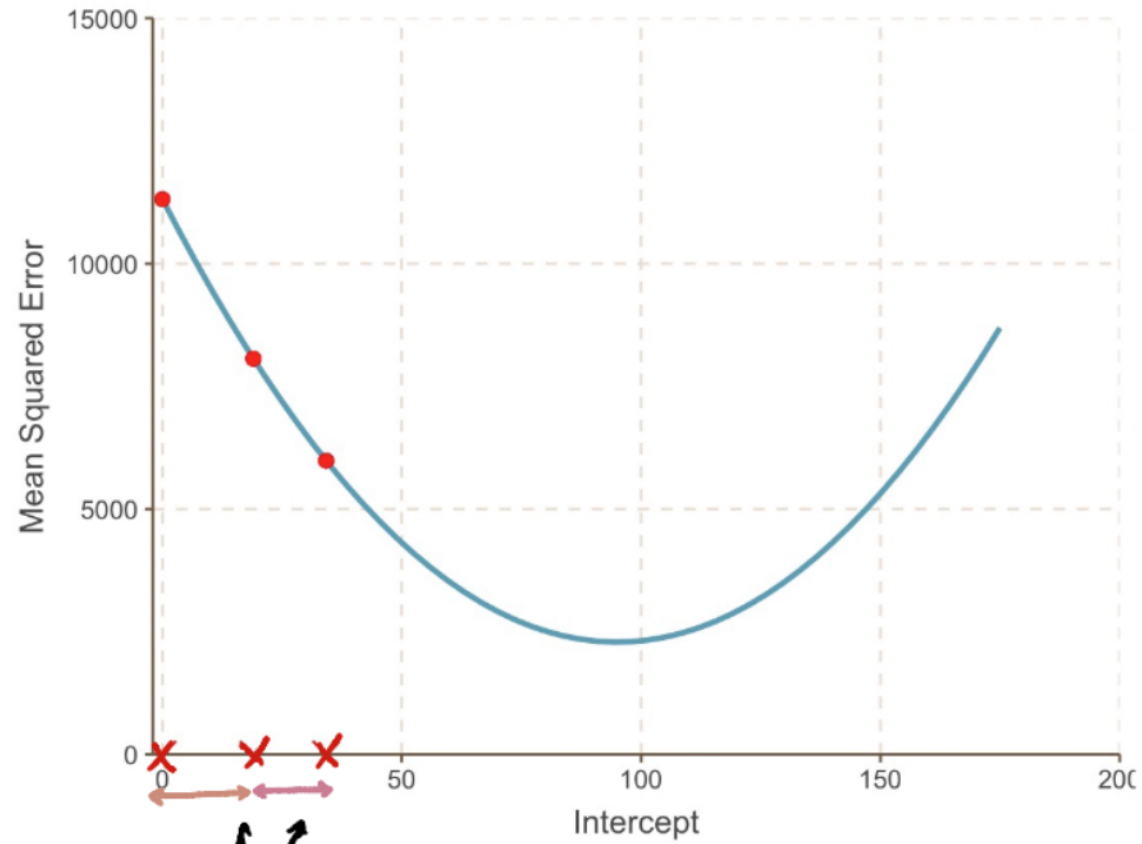




$$\begin{aligned}\text{Step Size} &= \text{Gradient} \times 0.1 \\ &= -152 \times 0.1 \\ &= \underline{-15.2}\end{aligned}$$

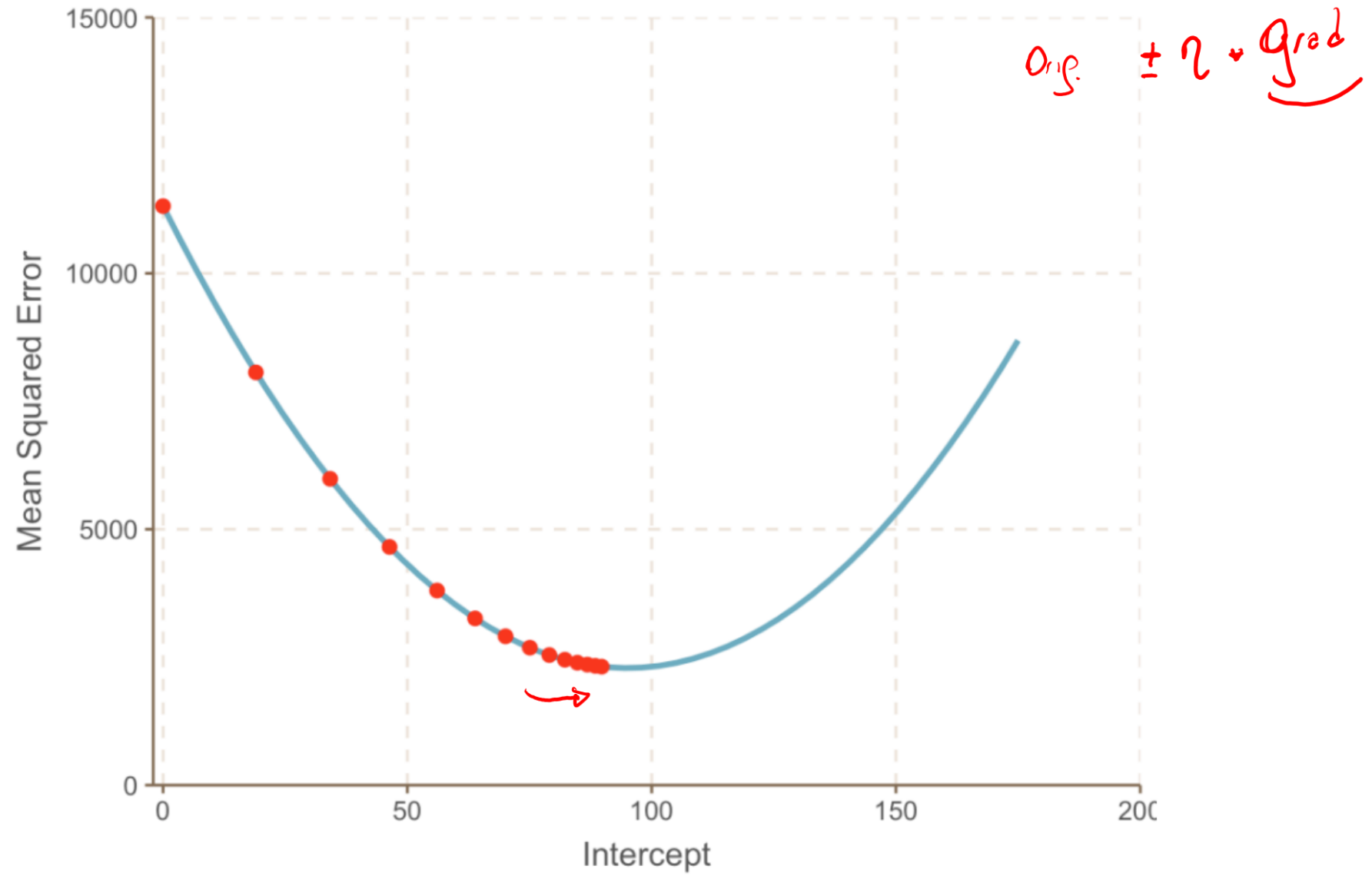
$$\begin{aligned}\text{New Intercept} &= \text{Old Intercept} - \text{Step Size} \\ &= 19 - (-15.2) \\ &= \underline{34.2}\end{aligned}$$





notice that  
the first step was  
slightly larger than  
the first





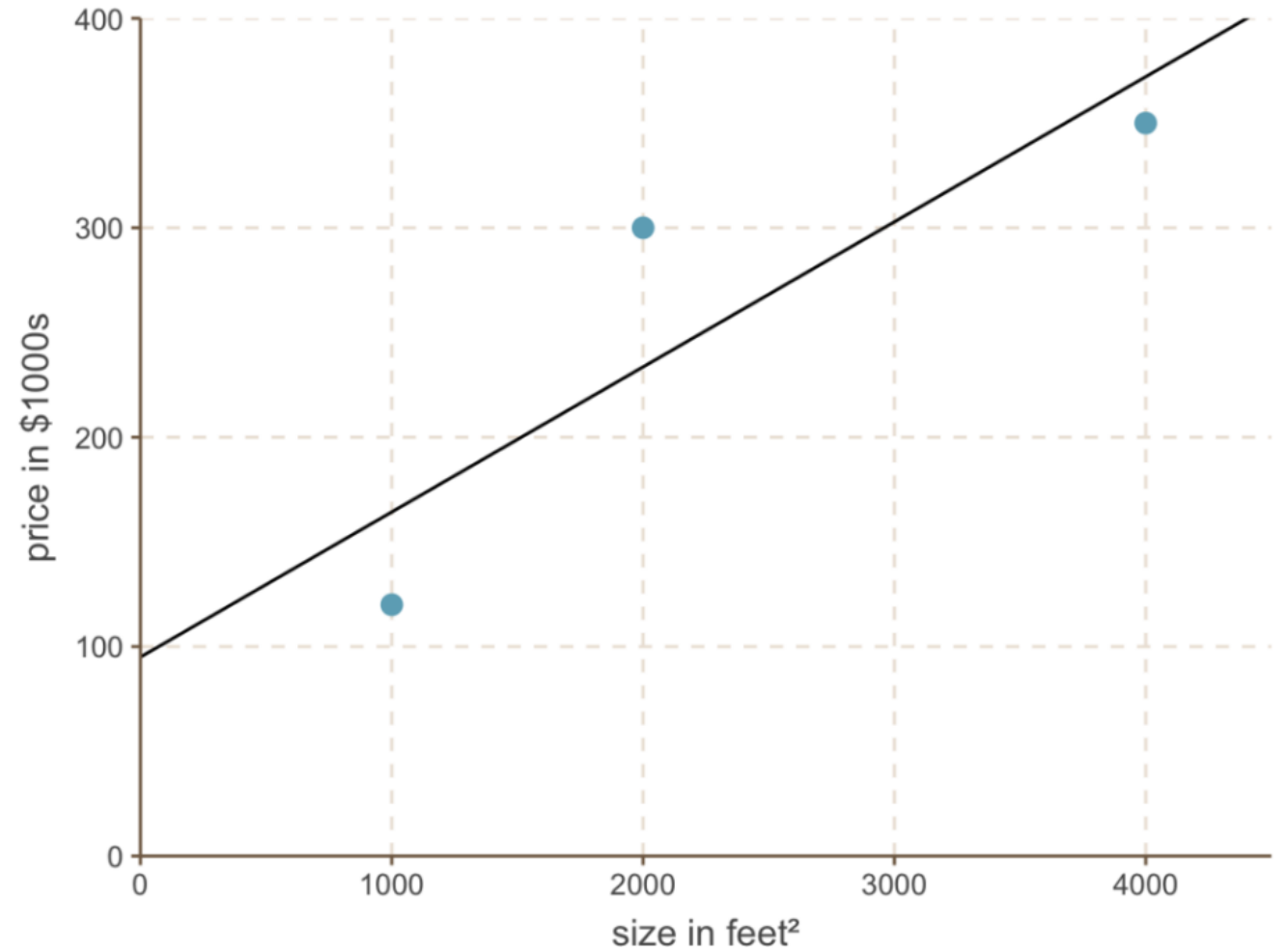


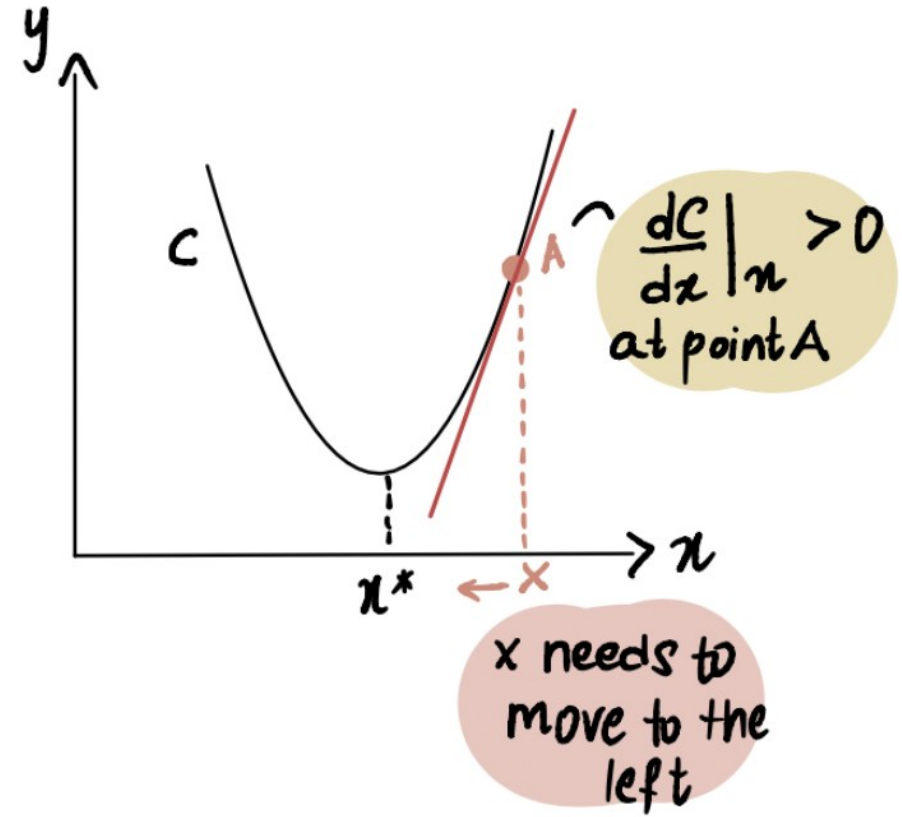
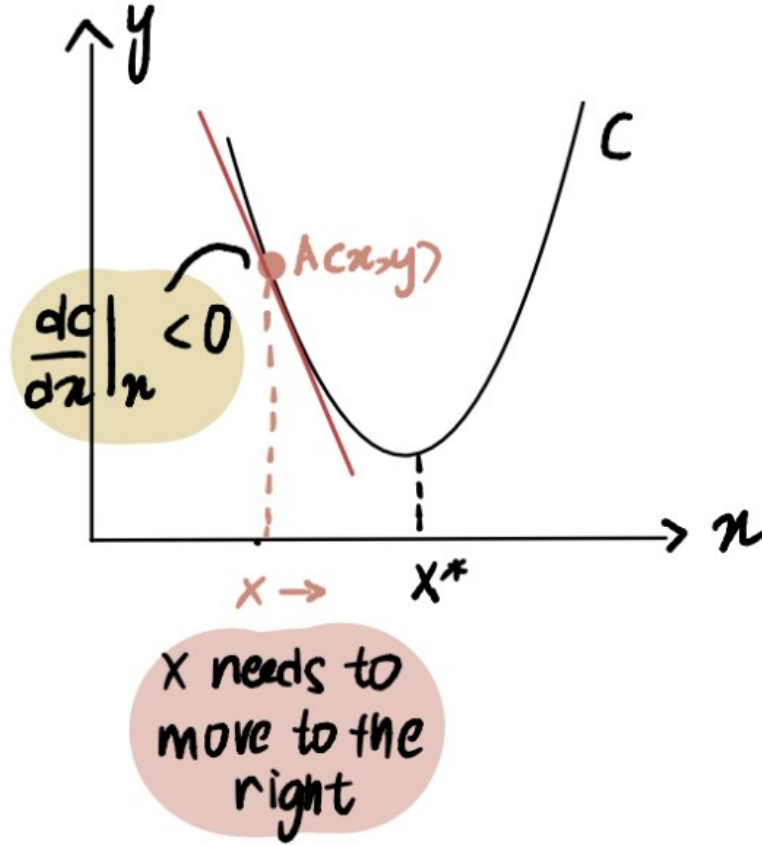
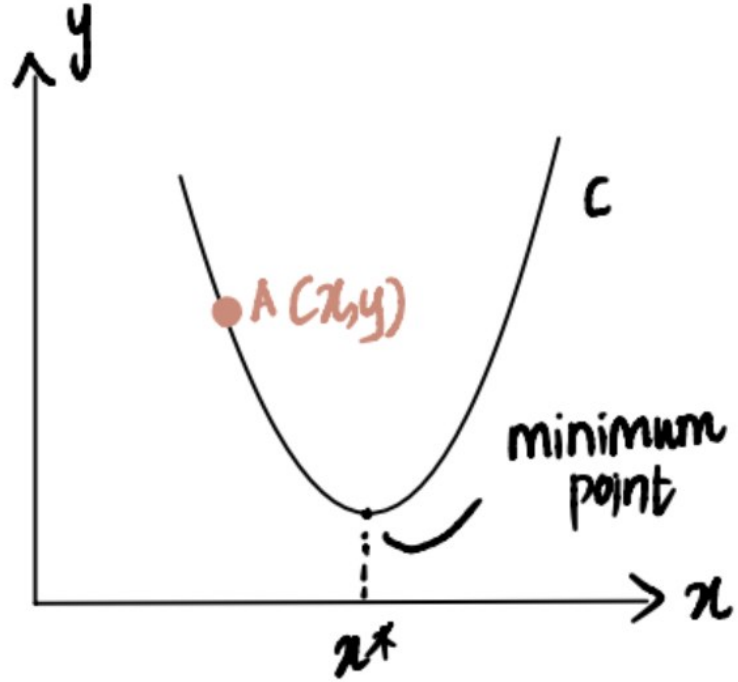
predicted house  
price

$$\hat{y} = 95 + 0.069 \times \text{size}$$

optimal  
intercept value  
we found

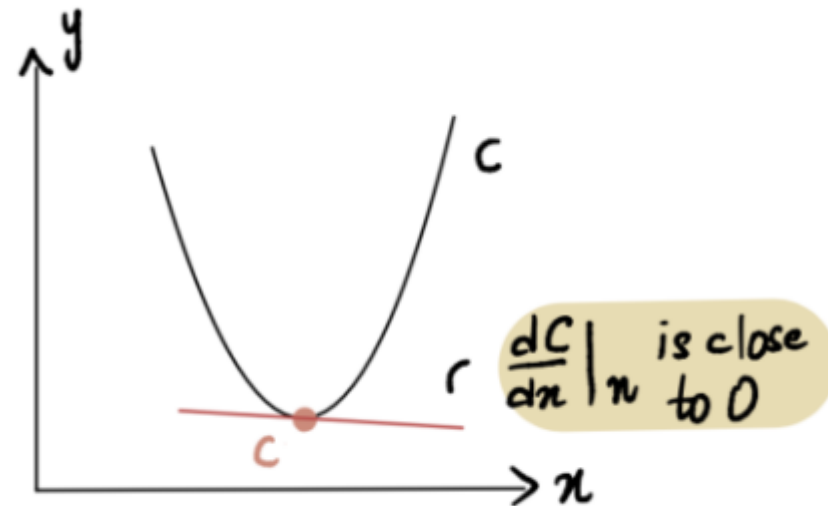
$$\begin{aligned}\hat{y} &= 95 + 0.069 \times 2400 \\ &= 260.6\end{aligned}$$

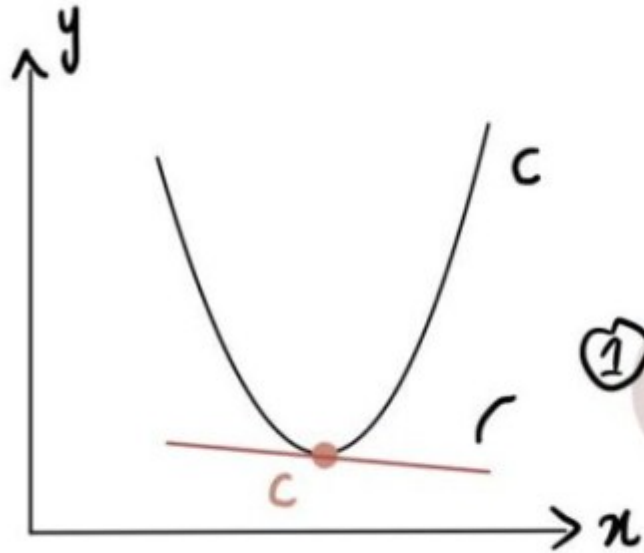




Step Size = Learning Rate  $\times$  Gradient

Step size will be close to 0 when gradient is close to 0

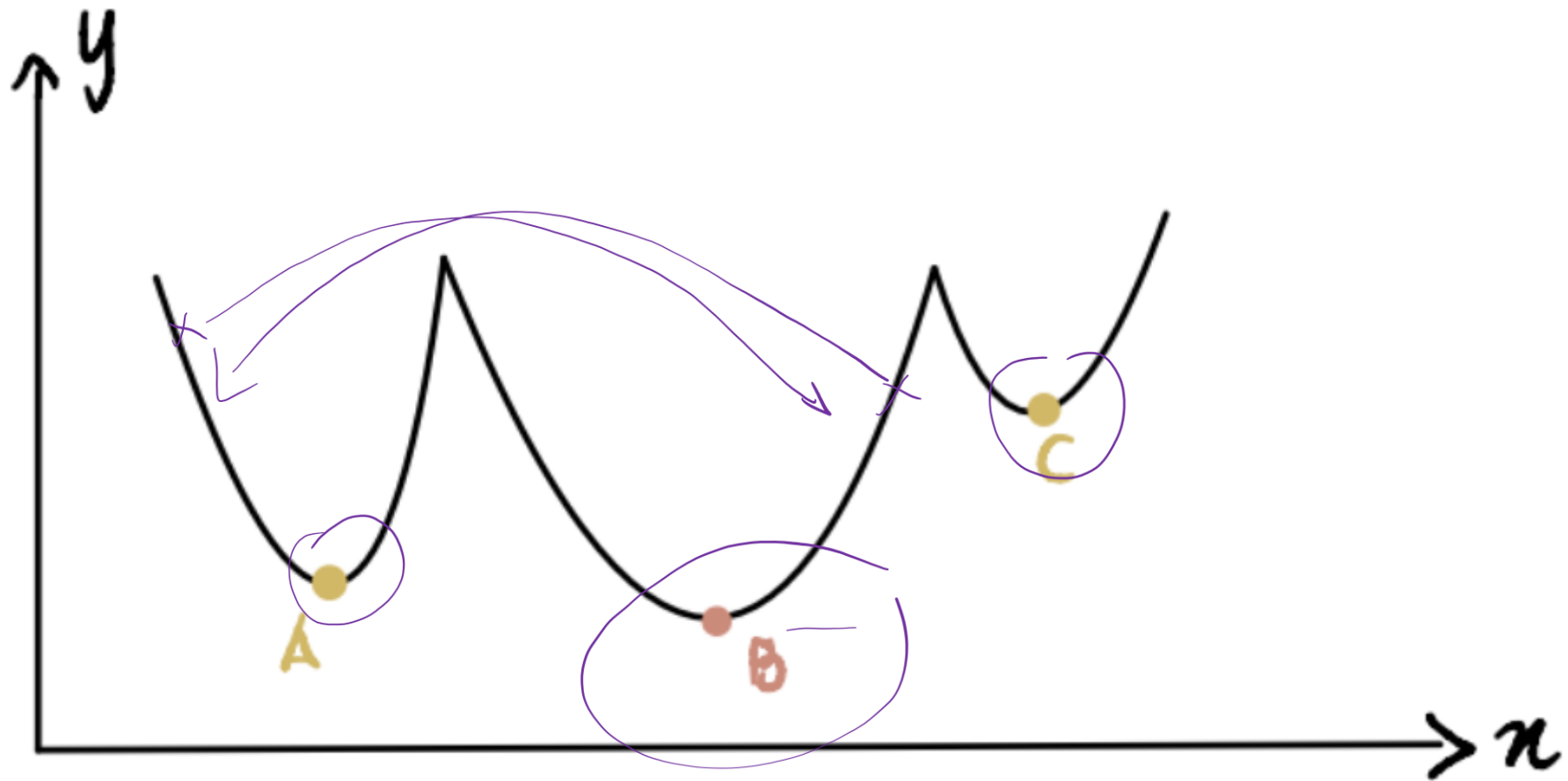


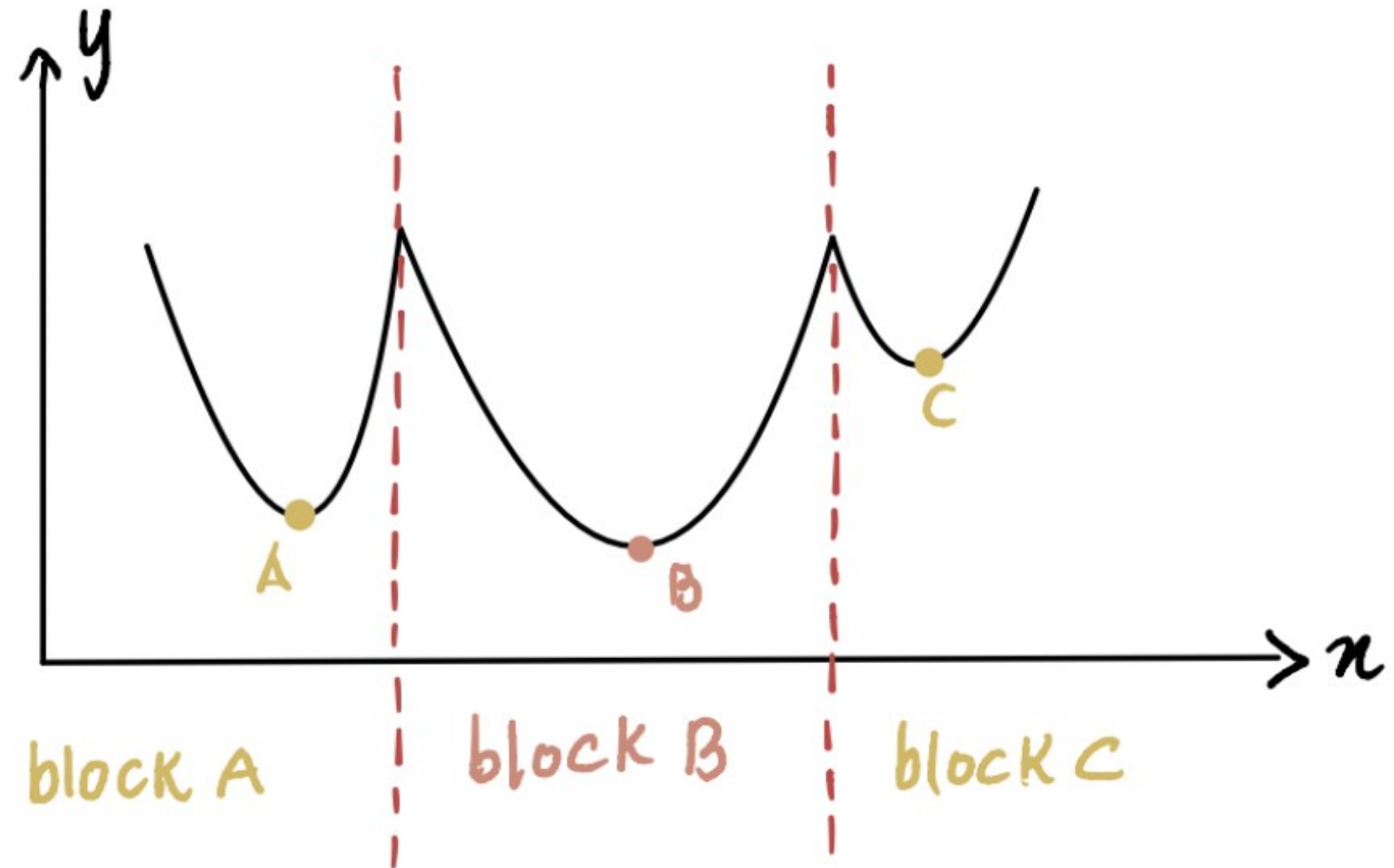


① if the gradient here = 0.009

② Step Size =  $0.009 \times 0.1$   
= 0.0009

③  $0.0009 < 0.001$ ,  
So Gradient Descent  
will stop





this symbol denotes partial derivative

$$\frac{\partial C(\pi_0, \pi_1)}{\partial \pi_0}, \quad \frac{\partial C(\pi_0, \pi_1)}{\partial \pi_1}$$

partial derivative with respect to  $\pi_0$       partial derivative with respect to  $\pi_1$

Step size for  $\pi_0$

$$\pi_0 = \pi_0 - \alpha \times \frac{\partial C(\pi_0, \pi_1)}{\partial \pi_0}$$

$$\pi_1 = \pi_1 - \alpha \times \frac{\partial C(\pi_0, \pi_1)}{\partial \pi_1}$$

Step size for  $\pi_1$

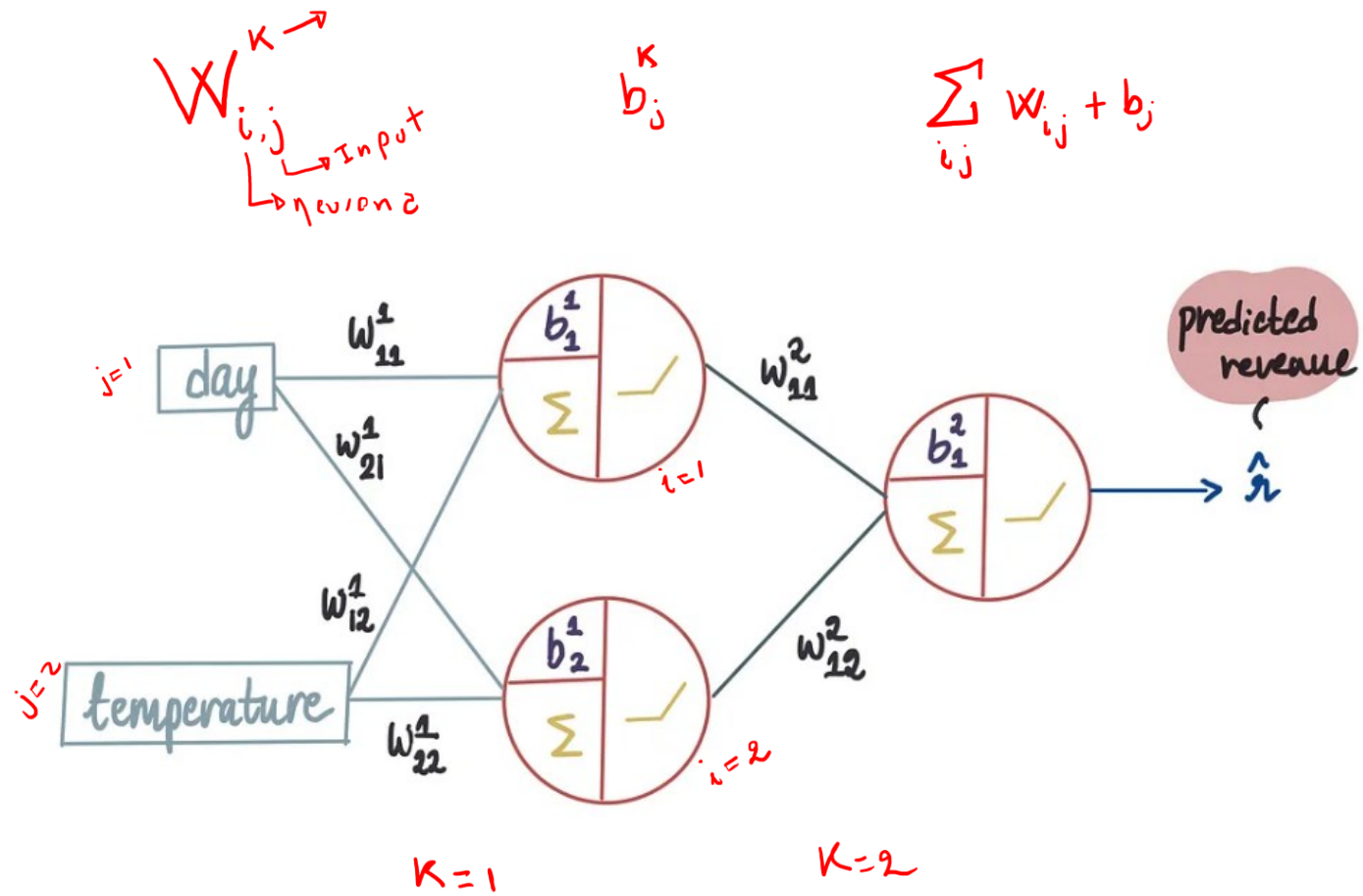
A Visual and Interactive Guide to the Basics of Neural  
Networks – Jay Alammam – Visualizing machine learning one  
concept at a time.

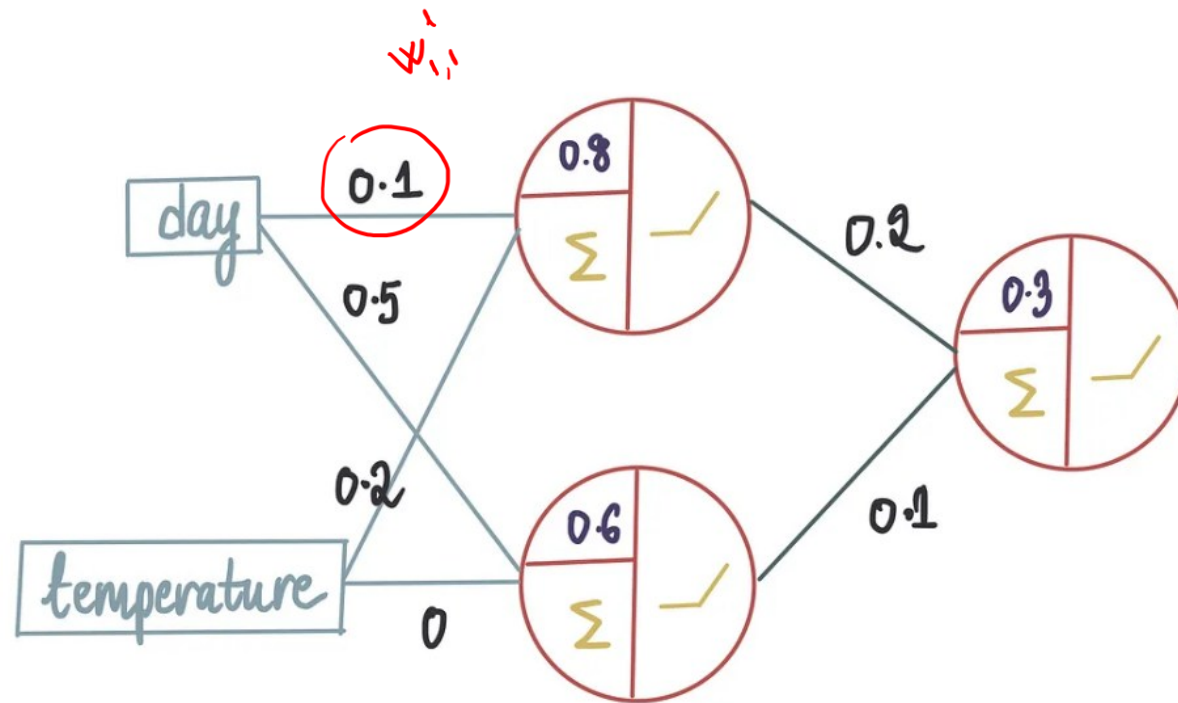


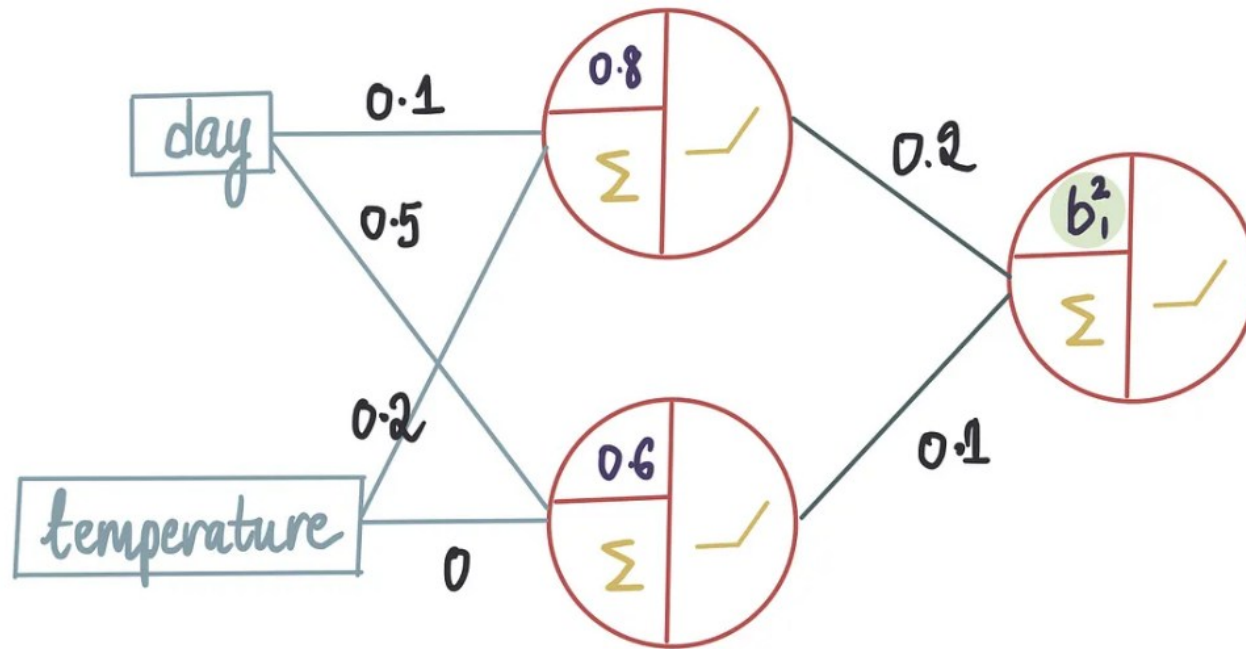
# **Introducción a Redes Neuronales**

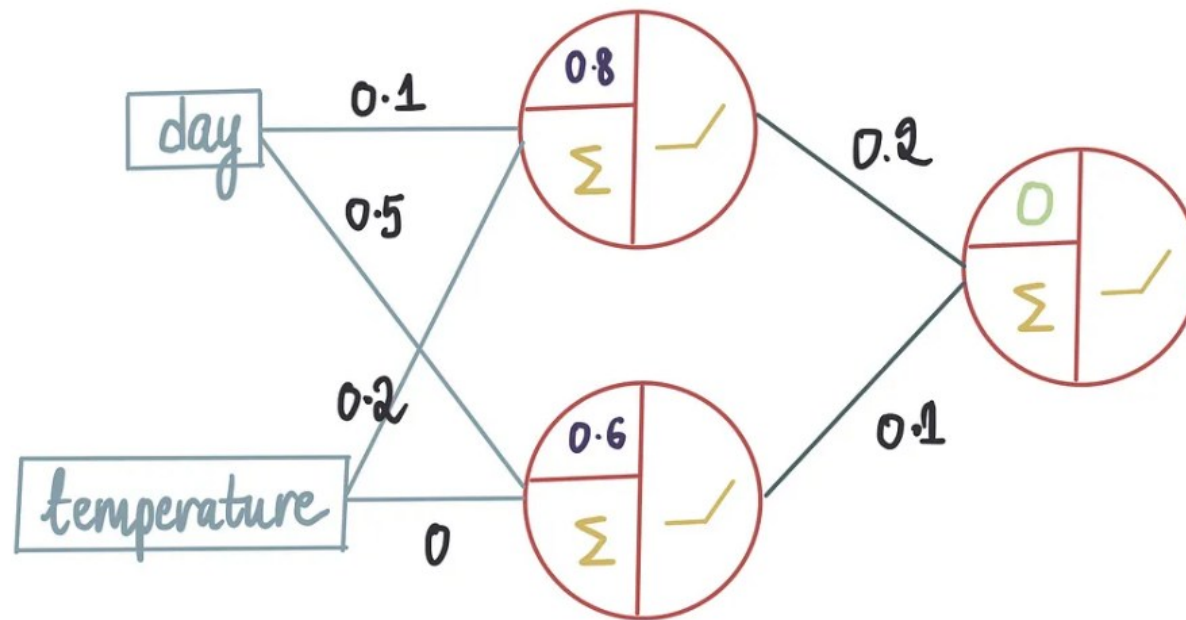
## **Back Propagation**

|                                                                   | day | temperature (°C) | revenue (in 1000s) |
|-------------------------------------------------------------------|-----|------------------|--------------------|
| <u>1</u> = Monday<br><u>2</u> = Tuesday<br>⋮<br><u>7</u> = Sunday | 2   | 22               | 1.51               |
|                                                                   | 6   | 33               | 2.22               |
|                                                                   | 1   | 20               | 1.37               |
|                                                                   | 3   | 25               | 1.77               |
|                                                                   | 2   | 24               | 1.64               |
|                                                                   | 5   | 30               | 2.04               |
|                                                                   | 7   | 35               | 2.42               |
|                                                                   | 4   | 28               | 1.90               |
|                                                                   | 3   | 26               | 1.75               |
|                                                                   | 1   | 21               | 1.45               |

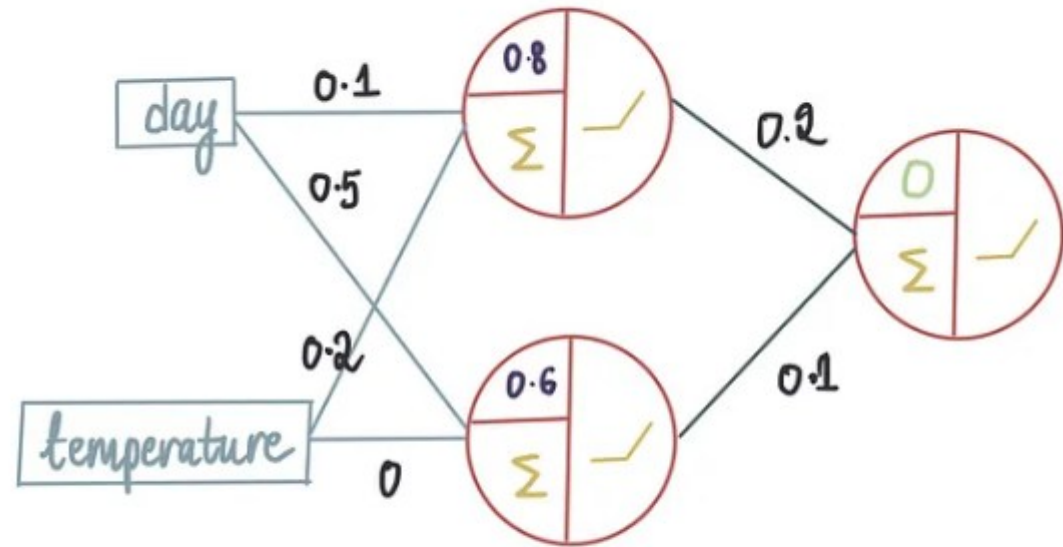








| day | temperature (°C) |
|-----|------------------|
| 2   | 22               |
| 6   | 33               |
| 1   | 20               |
| 3   | 25               |
| 2   | 24               |
| 5   | 30               |
| 7   | 35               |
| 4   | 28               |
| 3   | 26               |
| 1   | 21               |



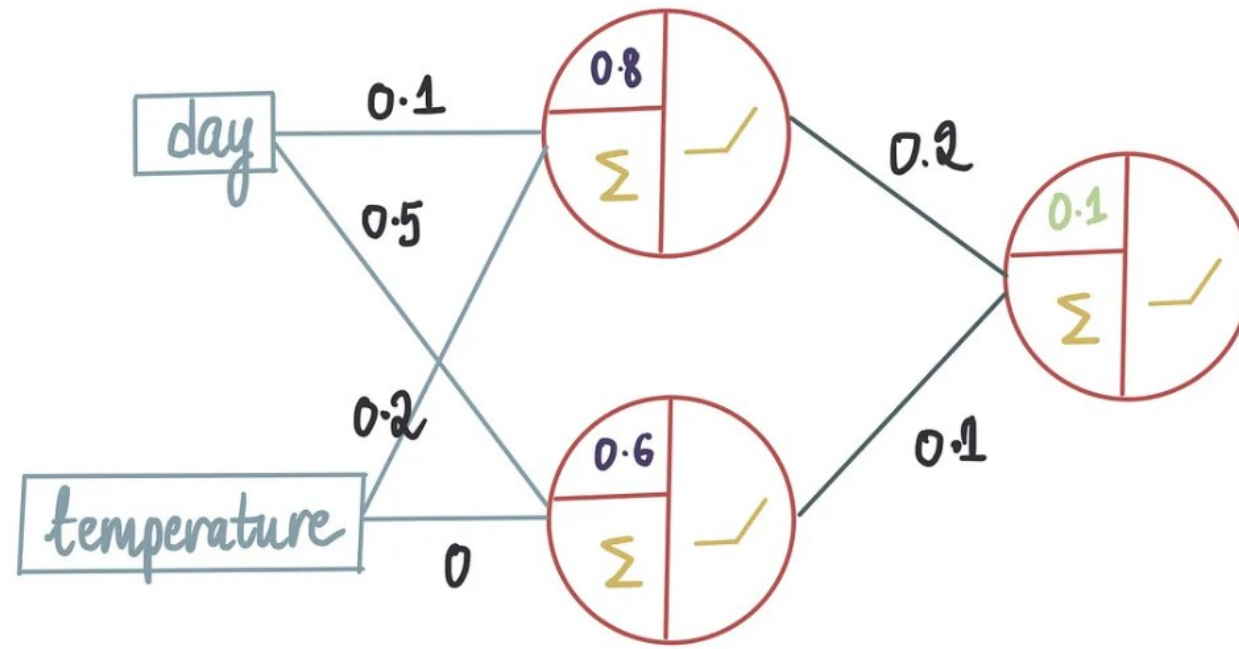
| day | temperature (°C) | revenue | predicted revenue ( $\hat{r}$ )<br>when bias = 0 |
|-----|------------------|---------|--------------------------------------------------|
| 2   | 22               | 1.51    | 1.24                                             |
| 6   | 33               | 2.22    | 1.96                                             |
| 1   | 20               | 1.37    | 1.09                                             |
| 3   | 25               | 1.77    | 1.43                                             |
| 2   | 24               | 1.64    | 1.32                                             |
| 5   | 30               | 2.04    | 1.77                                             |
| 7   | 35               | 2.42    | 2.11                                             |
| 4   | 28               | 1.90    | 1.62                                             |
| 3   | 26               | 1.15    | 1.47                                             |
| 1   | 21               | 1.45    | 1.23                                             |



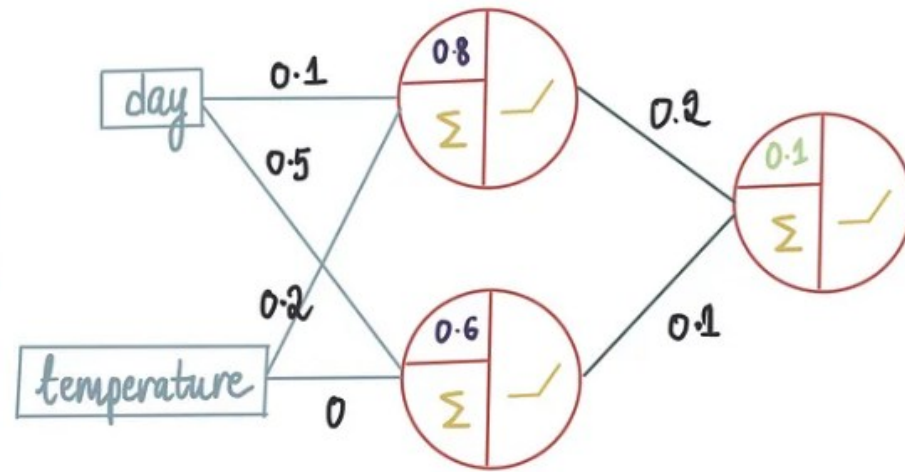
actual revenue      predicted revenue

$$MSE = \frac{1}{n} \sum (r - \hat{r})^2$$

$$\begin{aligned} MSE &= \frac{1}{10} [(1.51 - 1.24)^2 + (2.22 - 1.96)^2 + \dots \\ &\quad + (1.75 - 1.47)^2 + (1.45 - 1.13)^2] \\ &= 0.0851 \end{aligned}$$

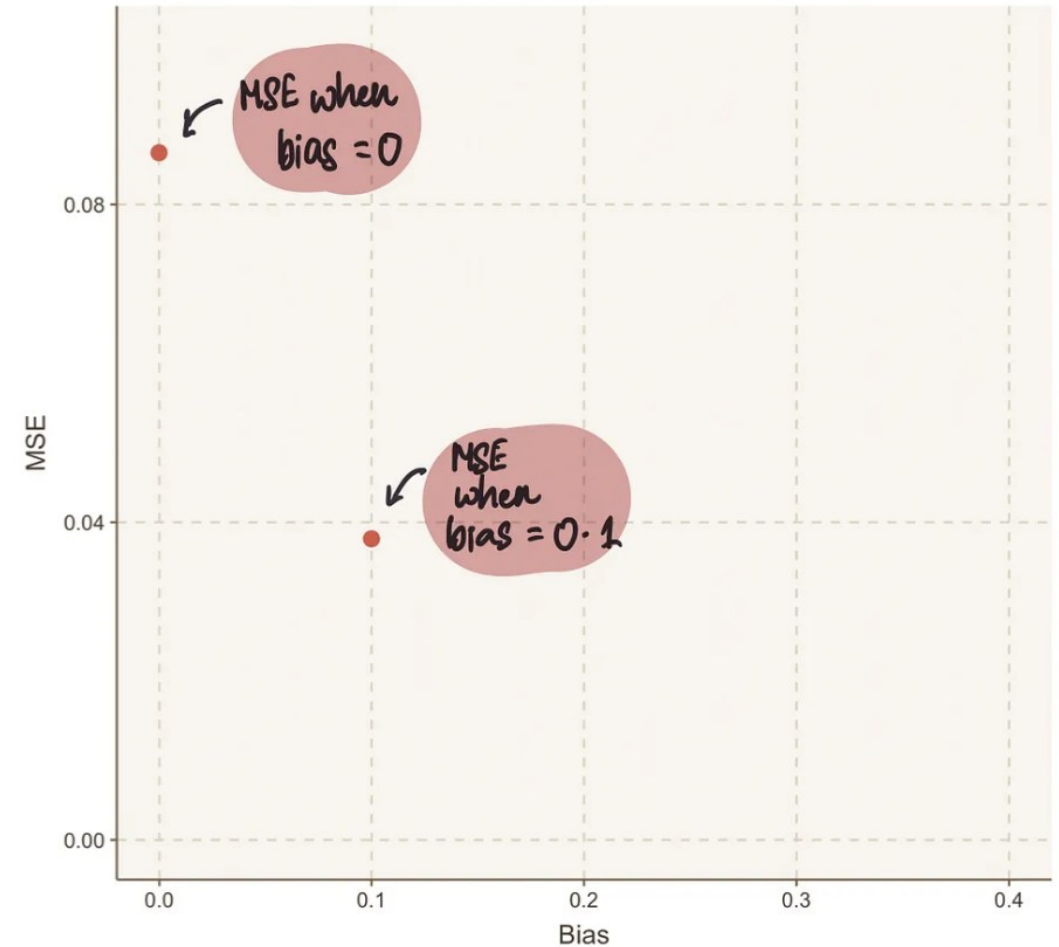


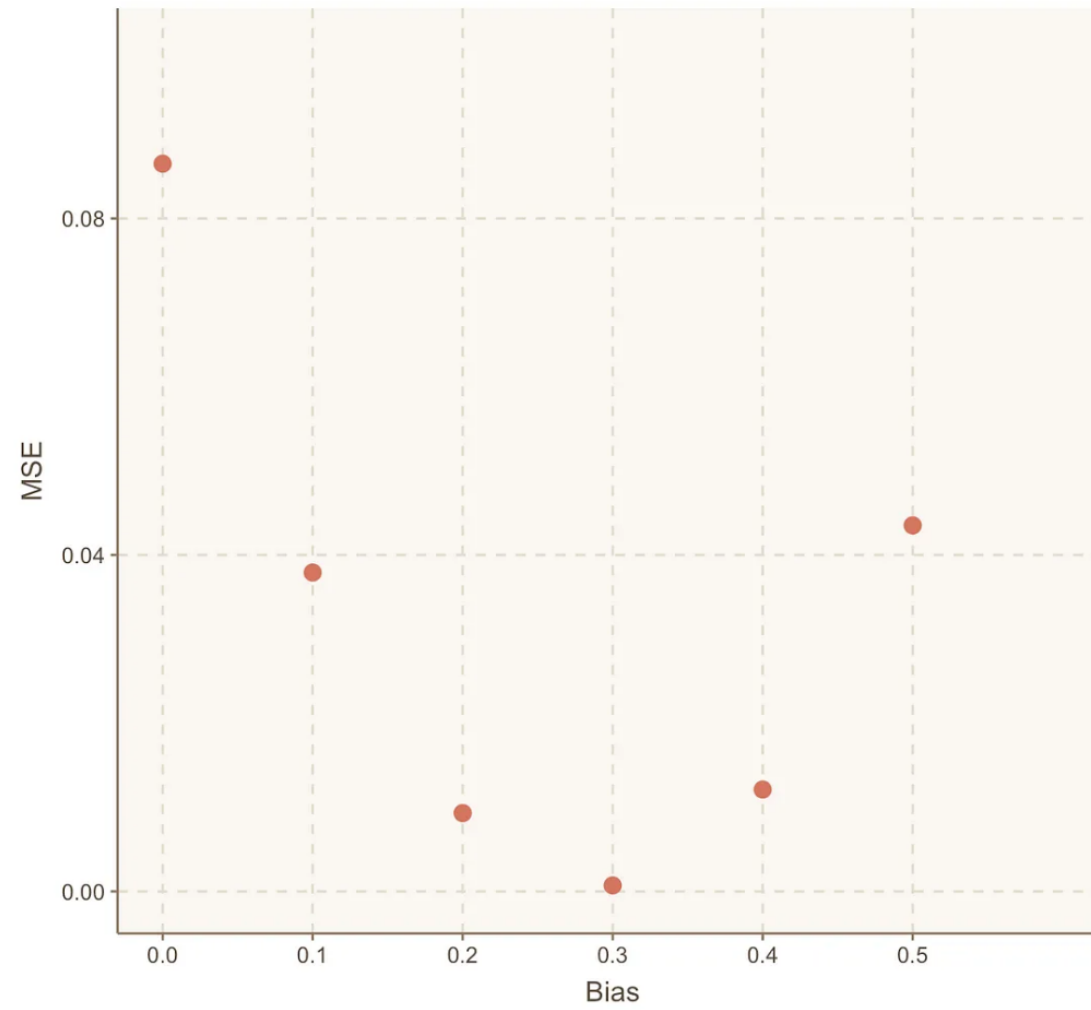
| day | temperature (°C) |
|-----|------------------|
| 2   | 22               |
| 6   | 33               |
| 1   | 20               |
| 3   | 25               |
| 2   | 24               |
| 5   | 30               |
| 7   | 35               |
| 4   | 28               |
| 3   | 26               |
| 1   | 21               |

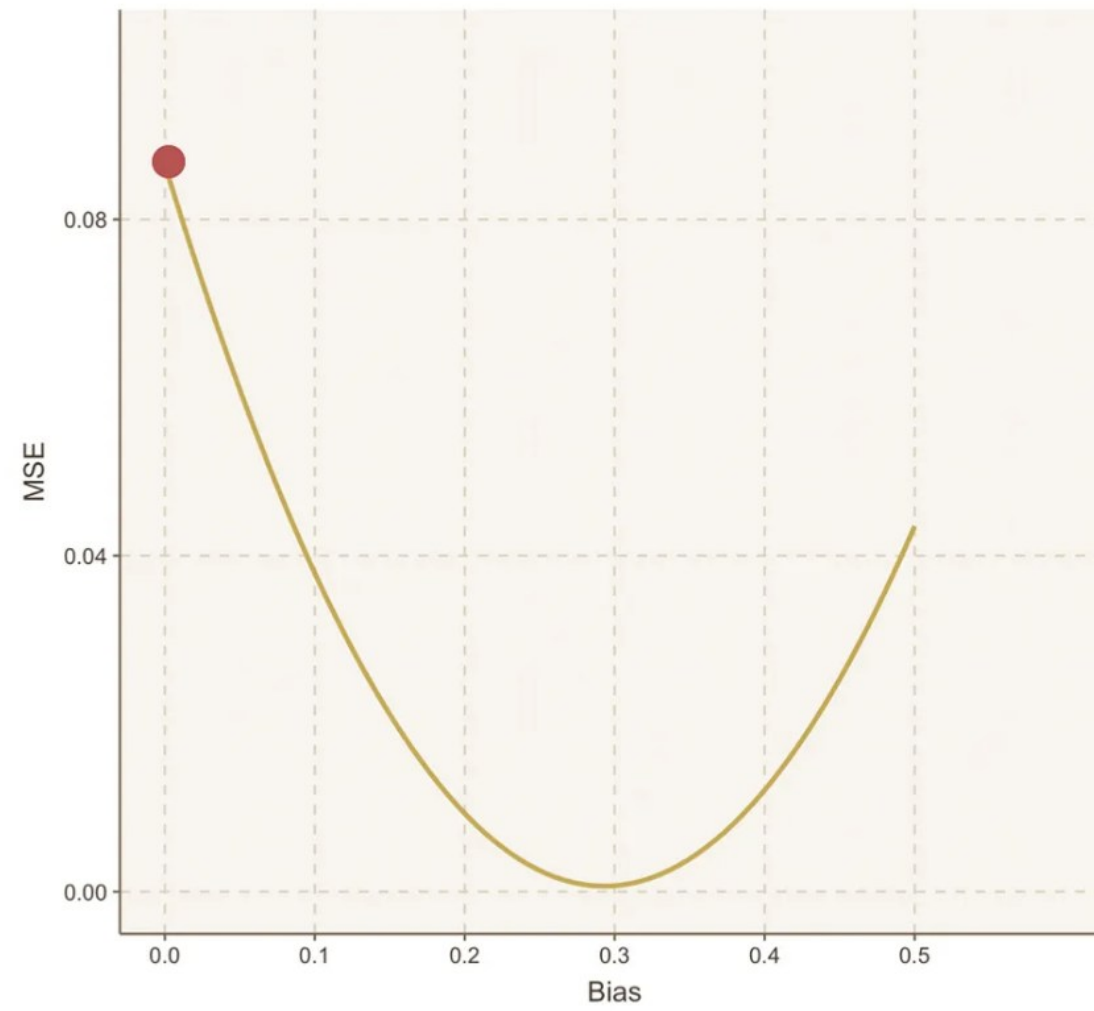


| day | temperature (°C) | revenue | predicted revenue<br>when bias = 0.1 |
|-----|------------------|---------|--------------------------------------|
| 2   | 22               | 1.51    | 1.34                                 |
| 6   | 33               | 2.22    | 2.06                                 |
| 1   | 20               | 1.37    | 1.19                                 |
| 3   | 25               | 1.77    | 1.53                                 |
| 2   | 24               | 1.64    | 1.42                                 |
| 5   | 30               | 2.04    | 1.87                                 |
| 7   | 35               | 2.42    | 2.21                                 |
| 4   | 28               | 1.90    | 1.72                                 |
| 3   | 26               | 1.75    | 1.57                                 |
| 1   | 21               | 1.45    | 1.23                                 |

$$\begin{aligned} \text{MSE} &= \frac{1}{10} [(1.51 - 1.34)^2 + (2.22 - 2.06)^2 + \dots \\ &\quad + (1.75 - 1.57)^2 + (1.45 - 1.23)^2] \\ &= 0.03791 \end{aligned}$$







$$\text{Step size} = \text{learning rate} \times \left. \frac{d\text{MSE}}{dbias} \right|_{bias=1}$$

*Salto* *r* *pendiente*

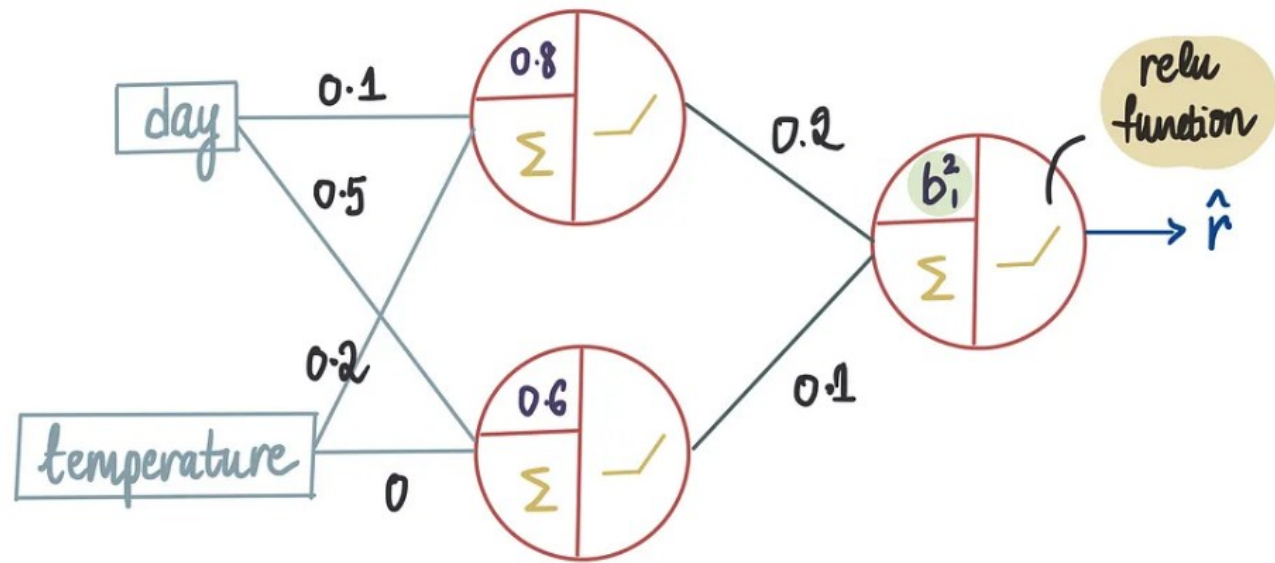


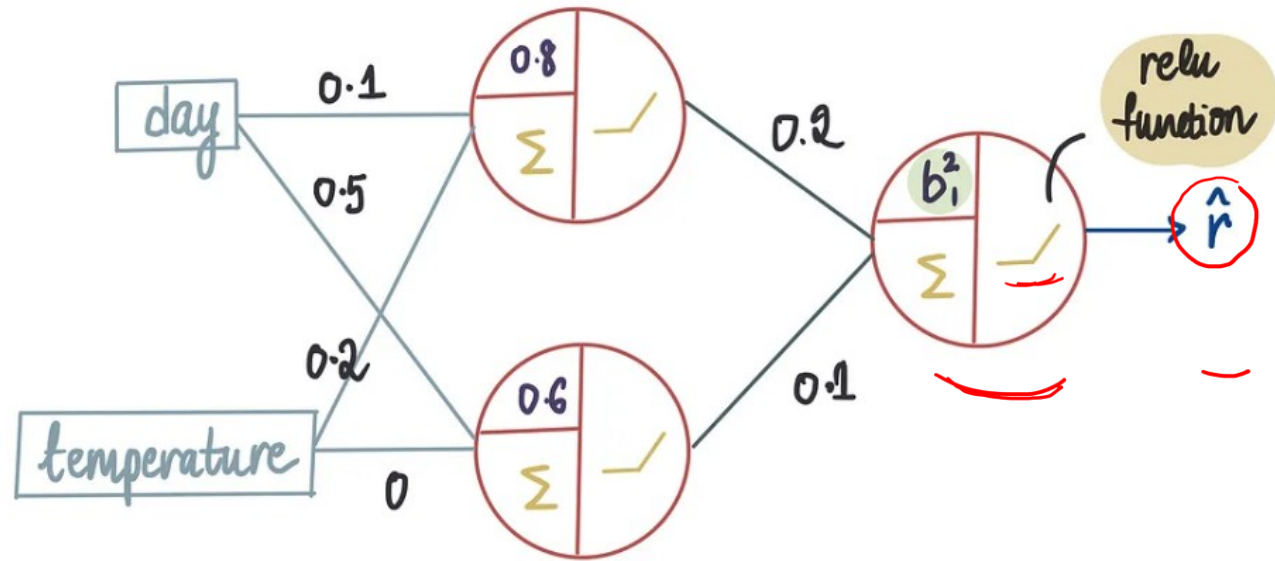
$$\text{Step size} = \text{learning rate} \times \left. \frac{d\text{MSE}}{dbias} \right|_{bias}$$

$$\text{MSE} = \frac{1}{n} \sum (r - \hat{r})^2$$

actual revenue

predicted revenue



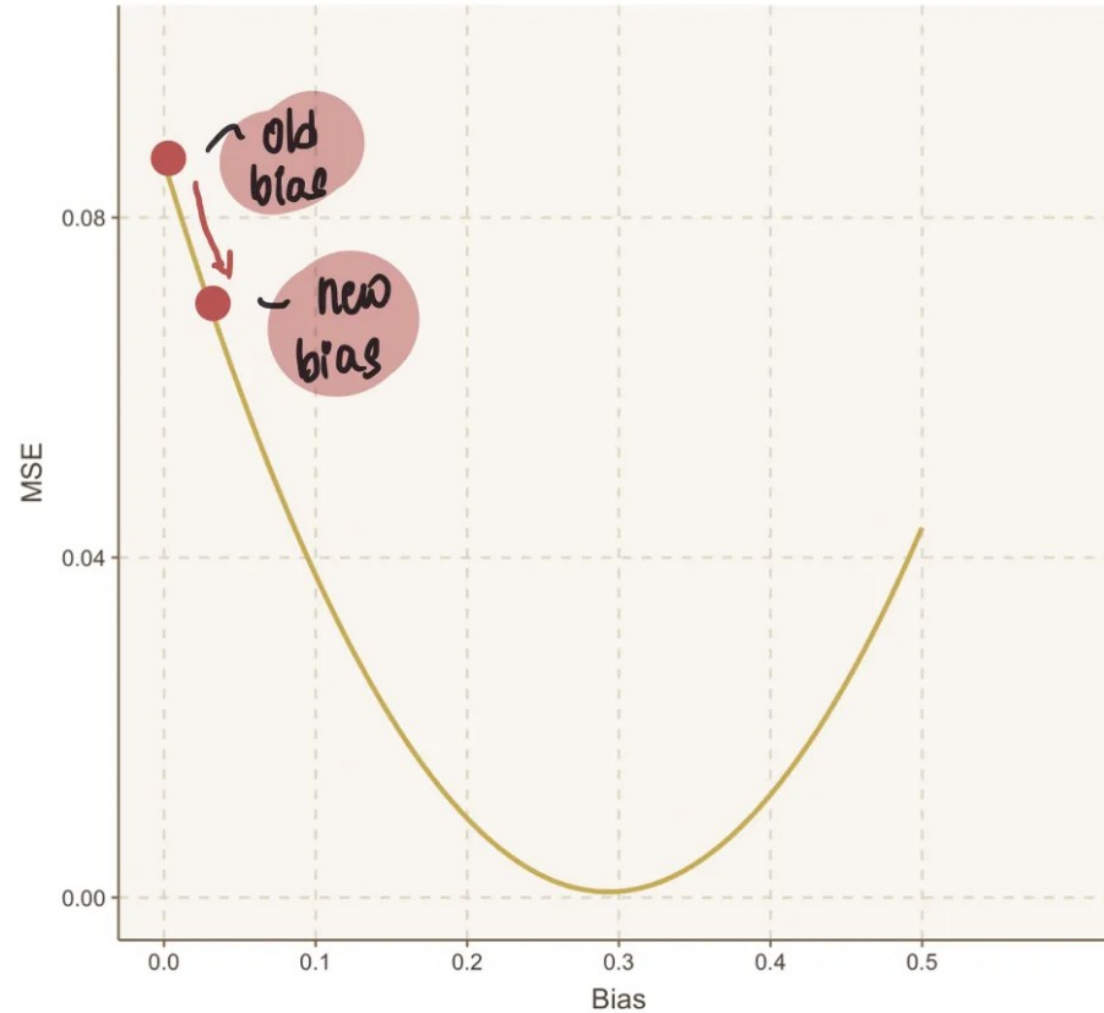


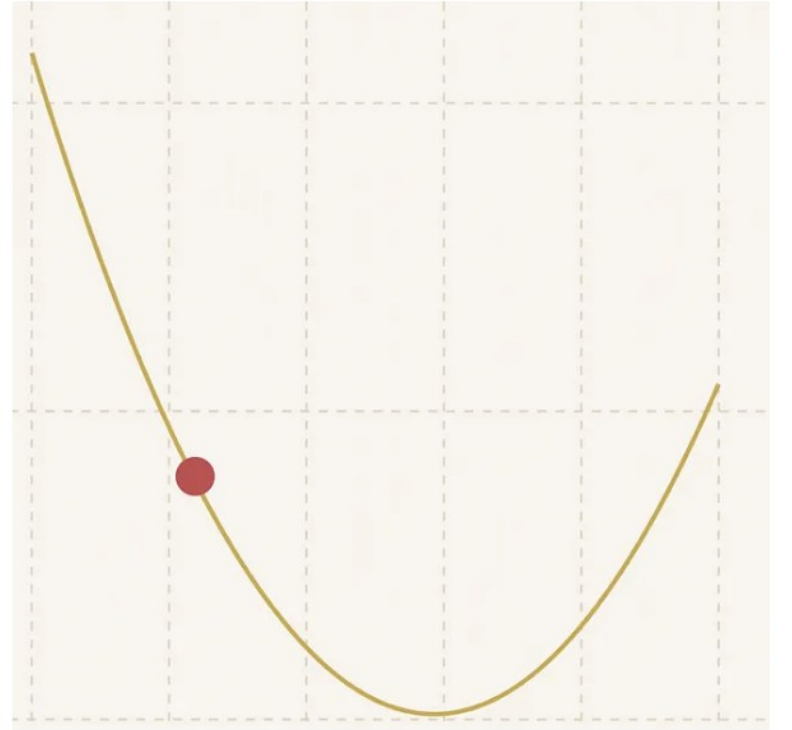
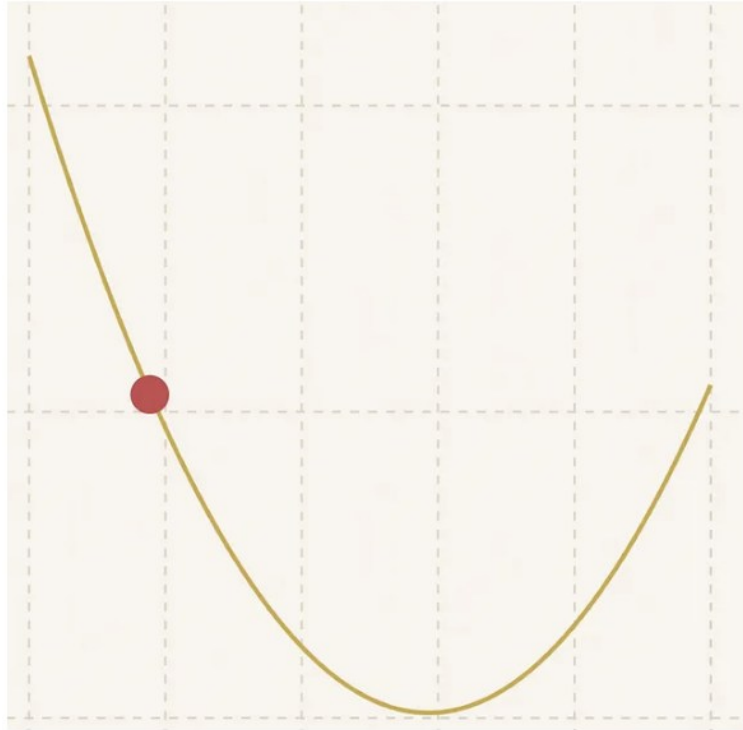
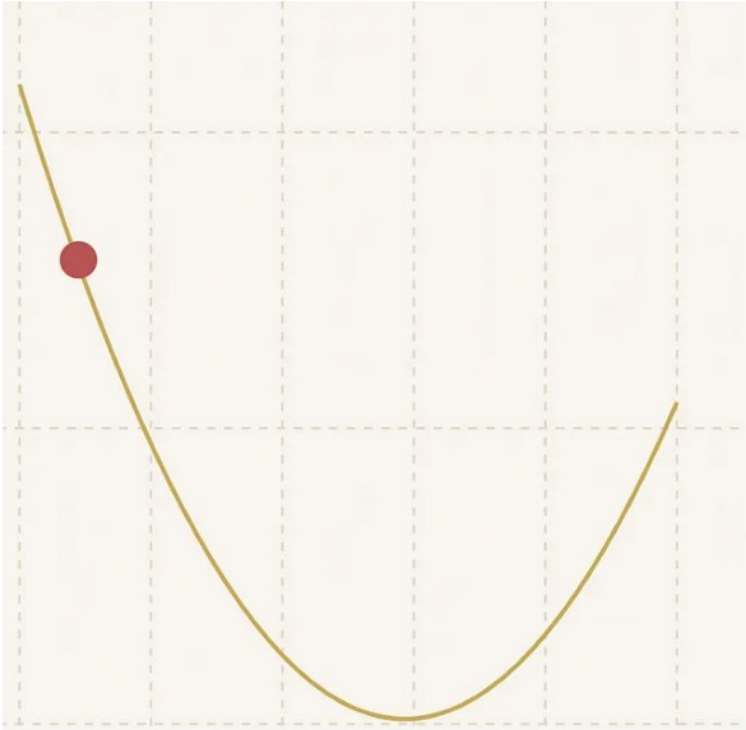
Error

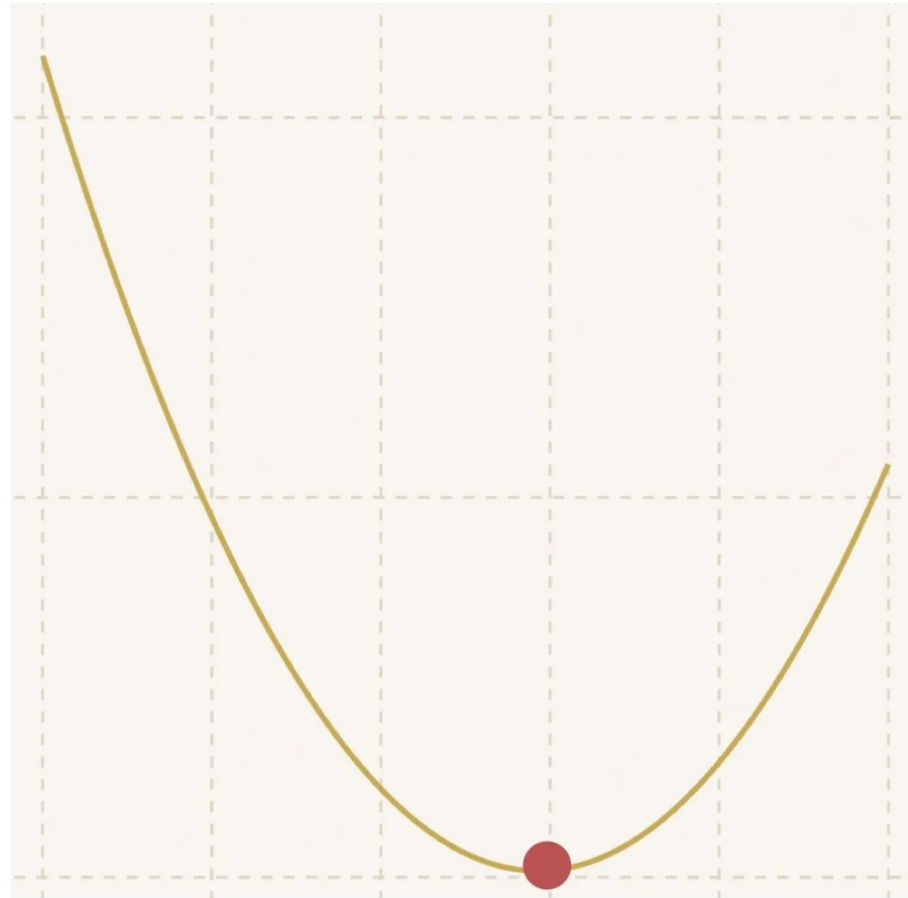
$$\frac{dMSE}{dbias} = \frac{dMSE}{d\hat{r}} \cdot \frac{d\hat{r}}{d\text{relu}} \cdot \frac{d\cdot\text{relu}}{dbias}$$

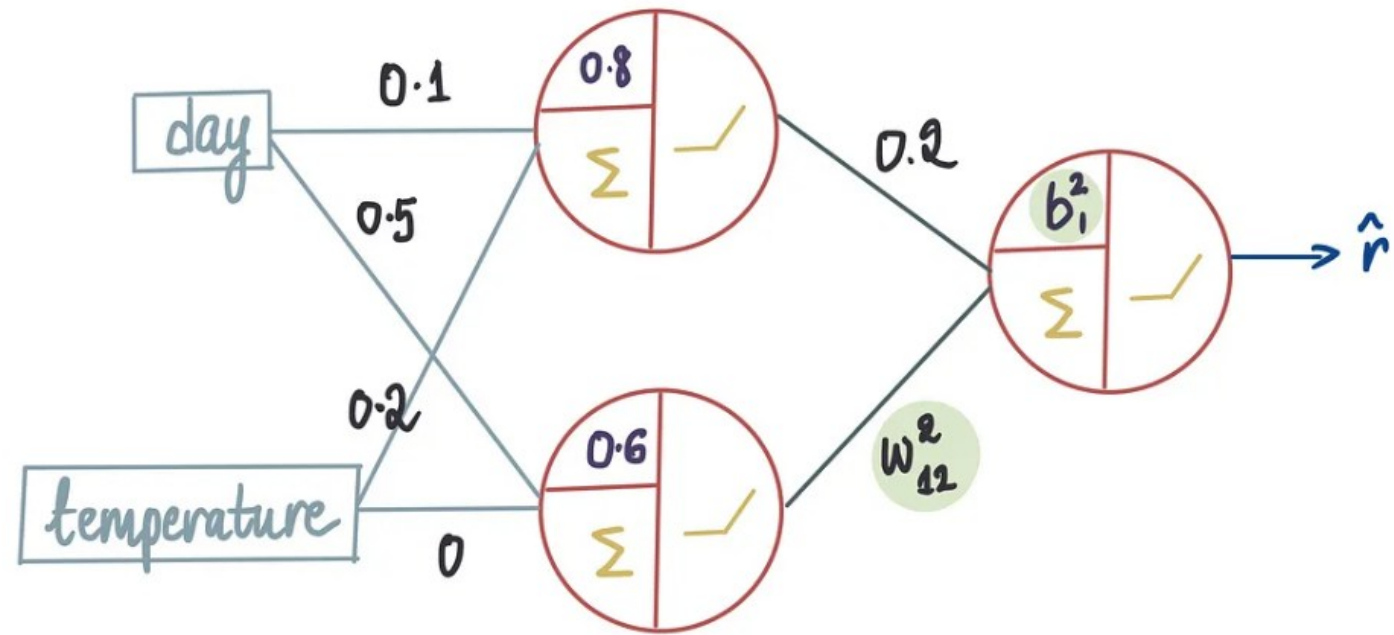
$$\begin{aligned}\text{new bias} &= \text{old bias} - \text{step size} \\ &= \text{old bias} - \text{learning rate} \times \left. \frac{d\text{MSE}}{dbias} \right|_{\text{old bias}}\end{aligned}$$

$$\begin{aligned}\text{new bias} &= \text{old bias} - \text{step size} \\ &= \text{old bias} - \text{learning rate} \times \left. \frac{d\text{MSE}}{dbias} \right|_{\text{old bias}}\end{aligned}$$

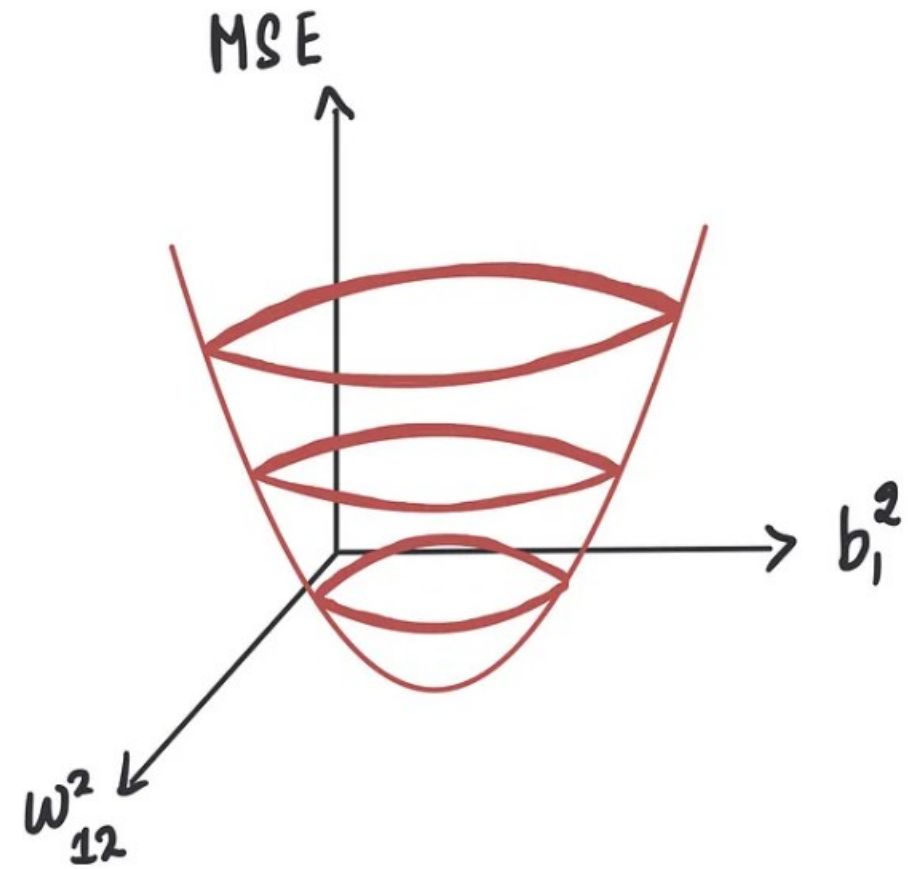












$$b_2^3 = b_2^3 - \eta \cdot \frac{\partial MSE}{\partial b_2^3}$$

$$\text{new } \underline{b_1^2} \text{ step size} = \overbrace{(\text{learning rate})}^{\eta} \cdot \frac{\partial MSE}{\partial b_1^2}$$

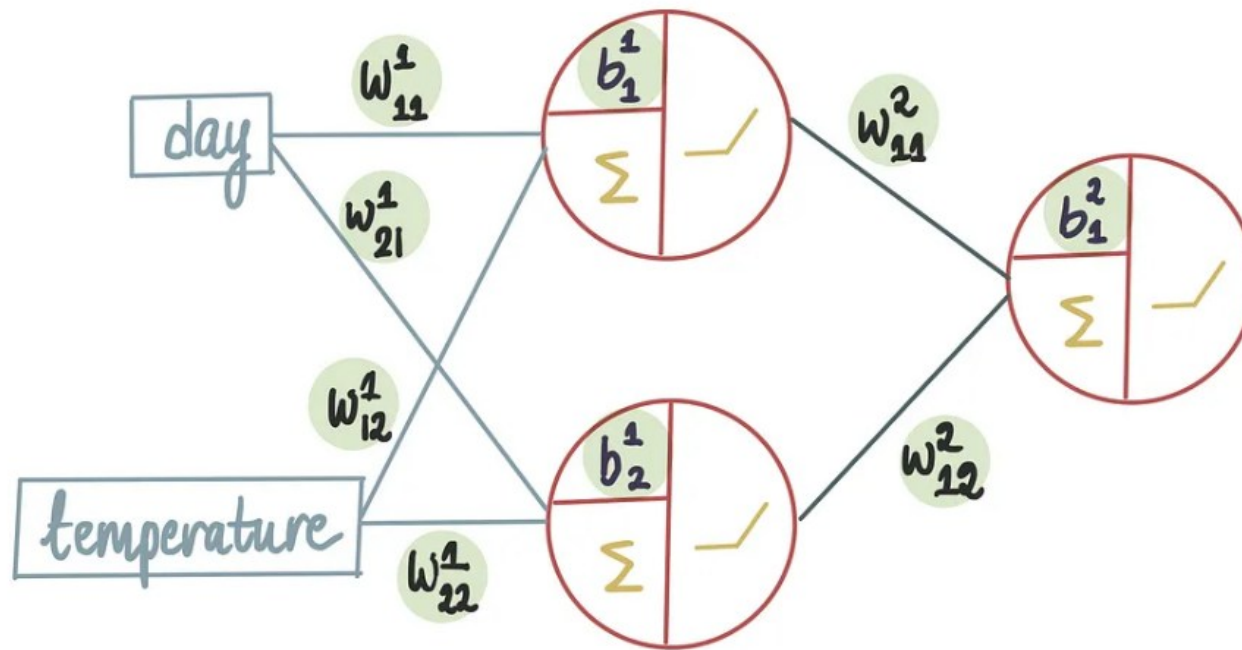
$$\text{new } w_{12}^2 \text{ step size} = (\text{learning rate}) \cdot \frac{\partial MSE}{\partial w_{12}^2}$$

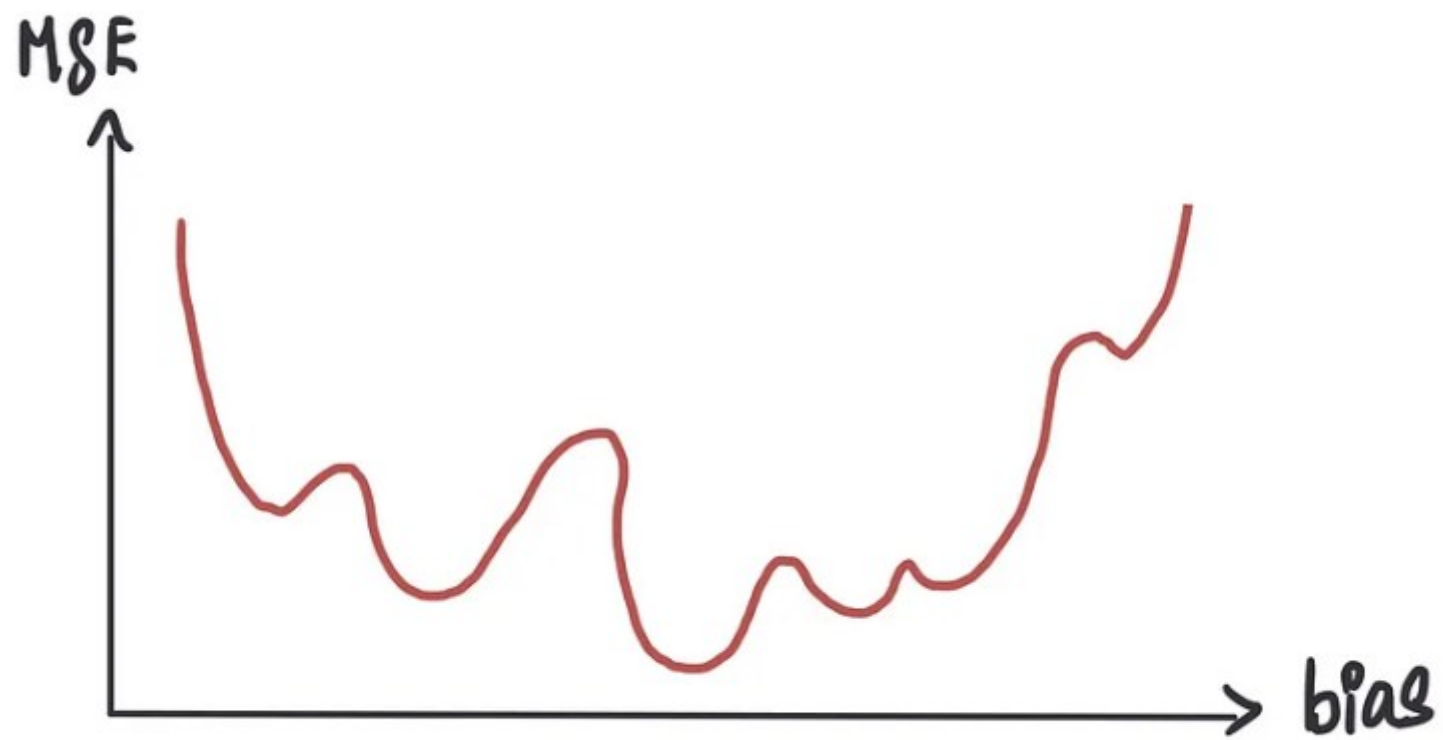
$$\frac{\partial \text{MSE}}{\partial b_1^2} = \frac{\partial \text{MSE}}{\partial \hat{r}} \cdot \frac{\partial \hat{r}}{\partial \text{relu}} \cdot \frac{\partial \text{relu}}{\partial b_1^2}$$

$$\frac{\partial \text{MSE}}{\partial w_{12}^2} = \frac{\partial \text{MSE}}{\partial \hat{r}} \cdot \frac{\partial \hat{r}}{\partial \text{relu}} \cdot \frac{\partial \text{relu}}{\partial w_{12}^2}$$

$$\text{new } b_1^2 = \text{old } b_1^2 - (\text{learning rate}) \cdot \frac{\partial \text{MSE}}{\partial b_1^2}$$

$$\text{new } w_{12}^2 = \text{old } w_{12}^2 - (\text{learning rate}) \cdot \frac{\partial \text{MSE}}{\partial w_{12}^2}$$





| day | temperature (°C) | revenue (in 1000s) |
|-----|------------------|--------------------|
| 2   | 22               | 1.51               |
| 6   | 33               | 2.22               |
| 1   | 20               | 1.37               |
| 3   | 25               | 1.77               |
| 2   | 24               | 1.64               |
| 5   | 30               | 2.04               |
| 7   | 35               | 2.42               |
| 4   | 28               | 1.90               |
| 3   | 26               | 1.15               |
| 1   | 21               | 1.45               |

## gradient descent

① initialize parameters

$b = 0$   
 $W_0^n = \frac{1}{n}$

② repeat until we reach optimal parameter values:

a) calculate step sizes

learning rate  $\cdot \frac{\partial \text{MSE}}{\partial b_1^2}$

uses MSE from all data points

b) update all the parameters

new  $b_1^2 = \text{old } b_1^2 - \text{step size of } b_1^2$

| day | temperature (°C) | revenue (in 1000s) |
|-----|------------------|--------------------|
| 2   | 22               | 1.51               |
| 6   | 33               | 2.22               |
| 1   | 20               | 1.37               |
| 3   | 25               | 1.77               |
| 2   | 24               | 1.64               |
| 5   | 30               | 2.04               |
| 7   | 35               | 2.42               |
| 4   | 28               | 1.90               |
| 3   | 26               | 1.15               |
| 1   | 21               | 1.45               |

data  
point i

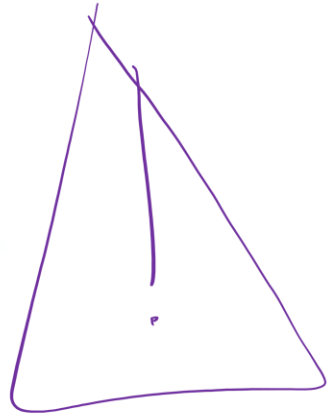
## Stochastic gradient descent

- ① initialize parameters
- ② repeat until we reach optimal parameter values:

a) calculate step sizes  
 (learning rate)  $\cdot \frac{\partial \text{MSE}_i}{\partial b_1^2}$  use MSE from data point i

b) update all the parameters

$$\text{new } b_1^2 = \text{old } b_1^2 - \text{step size of } b_1^2$$



|              | day | temperature (°C) | revenue (in 1000s) |
|--------------|-----|------------------|--------------------|
|              | 2   | 22               | 1.51               |
| batch<br>B \ | 6   | 33               | 2.22               |
|              | 1   | 20               | 1.37               |
|              | 3   | 25               | 1.77               |
|              | 2   | 24               | 1.64               |
|              | 5   | 30               | 2.04               |
|              | 7   | 35               | 2.42               |
|              | 4   | 28               | 1.90               |
|              | 3   | 26               | 1.15               |
|              | 1   | 21               | 1.45               |

## mini-batch gradient descent

① initialize parameters

② repeat until we reach optimal parameter values:

a) calculate step sizes

$$(\text{learning rate}) \cdot \frac{1}{B} \sum_{i=1}^B \frac{\partial \text{MSE}}{\partial b_1^2}$$

MSE using the datapoints in the batch

b) update all the parameters

$$\text{new } b_1^2 = \text{old } b_1^2 - \text{step size of } b_1^2$$